MAT2362 – Fall 2023

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Contents

1 Induction **4**
1.1 Simple induction 4

2 Propositional Logic **6**
2.1 Propositions and connectives 6
2.2 Equivalence of formulas 12
2.3 Proof of equivalence by algebraic manipulations 14
2.4 Contrapositive and converse of an implication 16
2.5 Elimination of Parentheses 17
2.6 Tautologies and contradictions 19

3 Methods of Proof **21**
3.1 Proving an implication $P \Rightarrow Q$ 21
3.2 Proving an equivalence 26

4 Using Quantifiers **28**
4.1 A concise prologue 28
4.2 Some notation 28
4.3 The existential quantifier 29
4.4 The universal quantifier 30
4.5 Variations 31
4.6 Negation of a quantifier 31
4.7 Free and bound variables 32
4.8 Formulas with several quantifiers 33
4.9 Order of quantifiers 34
4.10 Negation of a formula with several quantifiers 36
4.11 An application: continuity 36
4.12 Proving statements which contain quantifiers 38
4.13 Proving $\exists_{x\in A}\varphi(x)$ 38
4.14 Proving $\forall_{x\in A}\varphi(x)$ 39
4.15 Disproving statements which contain quantifiers 41

5 Naïve Set Theory **43**
5.1 What is a set, really? . . . 43
5.2 Sets of numbers . . . 45
5.3 Subsets defined by a property . . . 46
5.4 Intervals . . . 47
5.5 Sets of sets . . . 48
5.6 Power sets . . . 49
5.7 Indexed sets . . . 50
5.8 Operations with sets . . . 51
5.9 Union . . . 51
5.10 Intersection . . . 53
5.11 Difference . . . 55
5.12 Symmetric Difference . . . 56
5.13 Cartesian product . . . 58

6 Functions **60**
6.1 Definition of functions . . . 60
6.2 Composition of functions . . . 61
6.3 The graph of a function . . . 62
6.4 Further examples of functions . . . 63

7 Injections, Surjections, Bijections **65**
7.1 Injections . . . 65
7.2 Surjections . . . 66
7.3 A characterisation of injections and surjections . . . 68
7.4 Bijections . . . 68

8 Binary Operations **71**
8.1 Binary operations on sets . . . 71
8.2 Binary operations with extra properties . . . 71
8.3 Operations with functions . . . 74
8.4 An alternative characterization of injections and surjections . . . 75

9 Relations **77**
9.1 Basic properties of relations . . . 78
9.2 Equivalence relations . . . 79
9.3 Equivalence classes and partitions . . . 80
9.4 Canonical projection and induced map . . . 85
9.5 Partial orders . . . 87
9.6 Transfinite induction . . . 91

10 Cardinality **96**
10.1 Sets of equal cardinality 96
10.2 Comparing cardinalities 99

11 Countable Sets **102**
11.1 Definition of a countable set 102
11.2 Adjoining countably many elements 105

12 Axiomatic Set Theory **107**
12.1 Cantor's set theory and Russell's paradox 107
12.2 Russell's paradox 107
12.3 The ZFC universe 108
12.4 Some notation 108
12.5 The first three axioms of ZFC 109
12.6 Axiom of infinity 114
12.7 Natural numbers 115
12.8 Axiom of foundation 116

13 Cantor's Diagonal Argument **120**
13.1 Intervals and cardinality 120
13.2 Sets of strictly bigger cardinality than $\mathbb{N}$ 121
13.3 Cantor's diagonal argument and power sets 124
13.4 A Property of Infinite Sets 128

14 The Universal Debugger **131**
14.1 The problem statement 131

15 The Continuum Hypothesis **137**
15.1 A surprising answer 137

1 Induction

1.1 Simple induction

In this section we review a fundamental method of proof for mathematical statements, which is called *induction*. It is one of the most useful methods for proving statements. Let us begin this discussion with an example.

Exmaple 1.1.1. Prove that for every $n \in \mathbb{N}$, we have $0 + 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

Before proving this statement, we can start checking it for verious values of n:

$n = 0 \ \Rightarrow \ 0 = \frac{0(0+1)}{2}$.
$n = 1 \ \Rightarrow 0 + 1 = \frac{1(1+1)}{2}$.
$n = 2 \ \Rightarrow \ 0 + 1 + 2 = 3 = \frac{2(2+1)}{3}$.
$n = 3 \ \Rightarrow \ 0 + 1 + 2 + 3 = 6 = \frac{3(3+1)}{2}$.
$\cdots$

Even if you check the verasity of the statement for n up to 1000000, you have not yet proved it! To prove it just by case by case verification, you will need to do the above calculations for *every* n.

But here is another approach: Suppose that for every $n \in \mathbb{N}$, $P(n)$ denotes the following statement:

$$P(n) \ : \ \text{“}0 + 1 + 2 + \cdots + n = \frac{n(n+1)}{2}.\text{”}$$

We can easily check that $P(0)$ is true. Suppose that we have proved the following statement:

$$\forall_{n \in \mathbb{N}} \, (P(n) \ \Rightarrow \ P(n+1)). \tag{1}$$

Then from $P(0) = \mathbf{T}$ it follows that $P(1) = \mathbf{T}$, and in turn from $P(1) = \mathbf{T}$ it follows that $P(2) = \mathbf{T}$, and in turn from $P(2) = \mathbf{T}$ it follows that $P(3) = \mathbf{T}$, etc. Therefore we have logically proved the statement $P(n)$ for every $n \in \mathbb{N}$.

It now remains to prove statement (1). Here is how we can prove it:

Let $n \in \mathbb{N}$ and assume that $P(n)$ is true, i.e.,

$$0 + 1 + \cdots + n = \frac{n(n+1)}{2}.$$

Then:

$$\begin{aligned} 1 + \cdots + n + (n+1) &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{n^2 + n + 2(n+1)}{2} = \frac{n^2 + 3n + 2}{2} = \frac{(n+1)(n+2)}{2} \end{aligned}$$

which means that $P(n+1)$ is true.

More genrally, the method of induction can be stated as follows:

> To prove the statement
> $$\forall_{n\in\mathbb{N}}\, P(n)$$
> by induction, you need to do the following two things:
>
> a. Check that $P(0)$ is true,
>
> b. Prove that $\forall_{n\in\mathbb{N}}\, (P(n) \;\Rightarrow\; P(n+1))$.

Why is induction a rigorous method of proof? What is special about $\mathbb{N}$? Can we also prove statements of the form $\forall_{x\in\mathbb{R}}\, P(x)$ by induction? The answer is negative. Induction on $\mathbb{N}$ is possible because $(\mathbb{N}, \leq)$ is well-ordered. In fact a more general form of induction is possible on every well-ordered set. It is called the *transfinite* induction.

Exercise.

a. Prove that for every $n \in \mathbb{N}^+$, we have
$$1 + 3 + \cdots + (2n-1) = n^2.$$

b. Prove that for every natural number n and every real number $x \in [0,1]$, we have
$$(1-x)^n \leq 1 - nx + \frac{n(n-1)}{2}x^2.$$

c. Prove that for every integer $n > 3$ we have $n! > 2^n$, where
$$n! = n \times (n-1) \times \cdots \times 2 \times 1.$$

d. Prove that for every $n \in \mathbb{N}$ and $x > 1$, $(x-1)^n \geq x^n - nx^{n-1}$.

e. Prove that for every $n \in \mathbb{N}^+$, we have $1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2$.

f. The *Fibonacci sequence* is defined as follows: let $F_1 = F_2 = 1$, and
$$F_n = F_{n-1} + F_{n-2} \text{ for every integer } n > 2.$$
Let $\phi_+ = \frac{1+\sqrt{5}}{2}$ and $\phi_- = \frac{1-\sqrt{5}}{2}$. Prove that for every $n \in N^+$, we have
$$F_n = \frac{(\phi_+)^n - (\phi_-)^n}{\phi_+ - \phi_-} = F_n = \frac{(\phi_+)^n - (\phi_-)^n}{\sqrt{5}}$$

2 Propositional Logic

2.1 Propositions and connectives

Informally speaking, a proposition is a declarative sentence, i.e., one which is meant to convey some information (which may be correct or incorrect). A proposition is *not* an imperative or an interrogative sentence. More formally, we can say:

> A *proposition* is a phrase which is either true or false.

Exmaple 2.1.1. Here are some examples:

a. "*Canada lies to the north of the United States*" is a proposition, which is obviously true.

b. "*This course is being taught by Professor Baltazar*" is a proposition, certainly false!

c. "*5 times 3 equals 15*" is a mathematical proposition, which is true.

d. The mathematical relation "$2^5 - 1 = 15$" is a proposition, because it states an equality of two numbers, i.e., a statement whose verasity can be checked. Again, remember that a proposition can be either true or false (in this case, it *is* false).

e. However, the mathematical expression "$2^5 - 1$" is not a proposition, because it only describes a number. It does not not state any fact about that number. In other words, "$2^5 - 1$" is not a proposition because it is just a number, and a number is neither true nor false.

f. "*If $x \in \mathbb{R}$, then $x^2 \geq 0$*" is a proposition. It states a fact about *all* real numbers x. From highschool you know that this proposition is true.

g. "*If n is an odd integer, then n^2 is an odd integer*" is a proposition. It states a face about *all* integers n. Is it ture or false? We will answer this question later, when we study proof methods.

Remark. In mathematics, a proposition is sometimes called a *statement* or an *assertion*.

> When a proposition is true, we say that its *truth value* is **T**. When it is false, we say that its truth value is **F**.

We denote propositions by uppercase letters. For example, we write:

A = "5 is an odd number", or B = "$\sqrt{100} > 7$",
or C = "Every odd integer is divisible by 5".

The truth values of A, B, and C are **T**, **T**, and **F**. (Why?)

The five connectives of logic

From any list of propositions $A, B, C, \ldots$, you can form more complicated propositions using the following five *connectives*:

Conjuction ($\bigwedge$). If A and B are two propositions, then $A \wedge B$ is read "A and B". What is the truth value of $A \wedge B$? If both A and B are true, then the truth value of $A \wedge B$ is $\mathbf{T}$. If either A or B is false, then the truth value of $A \wedge B$ is $\mathbf{F}$. In other words, the truth value of $A \wedge B$ is obtained by the following table, which is called the *truth table* of $A \wedge B$.

A	B	$A \wedge B$
T	**T**	**T**
T	**F**	**F**
F	**T**	**F**
F	**F**	**F**

Exmaple 2.1.2. The proposition

$$P = \text{"5 is an odd number and } 2 \times 3 \text{ is equal to 5"}$$

is the conjunction of A = "5 is an odd number" and B = "2×3 is equal to 5". From the above truth table, we see that the truth value of P is $\mathbf{T} \wedge \mathbf{F} = \mathbf{F}$, i.e., P is false.

Disjunction ($\bigvee$). If A and B are two propositions, then $A \vee B$ is read "A or B". If either A or B is true, then the truth value of $A \vee B$ is $\mathbf{T}$. If both A and B are false, then the truth value of $A \vee B$ is $\mathbf{F}$. In other words, the truth table of $A \vee B$ is:

A	B	$A \vee B$
T	**T**	**T**
T	**F**	**T**
F	**T**	**T**
F	**F**	**F**

Exmaple 2.1.3. The proposition

$$P = \text{"5 is an odd number or } 2 \times 3 \text{ is equal to 5"}$$

is the disjunction of A = "5 is an odd number" and B = "2×3 is equal to 5". From the above truth table, we see that the truth value of P is $\mathbf{T} \vee \mathbf{F} = \mathbf{T}$, i.e., P is true.

Negation ($\neg$). If A is a proposition, then $\neg A$ is read "not A" or "it is not true that A". The truth table of $\neg A$ is:

A	$\neg A$
T	**F**
F	**T**

Exmaple 2.1.4. a. If A = "7 is an even number" then

$$\neg A = \text{"It is not true that 7 is an even number"}.$$

In this example, A is false and therefore $\neg A$ is true.

Remark. We may also reinterpret $\neg A$ as "7 is an odd number".

b. If A = "Every odd number is divisible by 3" then

$$\neg A = \text{"It is not true that every odd number is divisible by 3"}.$$

Of course there is a logically equivalent way to write $\neg A$:

$$\neg A = \text{"There exists at least one odd number which is not divisible by 3"}.$$

More on this later!

Implication ($\Rightarrow$) If A and B are propositions then the formula $A \Rightarrow B$ is called an *implication*. It is read "A implies B", or "if A then B". In this implication, A is called the *hypothesis*, and B is called the *conclusion*.

The truth table of $A \Rightarrow B$ is:

A	B	$A \Rightarrow B$
T	**T**	**T**
T	**F**	**F**
F	**T**	**T**
F	**F**	**T**

By looking at the truth table of $A \Rightarrow B$ we observe that:

a. If A is false then the implication $A \Rightarrow B$ is true (regardless of the truth value of B).

b. If B is true then the implication $A \Rightarrow B$ is true (regardless of the truth value of A).

Therefore the only way $A \Rightarrow B$ can be false is when A is true but B is false.

What you just saw may look strange at first glance: deducing a false statement from another false statement is considered a true implication! But the following two examples explain why this is so:

Exmaple 2.1.5. In a recent electoral campaign, a candidate's main campaign statement was the following:

If I am elected, then there will be an income tax reduction.

He/she was not elected, and a few months after the elections taxes have not been reduced yet. Can you accuse him/her of lying in his/her campaign?

Intuitively, you can argue that he/she cannot be accused of lying, because he/she didn't promise anything in case he/she was not elected! Therefore his/her main campaign statement is true. (If it were false then he/she would be lying, and vice versa.)

Mathematically, you can argue that both the hypothesis and the conclusion of his/her campaign statement are false. This means that his/her campaign statement is an implication of the form $\mathbf{F} \Rightarrow \mathbf{F}$. According to the truth table, the truth value of this implication is $\mathbf{T}$. Therefore his/her campaign statement is a true implication, i.e., he/she is not lying.

Exmaple 2.1.6. Is the proposition

$$P = \text{“}\textit{If } 2^4 = 15 \textit{ then 6 is an odd number}\text{”}$$

true or false?

Again, you can see that both the hypothesis and the conclusion of this proposition are false. Therefore it is an implication of the form $\mathbf{F} \Rightarrow \mathbf{F}$, and according to the truth table, its truth value is $\mathbf{T}$, i.e., P is true.

But is it logical to say that P is true? Yes, because the conclusion of P (i.e., that 6 is odd) has a disclaimer: only when $2^4 = 15$. In other words, when the hypothesis of P is false, P does not claim anything about 6.

Exmaple 2.1.7. *(just for fun)* Is the following proposition true or false?

If all the world were paper,
And all the seas were ink,
And all the trees were bread and cheese,
Then we had nothing to drink!

Exmaple 2.1.8. Is the proposition

$$P = \text{“}\textit{If } x > 2 \textit{ then } x^2 > 2\text{”}$$

true or false?

The hypothesis of P is $A =$ “$x > 2$” and its conclusion is $B =$ “ $x^2 > 2$”.

- If the hypothesis is false (i.e., $x \leq 2$), then we have an implication of the form $\mathbf{F} \Rightarrow B$, which is always true.
- If the hypothesis is true, (i.e., $x > 2$), then $x^2 > 4 > 2$. This means that the conlcusion also holds, and therefore we have an implication of the form $\mathbf{T} \Rightarrow \mathbf{T}$, which is true.

The above reasoning proves that P is true.

Equivalence ($\Leftrightarrow$) If A and B are two propositions, the formula $A \Leftrightarrow B$ is read "A if and only if B" or "A iff B". Its truth table is:

A	B	$A \Leftrightarrow B$
T	**T**	**T**
T	**F**	**F**
F	**T**	**F**
F	**F**	**T**

Therofore:

a. If A and B are both true, or both false, then $A \Leftrightarrow B$ is also true.

b. If one of A and B is true while the other one is false, then $A \Leftrightarrow B$ is false.

Exercise. Are the following propositions true or false? Answer by **T** or **F**.

(1) 23 = 8 and $5 \times 6 = 40$.

(2) 2 + 3 = 6 or 2 + 7 = 9.

(3) If 2 + 7 = 9 then 2 + 3 = 9.

(4) 2 + 7 = 5 if and only if 2 + 3 = 9.

(5) If 1027 is a positive integer then 1 + 1 = 2.

(6) If $17^{231} + 2$ is a prime number then 3 + 4 = 7.

(7) If 3 + 4 = 13 then $17^{231} + 2$ is a prime number.

(8) If $17^{231} + 2$ is prime number or 3 + 4 = 7, then 25 = 8.

(9) If $17^{231} + 2$ is a prime number then $17^{231} + 2$ is a prime number.

(10) If ($17^{231} + 2$ is a prime number if and only if 1 + 1 = 2), then $17^{231} + 2$ is a prime number.

Atomic formulas and compound formulas.

From now on we call the mathematical expressions formed by propositions and connectives *formulas*. A formula which contains at least one connective is called a *compound formula*. If a formula contains no connectives then it is called an *atomic formula*, or sometimes simply an *atom*.

Exmaple 2.1.9. If A, B and C are atoms then we can make compound formulas such as:

$$(A \wedge B) \Rightarrow C, \quad A \vee (B \Rightarrow C), \quad (A \vee B) \wedge (A \vee C),$$

and of course many more.

By definition, the formulas **T** and **F** are atoms. Note that if A is an atom then $\neg A$ is not an atom. If A is an atom then the formula $A \vee \mathbf{F}$ is a compound formula, although its truth table is the same as A.

Exercise. For two given atoms P and Q, write a compound formula which is true when exactly one of P and Q is true and the other one is false. (Note that the answer is not $P \vee Q$!)

The principal connective

Every compound formula has a *principal connective*, which is exactly the point that we can break the formula into two shorter formulas (which may be compound or atomic formulas). For example, consider the formula

$$(A \vee C) \Rightarrow (B \vee \neg C).$$

Then the connective "$\Rightarrow$" is the principal connective of this formula, and the above formula is the implication with hypothesis $A \vee C$ and conclusion $B \vee \neg C$. The last two formulas are called the *immediate components* of $(A \vee C) \Rightarrow (B \vee \neg C)$.

Every compound formula has a *unique* principal connective.

If the principal connective of a formula is a negation, then there is just one immediate component. For example, the principal connective of

$$\neg((A \Rightarrow B) \wedge C)$$

is "$\neg$", and its only immediate component is $(A \Rightarrow B) \wedge C$. here are some more examples:

Formula	Principal connective	Immediate components	
$(A \wedge B) \vee C$	$\vee$	$A \wedge B$	C
$\neg(A \vee B) \Rightarrow (\neg A \wedge D)$	$\Rightarrow$	$\neg(A \vee B)$	$\neg A \wedge D$
$\neg((A \wedge B) \Rightarrow C)$	$\neg$	$(A \wedge B) \Rightarrow C$	

The point of introducing the principal connective is to write the truth table of a compound formula. The next example ilustrates how this can be done.

Exmaple 2.1.10. Suppose that we want to write the truth table of

$$\neg(A \vee B) \Rightarrow (A \wedge C).$$

To this end, we should compute the truth value of the above formula for every choice of truth values of A, B, and C. For example, let $A = \mathbf{T}$, $B = \mathbf{F}$, and $C = \mathbf{T}$. The immediate components of the above formula are

$$\neg(A \vee B) \quad \text{and} \quad (A \wedge C).$$

We calculate the truth values of these components.

- For $\neg(A \vee B)$ we have $\neg(\mathbf{T} \vee \mathbf{F}) = \neg\mathbf{T} = \mathbf{F}$.
- For $(A \wedge C)$ we have $\mathbf{T} \wedge \mathbf{T} = \mathbf{T}$.

Consequently, $\neg(A \vee B) \Rightarrow (A \wedge C)$ is the same as $\mathbf{F} \Rightarrow \mathbf{T}$, whose truth value is $\mathbf{T}$.

What we just did was to calculate the third row of the following truth table:

A	B	C	$\neg(A \vee B) \Rightarrow (A \wedge C)$
T	**T**	**T**	**T**
T	**T**	**F**	**T**
T	**F**	**T**	**T**
T	**F**	**F**	**T**
F	**T**	**T**	**T**
F	**T**	**F**	**T**
F	**F**	**T**	**F**
F	**F**	**F**	**F**

Exercise. Write the truth table of $P \vee ((R \wedge \neg Q) \Rightarrow P)$.

Remark. As you saw in the above example, in the calculation of the truth value of a compound formula, the principal connective is the last one that should be considered.

2.2 Equivalence of formulas

Two formulas are called *equivalent* if their truth tables are the same. (This means that for every choice of truth values of their atoms, the truth values of the two formulas are the same.)

To indicate that two formulas φ and ψ are equivalent, we write $\varphi \equiv \psi$.

Exmaple 2.2.1. The formulas $A \wedge B$ and $B \wedge A$ are equivalent, because their truth tables are the same:

A	B	$A \wedge B$
T	**T**	**T**
T	**F**	**F**
F	**T**	**F**
F	**F**	**F**

A	B	$B \wedge A$
T	**T**	**T**
T	**F**	**F**
F	**T**	**F**
F	**F**	**F**

Therefore we can write $A \wedge B \equiv B \wedge A$.

Exmaple 2.2.2. The formulas $A \Rightarrow B$ and $B \vee \neg A$ are equivalent, since their truth tables are the same:

A	B	$A \Rightarrow B$
T	**T**	**T**
T	**F**	**F**
F	**T**	**T**
F	**F**	**T**

A	B	$B \vee \neg A$
T	**T**	**T**
T	**F**	**F**
F	**T**	**T**
F	**F**	**T**

We can also write $A \Rightarrow B \equiv B \vee \neg A$.

Exmaple 2.2.3. The tables

A	B	$A \Rightarrow B$
T	**T**	**T**
T	**F**	**F**
F	**T**	**T**
F	**F**	**T**

and

A	B	$B \Rightarrow A$
T	**T**	**T**
T	**F**	**T**
F	**T**	**F**
F	**F**	**T**

show that the formulas $A \Rightarrow B$ and $B \Rightarrow A$ are not equivalent (for example look at the second rows of tables). We can also write $A \Rightarrow B \not\equiv B \Rightarrow A$.

Exmaple 2.2.4. The following two truth tables show that $\neg\neg A$ is equivalent to A:

A	A
T	**T**
F	**F**

and

A	$\neg\neg A$
T	**T**
F	**F**

Exmaple 2.2.5. Truth tables

A	$A \wedge \neg A$
T	**F**
F	**F**

and

A	**F**
T	**F**
F	**F**

show that $A \wedge \neg A \equiv \mathbf{F}$.

Exmaple 2.2.6. The truth tables

A	B	$(A \wedge \neg A) \vee B$
T	**T**	**T**
T	**F**	**F**
F	**T**	**T**
F	**F**	**F**

and

A	B	B
T	**T**	**T**
T	**F**	**F**
F	**T**	**T**
F	**F**	**F**

show that $(A \wedge \neg A) \vee B \equiv B$.

The most commonly used equivalences are shown in Table 1 below:

	Equivalence			Name or comment
(1)	$P \vee Q$	$\equiv$	$Q \vee P$	Commutativity of $\vee$
(2)	$P \wedge Q$	$\equiv$	$Q \wedge P$	Commutativity of $\wedge$
(3)	$(P \vee Q) \vee R$	$\equiv$	$P \vee (Q \vee R)$	Associativity of $\vee$
(4)	$(P \wedge Q) \wedge R$	$\equiv$	$P \wedge (Q \wedge R)$	Associativity of $\wedge$
(5)	$P \vee P$	$\equiv$	P	Idempotence
(6)	$P \wedge P$	$\equiv$	P	Idempotence
(7)	$P \vee \mathbf{F}$	$\equiv$	P	$\mathbf{F}$ is neutral for $\vee$
(8)	$P \wedge \mathbf{T}$	$\equiv$	P	$\mathbf{T}$ is neutral for $\wedge$
(9)	$P \vee \mathbf{T}$	$\equiv$	$\mathbf{T}$	$\mathbf{T}$ is absorbent for $\vee$
(10)	$P \wedge \mathbf{F}$	$\equiv$	$\mathbf{F}$	$\mathbf{F}$ is absorbent for $\wedge$
(11)	$\neg\neg P$	$\equiv$	P	Double negation
(12)	$P \vee \neg P$	$\equiv$	$\mathbf{T}$	Excluded middle
(13)	$P \wedge \neg P$	$\equiv$	$\mathbf{F}$	Contradiction
(14)	$\neg(P \vee Q)$	$\equiv$	$\neg P \wedge \neg Q$	De Morgan's law
(15)	$\neg(P \wedge Q)$	$\equiv$	$\neg P \vee \neg Q$	De Morgan's law
(16)	$P \wedge (Q \vee R)$	$\equiv$	$(P \wedge Q) \vee (P \wedge R)$	Distributivity of $\wedge$ over $\vee$
(17)	$P \vee (Q \wedge R)$	$\equiv$	$(P \vee Q) \wedge (P \vee R)$	Distributivity of $\wedge$ over $\vee$
(18)	$P \Rightarrow Q$	$\equiv$	$\neg P \vee Q$	
(19)	$P \Leftrightarrow Q$	$\equiv$	$(P \Rightarrow Q) \wedge (Q \Rightarrow P)$	
(20)	$P \Leftrightarrow Q$	$\equiv$	$(P \wedge Q) \vee (\neg P \wedge \neg Q)$	

Table 1.

Exercise. Check the above equivalences using truth tables.

> The rules in the above table are among the basic rules of inference. Ponder them deeply! When we want to *prove* mathemetical facts, we will be using them all the time. Well, if you really think about it, you are actually using them in your daily conversations when you want to "argue".

2.3 Proof of equivalence by algebraic manipulations

Suppose you are given two formulas φ and ψ, and you want to check whether or not they are equivalent. You can always use truth tables but it usually takes a long time and it is easy to make mistakes. Another more efficient method you can pursue is to use the equivalences in Table 1. The next examples show how this can be done.

Exmaple 2.3.1. Suppose we want to prove that $\neg(A \Rightarrow B) \equiv A \wedge \neg B$.

We can write:

$$\begin{aligned} \neg(A \Rightarrow B) &\overset{(18)}{\equiv} \neg(B \vee \neg A) \\ &\overset{(14)}{\equiv} \neg B \wedge \neg\neg A \\ &\overset{(11)}{\equiv} \neg B \wedge A \\ &\overset{(2)}{\equiv} A \wedge \neg B \end{aligned}$$

Exmaple 2.3.2. Suppose we want to prove that $P \wedge (P \vee Q) \equiv P$.

We can write:

$$\begin{aligned} P \wedge (P \vee Q) &\overset{(7)}{\equiv} (P \vee \mathbf{F}) \wedge (P \vee Q) \\ &\overset{(17)}{\equiv} P \vee (\mathbf{F} \wedge Q) \\ &\overset{(2)}{\equiv} P \vee (Q \wedge \mathbf{F}) \\ &\overset{(10)}{\equiv} P \vee \mathbf{F} \\ &\overset{(7)}{\equiv} P \end{aligned}$$

Exercise. Use algebraic manipulations to prove that $P \vee (P \wedge Q) \equiv P$.

The equivalences

$$P \wedge (P \vee Q) \equiv P$$

and

$$P \vee (P \wedge Q) \equiv P$$

are sometimes called *absorption laws*.

Exmaple 2.3.3. Suppose that x is a given number. Consider the following two assertions about x:

$$P = \text{“}\textit{If } x \neq 0 \textit{ then } x > 1.\text{”}$$

and

$$Q = \text{“}x > 1 \textit{ or } x = 0.\text{”}$$

We can use algebraic manipulations to see that $P \equiv Q$:

Consider the following two statements:

$$A = \text{“}x \neq 0\text{”} \qquad \text{and} \qquad B = \text{”}x > 1\text{”}.$$

Then P is basically $A \Rightarrow B$ and Q is basically $B \vee \neg A$. From the equivalence (18) of the above table it follows that $P \equiv Q$.

Exmaple 2.3.4. Consider the proposition

$$P = \text{“}a \text{ is odd and } b \text{ is divisible by 3”}.$$

What is the negation of P?

This is an exmaple of De Morgan's law (i.e., the equivalence (20) of the above table): If we set

$$A = \text{“}a \text{ is odd”} \qquad \text{and} \qquad B = \text{“}b \text{ is divisible by 3”}$$

then $P \equiv A \wedge B$. By De Morgan's law, $\neg P \equiv \neg A \vee \neg B$, i.e.,

$$\neg P = \text{“}a \text{ is not odd } \textit{or } b \text{ is not divisible by 3”}.$$

Exercise.

(1) What is the negation of “$a \geq 0$ or $b < 0$”?

(2) What is the negation of “$x \neq 0$ or $x + y > 0$”?

2.4 Contrapositive and converse of an implication

Suppose you have an implication “$P \Rightarrow Q$”. You can make two new implications from it, as follows:

> The *contrapositive* of an implication “$P \Rightarrow Q$” is “$\neg Q \Rightarrow \neg P$”. The *converse* of an implication “$P \Rightarrow Q$” is “$Q \Rightarrow P$”.

The truth tables of $P \Rightarrow Q$, its contrapositive, and its converse are:

P	Q	$P \Rightarrow Q$
T	**T**	**T**
T	**F**	**F**
F	**T**	**T**
F	**F**	**T**

and

P	Q	$\neg Q \Rightarrow \neg P$
T	**T**	**T**
T	**F**	**F**
F	**T**	**T**
F	**F**	**T**

and

P	Q	$Q \Rightarrow P$
T	**T**	**T**
T	**F**	**T**
F	**T**	**F**
F	**F**	**T**

From these tables you can see that:

> The contrapositive of $P \Rightarrow Q$ is equivalent to $P \Rightarrow Q$, but the converse of $P \Rightarrow Q$ is *not* equivalent to $P \Rightarrow Q$. In other words:
>
> $$P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P \quad \text{but} \quad P \Rightarrow Q \not\equiv Q \Rightarrow P.$$

Exmaple 2.4.1. Suppose that a and b are two real numbers. Then the two sentences:

(i) *If* $a = b$, *then* $a^2 = b^2$

(ii) *If* $a^2 \neq b^2$, *then* $a \neq b$

are two equivalent ways of expressing the same fact about a and b. In other words, if (i) is true then (ii) is true and vice versa. Note that the second sentence is the contrapositive of the first one. However, the converse of (i) is:

(iii) If $a^2 = b^2$, then $a = b$.

and (iii) is *not* equivalent to (i). For example, if $a = 1$ and $b = -1$, then (i) is true but (iii) is false.

Exercise. For each of the following propositions, write the contrapositive and converse.

(1) If $x = 0$ then $xy = 0$.

(2) If $s > 0$ then $s^2 > 0$.

(3) If d divides 6 then d divides 30.

(4) If the matrix A is invertable, then it is a square matrix and it does not have any zero rows.

(5) If two integers are even, then their sum is even.

(6) If a series $\sum_{i=1}^{\infty} a_n$ converges, then the sequence $\{a_n\}_{n=1}^{\infty}$ converges and $\lim_{n\to\infty} a_n = 0$.

(7) If $ab = 0$, then $a = 0$ or $b = 0$.

(8) If $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are convergent sequences, then $\{a_n + b_n\}_{n=1}^{\infty}$ is convergent.

2.5 Elimination of Parentheses

Why is $8 \div 4 \div 2$ ambiguous, while $8 \times 4 \times 2$ is not? The point is that the expression $8 \div 4 \div 2$ can be interpreted in two different ways: $(8 \div 4) \div 2 = 1$ and $8 \div (4 \div 2) = 4$, i.e., there are two ways to put parentheses to determine the order of operators. However, for $8 \times 4 \times 2$, both of these ways result in the same value : $(8 \times 4) \times 2 = 8 \times (4 \times 2) = 64$.

More formally, we say that multiplication is *associative*, whereas division is not:

$$(a \times b) \times c = a \times (b \times c) \quad \text{but} \quad (a \div b) \div c \neq a \div (b \div c)$$

How about the connectives of logic? Using truth tables one check that the connectives $\vee, \wedge$, and $\Leftrightarrow$ are associative, but $\Rightarrow$ is not. Therefore we have:

$$\begin{array}{rcl} (P \wedge Q) \wedge R & \equiv & P \wedge (Q \wedge R) \\ (P \vee Q) \vee R & \equiv & P \vee (Q \vee R) \\ (P \Leftrightarrow Q) \Leftrightarrow R & \equiv & P \Leftrightarrow (Q \Leftrightarrow R) \end{array}$$

But $(P \Rightarrow Q) \Rightarrow R \not\equiv P \Rightarrow (Q \Rightarrow R)$. For example,

$$(\mathbf{F} \Rightarrow \mathbf{T}) \Rightarrow \mathbf{F} \ \equiv \ \mathbf{T} \Rightarrow \mathbf{F} \ \equiv \ \mathbf{F}$$

but

$$\mathbf{F} \Rightarrow (\mathbf{T} \Rightarrow \mathbf{F}) \ \equiv \ \mathbf{F} \Rightarrow \mathbf{F} \equiv \ \mathbf{T}.$$

Another rule which we stipulate to help us avoid parentheses is that:

negation should precede any other operation.

Therefore $\neg P \vee Q$ means $(\neg P) \vee Q$ and not $\neg(P \vee Q)$. Note that we have already used the latter rule without mentioning it : when we said that the contrapositive of $P \Rightarrow Q$ is $\neg Q \Rightarrow \neg P$, we were avoiding extra parenthesis; otherwise we should have written $(\neg Q) \Rightarrow (\neg P)$.

Similarly,

$P \vee \neg Q \equiv P \vee (\neg Q) \quad , \quad \neg P \Rightarrow R \equiv (\neg P) \Rightarrow R \quad$ and $\quad \neg P \Leftrightarrow (P \vee Q) \equiv (\neg P) \Leftrightarrow (P \vee Q)$.

Exmaple 2.5.1. Associativity of $\vee$ implies that the formulas

$$(A \vee B) \vee (C \vee \neg D) \ , \ ((A \vee B) \vee C) \vee \neg D \ , \ A \vee (B \vee (C \vee \neg D))$$

and any other way of parenthesizing of $A \vee B \vee C \vee \neg D$ are equivalent. For conciseness we sometimes eliminate the parentheses. (But we only do so if the formula does not become ambiguous.)

Exmaple 2.5.2. The formulas

$$(P \vee Q) \Leftrightarrow (Q \Rightarrow R) \Leftrightarrow (R \wedge R) \Leftrightarrow (T \Rightarrow P)$$

and

$$(P \vee Q) \wedge (Q \Rightarrow R) \wedge (R \wedge R) \wedge (T \Rightarrow P)$$

are not ambiguous.

Exercise. Which of the following formulas are ambiguous? Which of them are not? Justify your answers.

(1) $P \Rightarrow Q \vee R$

(2) $(P \Rightarrow Q) \vee (R \Leftrightarrow P) \vee (R \wedge \neg P)$

(3) $(\neg P \Rightarrow Q) \wedge (P \vee (R \wedge \neg S)) \wedge (S \Rightarrow R)$

(4) $\neg P \Rightarrow (P \vee Q) \Rightarrow \neg R$

(5) $\neg P \Leftrightarrow (P \vee Q) \Leftrightarrow \neg R$

2.6 Tautologies and contradictions

The concepts of tautology and contradiction play fundamental roles in mathematical proofs.

> A *tautology* is a formula whose truth value is always **T**.
> A *contradiction* is a formula whose truth value is always **F**.

In other words, a formula is a tautology if in the rightmost column of its truth table all you see is **T**'s, and it is a contradiction if in the rightmost column of its truth table all you see is **F**'s. Note that most formulas are neither tautologies nor contradictions.

Exmaple 2.6.1. The truth table

P	$P \vee \neg P$
T	**T**
F	**T**

shows that the formula $P \vee \neg P$ is a tautology. Note that $P \vee \neg P \equiv \mathbf{T}$ is rule (12) of Table 1. For a similar reason, $P \vee \mathbf{T}$ is a tautology.

Exmaple 2.6.2. Example 2.6.1 above implies that the formula $(A \vee B) \vee \neg(A \vee B)$ is a tautology (just set $P = A \vee B$). For your own practice, check directly that $(A \vee B) \vee \neg(A \vee B)$ is a tautology, i.e., write its truth table.

Exmaple 2.6.3. The truth table

A	B	$(A \wedge B) \Rightarrow (A \vee B)$
T	**T**	**T**
T	**F**	**T**
F	**T**	**T**
F	**F**	**T**

shows that the formula $(A \wedge B) \Rightarrow (A \vee B)$ is a tautology. What does this formula mean? Let us suppose

A = "*Your dad was born in Ottawa*" and B = "*your mom was born in Ottawa*"

Then, in English, the formula $(A \wedge B) \Rightarrow (A \vee B)$ essentially says:

> *If both your mom and your dad were born in Ottawa the at least one of them was born in Ottawa.*

Can you explain (in English) why this statement is a tautology?

Exmaple 2.6.4. The truth table

P	$P \wedge \neg P$
T	**F**
F	**F**

shows that the formula $P \wedge \neg P$ is a contradiction. Note that $P \wedge \neg P \equiv \mathbf{F}$ is rule (13) of the table of equivalences. For a similar reason, $P \wedge \mathbf{F}$ is a contradiction.

Exercise. Is the formula $P \Rightarrow \neg P$ a contradiction?

The formula **T** is a tautology, and the formula **F** is a contradiction. Another way of describing tautologies and contradictions is as follows:

> A tautology is a formula which is equivalent to **T**.
> A contradiction is a formula which is equivalent to **F**.

Exercise. Using truth tables, determine whether each of the following formulas is a tautology, a contradiction, or none.

(1) $(X \wedge Y) \Leftrightarrow \neg(X \Rightarrow \neg Y)$

(2) $(X \Rightarrow Y) \vee (Y \Rightarrow X)$

(3) $(X \vee Y) \Leftrightarrow \neg(X \Rightarrow \neg Y)$

(4) $(A \Rightarrow C) \Rightarrow ((B \Rightarrow C) \Rightarrow ((A \vee B) \Rightarrow C))$

(5) $(A \Rightarrow (B \vee C)) \Rightarrow ((A \Rightarrow B) \vee (A \Rightarrow C))$

3 Methods of Proof

We are now going to discuss general methods of proof in mathematics. Note that these are just general schemes : in practice, each mathematical proof has its own subtleties and requires new ideas. (Statements whose proofs require 100 pages of delicate arguments are becoming more and more common in contemporary mathematics!) These general schemes do nothing but help us with the logical correctness of the argument. But when mathematicians "prove" a statement is an area of mathematics, they actually need to use several techniques from that area.

So, why are we going to study these proof schemes if they are not going to be the techniques needed for proving actual mathemetical statements? The reason is that they are used in literally *every* mathematical proof. It is very important to learn these fundamental logical outlines.

3.1 Proving an implication $P \Rightarrow Q$

The mathematical statements that we prove are usually in the form of an implication $P \Rightarrow Q$. What are we supposed to do to prove such a statement? We should check that the truth value of this proposition is always **T**. There are a number of ways to do so, such as:

- Direct proof
- Proof by transposition
- Proof by exhaustion
- Proof by contradiction

Direct proof. It is better to demonstrate the process via an exmple. Let us see how we can prove the following statement:

Suppose n is an integer. If n is odd, then n^2 is odd.

This statement is an implication $P \Rightarrow Q$ where

P = "n *is odd*" and Q = "n^2 *is odd*".

To use the direct proof method, we assume that P is true, and with some methematical arguments conclude that Q is true as well.

Assertion: $P \Rightarrow Q$

Proof. Assume P is true. Then blahblahblah ...

$\vdots$

... and this implies that Q is true!!

Let us apply the direct proof method to the statement we want to prove:

Assertion: If n is odd, then n^2 is odd.
Proof. Assume n is odd.

This means that $n = 2k + 1$ for some integer k.
Now $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$.
Therefore $n^2 = 2r + 1$ where $r = 2k^2 + 2k$ is an integer.
This implies that n^2 is an odd integer.

Remark 3.1.1. Once you prove a statement, you have also proved any other statement that is equivalent to it. For instance, we just proved that *if n is odd, then n^2 is odd*, and this means that we have proved its contrapositve too, i.e., we have proved that:

If n^2 is even then n is even.

Exercise. Prove the following statements by the direct proof method.

(1) Suppose A is a 3×3 matrix with real entries. If $A^2 = -A^4$ then $\det A = 0$.

(2) Suppose $x \in \mathbb{R}$. If $-1 < x < 0$ then $x^3 > x$.

Proof by transposition. In this method of proof, instead of proving $P \Rightarrow Q$ directly, you prove its contrapositive, i.e., you prove $\neg Q \Rightarrow \neg P$ by a direct proof. (Remeber that $P \Rightarrow Q$ is equivalent to $\neg Q \Rightarrow \neg P$.) The proof should essentially look like this:

Assertion: $P \Rightarrow Q$
Proof. Assume $\neg Q$ is true (in other words, Q is false). Then blahblahblah ...
$\vdots$
... and this implies that $\neg P$ is true (i.e., P is false)!!

For example, let us see how we can prove the following statement by this method:

Suppose n is an integer. If n^3 is odd then n is odd as well.

We can write the above statement as $P \Rightarrow Q$, where

P = "n^3 *is odd*" and Q = "n *is odd*".

Assertion: *If n^3 is odd, then n is odd as well.*

Proof. Assume n is not odd.
Therefore n is even, i.e., $n = 2m$ for some integer m.
Now $n^3 = (2m)^3 = 8m^3 = 2(4m^3)$.
Setting $r = 4m^3$, we see that r is an integer and $n^3 = 2r$,
which means that n^3 is even, i.e., n^3 is not odd.

Exercise. Suppose that a and b are integers. Use the method of proof by transposition to prove the following statements.

(1) If $a + b$ is odd, then al least one of a and b is odd.

(2) If ab is even then at least one of a and b is even.

Proof by exhaustion. Suppose you want to prove a statement Q, and you also know that a statement $P_1 \vee P_2$ is always true. This means that at any time at least one of P_1 and P_2 is true. Then you can break the proof into two cases:

- Assuming P_1 is true, and deducing Q is true using this assumption.
- Assuming P_2 is true, and deducing Q is true using this assumption.

Your proof should look like this:

Assertion: Q

Proof. We consider two cases:

Case 1. Assume P_1 is true. Then blahblah...

$\vdots$

... and this implies that Q is true.

Case 2. Assume P_2 is true. Then blahblah...

$\vdots$

... and this implies that Q is true.

This completes the proof of Q!!

Exmaple 3.1.2. Let us prove the following statement:

$$\textit{If } x \in \mathbb{R} \textit{ then } 0 \leq \frac{x + |x|}{2} \leq |x|$$

Here the key point is that $|x| = x$ when $x \geq 0$ and $|x| = -x$ when $x < 0$. Whatever x is, either $x \geq 0$ or $x < 0$. Therefore we can consider two case:

- **Case 1.** Assume $x \geq 0$: then $\dfrac{x+|x|}{2} = \dfrac{x+x}{2} = x = |x|$, and it follows that

$$0 \leq \frac{x+|x|}{2} \leq |x|.$$

- **Case 2.** Assume $x < 0$: then $\dfrac{x+|x|}{2} = \dfrac{x+(-x)}{2} = 0$, and it follows that

$$0 \leq \frac{x+|x|}{2} \leq |x|.$$

Exmaple 3.1.3. Let us prove the following statement:

Suppose that n is an integer. If n is not divisible by 3 then $n^2 - 1$ is divisible by 3.

When n is not divisible by 3, it is of the form $n = 3k + 1$ or $3k + 2$ for some integer k.

- **Case 1.** Assume n is of the form $n = 3k + 1$ for some integer k. Then

$$n^2 - 1 = 9k^2 + 6k + 1 - 1 = 9k^2 + 6k = 3(3k^2 + 2k).$$

  Setting $r = 3k^2 + 2k$, we see that $n^2 - 1 = 3r$ and r is an integer. Therefore $n^2 - 1$ is divisible by 3.

- **Case 2.** Assume n is of the form $n = 3k + 2$ for some integer k. Then

$$n^2 - 1 = 9k^2 + 12k + 4 - 1 = 9k^2 + 12k + 3 = 3(3k^2 + 4k + 1).$$

  Setting $r = 3k^2 + 4k + 1$, we see that $n^2 - 1 = 3r$ and r is an integer. Therefore $n^2 - 1$ is divisible by 3.

We have just shown that in both cases $n^2 - 1$ is divisible by 3, which completes the proof of the initial statement.

Exmaple 3.1.4. Suppose a and b are two integers. We want to prove the following statement about a and b:

$$P = \text{“}\textit{If } ab \textit{ is odd then both } a \textit{ and } b \textit{ are odd}\text{”}.$$

To prove this statement, we need to invoke *two* of the above proof schemes: proof by transposition and proof by exhaustion.
Step 1. We begin by proof by transposition, i.e., proving the contrapositive of P:

If a is even or b is even then ab is even.

Step 2. Next we consider two cases:

- Assume a is even. Then $a = 2k$ for some integer k. Therefore $ab = 2kb = 2(kb)$ where kb is an integer. Therefore ab is even.
- Assume b is even. Then $b = 2r$ for some integer r. Therefore $ab = a(2r) = 2(ar)$ where ar is an integer. Therefore ab is even.

The above arguments imply that ab is even. The proof is complete.

Exercise. Prove the following statements.

(1) If $x \in \mathbb{R}$ then $x|x| \leq x^2$.

(2) If n is an integer then $n(n+1)(n+2)$ is divisible by 3.

(3) If n is an odd integer, then $n^2 - 1$ is divisible by 4.

(4) **Bonus question.** If n is an odd integer, then $n^2 - 1$ is divisible by 8.

Proof by contradiction. This method of proof is somewhat similar to proof by transposition. In this method, to prove a statement P, we assume that P is false, and from this assumption we deduce a statement which is impossible. The proof should look like this:

Assertion: P

Proof. Assume that P is false, i.e., $\neg P$ is true. Then blahblah...

$\vdots$

...and this is a contradiction! Therefore P cannot be false.

Exmaple 3.1.5. Let us use this method to prove the following statement:

P = *"If n is an integer then $n(n+1)$ is even"*.

To prove P by contradiction, we assume, on the contrary, that it is false. This means that for some integer n, $n(n+1)$ is odd. Now we use the statement proved in Example 3.1.4:

If ab is odd then both a and b are odd.

applying this statement with $a = n$ and $b = (n+1)$ we deduce that both n and $n+1$ are odd. But this is absurd! We have just shown that assuming that P is false leads to a contradiction. Therefore P should be true.

Exmaple 3.1.6. We are now going to prove a famous mathematical statement:

P = *" $\sqrt{2}$ is an irrational number. In other words, it is impossible to find two integers p and q such that $\sqrt{2} = \frac{p}{q}$ ".*

To prove this theorem, we use the method of proof by contradiction. The negation of the above statement is:

$$\neg P = \text{“}\textit{It is possible to find integers } p \textit{ and } q \textit{ such that } \frac{p}{q} = \sqrt{2}\text{ ”}.$$

We are now supposed to use this hypothesis and deduce something which is impossible. How do we do so?

Let p and q be two integers such that

$$\frac{p}{q} = \sqrt{2}. \tag{2}$$

Clearly we should have $p \neq 0$ and $q \neq 0$. We can assume that the fraction $\frac{p}{q}$ is reduced, i.e., that the only common divisor of p and q is 1. We now rewrite (2) as

$$p^2 = 2q^2. \tag{3}$$

It follows that p^2 is an even number. We now use the following statement, which was proved in Remark 3.1.1:

$$(\star) \qquad \text{If } n^2 \text{ is even then } n \text{ is even.}$$

Setting $n = p$, it follows that p is even. Since the fraction $\frac{p}{q}$ is supposed to be reduced, q cannot be even, i.e., it follows that q is odd.

At the same time, since p is even, we can write $p = 2p'$ for some integer p'. Substituting in equation (3) we obtain

$$2q^2 = p^2 = (2p')^2 = 4p'^2, \quad \text{i.e.,} \quad q^2 = 2p'^2.$$

The last equality implies that q^2 is even, and setting $n = q$ in the statement $(\star)$ we conclude that q is even.

We have just proved that q should be both odd and even, which is absurd! Thus, assuming $\neg P$ is true, we have reached a contradiction. This means that $\neg P$ is false, i.e., P is true.

3.2 Proving an equivalence

Sometimes we are interested in proving that two statements P and Q are equivalent, i.e., $P \Leftrightarrow Q$. In other words, we want to prove that

$$P \textit{ is true if } Q \textit{ is true, and vice versa.}$$

This is sometimes expressed as

$$P \textit{ is true if and only if } Q \textit{ is true}$$

or

P is true iff Q is true.

To prove $P \Leftrightarrow Q$, we can use the equivalence $P \Leftrightarrow Q \equiv (P \Rightarrow Q) \wedge (Q \Rightarrow P)$. This means that proving $P \equiv Q$ amounts to proving two impications: $P \Rightarrow Q$ and $Q \Rightarrow P$. Each of them can be proved by means of any of the general proof methods discussed above.

4 Using Quantifiers

4.1 A concise prologue

In these notes, we will be using the notion of "sets". For the time being, our approach to the notion of sets is intuitive, i.e., at the moment we are not concerned about mathematical abstraction and rigour. But in a few weeks, we will be! In fact we will see that from a mathematical perspective, the concept of sets is a very delicate one, which, if only loosely defined, leads to jarring paradoxes like *Russell's Paradox*! But more on this later!

In these notes, a *set* simply means "a collection of objects". Of course with such a general definition, any collection of physical objects can be considered as a set! Nevertheless, for us the objects of main interest are only *mathematical objects*, like numbers, functions, etc.

Why are we going to begin our discussion of sets without mathematical precision although we know that it will lead to serious problems? For two reasons:

a. This is the way things happened historically. While the modern study of set theory was esentially established in 1870's, Russell's paradox was only discovered in 1901.

b. We will see that it will take a lot of effort to amend set theory to avoid paradoxes. Therefore it is important to first become familiar with sets and what we expect from them. Before we accept sets as concrete objects which are fundamental to mathematics, it doesn't make sense to try to study them abstractly.

4.2 Some notation

Recall that:

- $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$. These numbers are called *natural numbers*.
- $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$. These numbers are called *integers*
- $\mathbb{N}^+ = \{1, 2, 3, \ldots\}$. These numbers are called *positive integers*.
- $\mathbb{R}$ is the set of real numbers. The set of *positive real numbers* (i.e, those $x \in \mathbb{R}$ for which we have $x > 0$), is denoted by $\mathbb{R}^+$.
- $\mathbb{Q}$ is the set of *rational* numbers. A real number x is said to be rational if we can find two integers m and n such that $x = \dfrac{m}{n}$. For example, the numbers $2, -3, \dfrac{1}{5}$, and 1.345 are rational. However, in the notes on propositional logic we proved that $\sqrt{2}$ is *irrational*, i.e., it is not rational.

In these notes we denote sets by capital italic letters, e.g., $A, B, C, ...$ (Recall that we use the same letters for denoting propositions, and matrices. But when we write a capital letter, it should be clear from the context whether it is a set, a proposition, a matrix, or something else.

4.3 The existential quantifier

The symbol $\exists$ is called the *existential quantifier*. It means *"there exists..."*. In other words, if A is a set, then an expression of the form

$$\exists_{a \in A} \, (\dots)$$

means the following sentence in English:

There exists at least one element a of A which satisfies property (...).

Exmaple 4.3.1. The expression

$$\exists_{x \in \mathbb{N}} \, (x \text{ is even})$$

means the folowing sentence in English:

There exists at least one element x of $\mathbb{N}$ which satisfies "x is even".

Clearly, the last sentence is proposition. It is easy to see that it is true, because for example $x = 2$ is a natural number which is even.

Can we say this proposition in a simpler way? Well, it seems that the variable x can be avoided, i.e., the above proposition is equivalent to the following:

There exists at least one element of $\mathbb{N}$ which is even.

Of course the equivalent of the two propositions implies that the latter statement is also true.

You should also note that the variable x does not play any specific role: if we change it to any other letter, we get an equivalent statement. For example, the above statements are equivalent to:

$$\exists_{a \in \mathbb{N}} \, (a \text{ is even}).$$

Exmaple 4.3.2. The mathematical expression

$$\exists_{n \in \mathbb{N}} \, (n < 0)$$

means the following sentence in English:

There exists at least one element of $\mathbb{N}$ which is less than zero.

Clearly, the latter proposition is false.

4.4 The universal quantifier

The symbol $\forall$ is called the *universal quantifier*. It means *"for every..."*, or *"for all..."*. In other word, if A is a set, then the expression

$$\forall_{a \in A} \, (\dots)$$

means the following essentially equivalent sentences in English:

- *For every element a of A, the property (...) holds.*
- *For all elements a of A, the property (...) holds.*
- *Every element a of A satisfies the property (...).*
- *All of the elements a of A satisfy the property (...).*

Exmaple 4.4.1. The expression

$$\forall_{x \in \mathbb{Z}} \, (x^2 \geq 0)$$

means the following sentence in English:

For every element x of $\mathbb{Z}$, the condition $x^2 \geq 0$ is satisfied.

From high school mathematics you know that the above proposition is true. Also note that one can also state this fact in a more concise form:

The square of any integer is nonnegative.

Exmaple 4.4.2. The expression

$$\forall_{x \in \mathbb{Z}} \, (x^2 > 0)$$

means the following sentence in English:

For every element x of $\mathbb{Z}$, the condition $x^2 > 0$ is satisfied.

It is easy to see that this proposition is false, because $x = 0$ is an element of $\mathbb{Z}$ which does not satisfy $x^2 > 0$.

What we saw above can be summarized as follows:

> If φ is a property involving a variable x, then we can use quantifiers to make propositions:
>
> - $\forall_{x \in A} \, \varphi$ *: every element of A satisfies property φ.*
> - $\exists_{x \in A} \, \varphi$ *: There exists at least one element of A which satisfies φ.*
>
> Sometimes φ is called a *predicate*.

4.5 Variations

Note that the following two sentences are *not* equivalent:

- $\forall_{x\in A}\ \varphi$ *: for every element of a set* A*, property* φ *holds.*
- $\forall_x\ \varphi$ *: for every object of the universe, property* φ *holds.*

The second sentence may seem absurd: what property is satisfied by all objects in the universe (except, probably, for existence)?! But by universe here we do not really mean the galaxies. In mathematics, a universe can be predetermined. This means that we may have stipulated that when we say "universe", we mean a predetermined collection of objects. Such a universe can be as small as the set of numbers.

Similarly, in general the formulas $\exists_{x\in A}\ \varphi$ and $\exists_x\ \varphi$ are not equivalent (unless we have stipulated that our universe is identical to A).

However, the formulas

$$\forall_{x\in A}\ \varphi \quad \text{and} \quad \forall_x\ (x \in A \Rightarrow \varphi)$$

are equivalent. Similarly, the formulas

$$\exists_{x\in A}\ \varphi \quad \text{and} \quad \exists_x\ (x \in A\ \wedge\ \varphi)$$

are equivalent. In other words:

$$\forall_{x\in A}\varphi\ \equiv\ \forall_x\ (x \in A \Rightarrow \varphi)$$

and

$$\exists_{x\in A}\varphi\ \equiv\ \exists_x\ (x \in A\ \wedge\ \varphi)$$

For instance, instead of $\forall_{x\in\mathbb{R}}\ (x^2+1>0)$ we can write $\forall_x\ (x\in\mathbb{R}\Rightarrow x^2+1>0)$ and instead of $\exists_{m\in\mathbb{N}}\ (2^m=4m)$ we can write $\exists_m\ (m\in\mathbb{N}\ \wedge\ 2^m=4m)$.

4.6 Negation of a quantifier

Suppose you have a proposition

$$P = \text{"}\forall_{x\in A}\varphi\text{"}.$$

What is $\neg P$? The rule is the following:

$$\neg\forall_{x\in A}\ \varphi \equiv \exists_{x\in A}\ \neg\varphi \quad \text{and} \quad \neg\exists_{x\in A}\ \varphi \equiv \forall_{x\in A}\ \neg\varphi$$

Exmaple 4.6.1. Here are two examples of negation:

a. The negation of $\forall_{x\in\mathbb{R}}\ (x^2 > 0)$ is:

$$\begin{aligned} \neg\forall_{x\in\mathbb{R}}\ (x^2 > 0) &\equiv \exists_{x\in\mathbb{R}}\ \neg(x^2 > 0) \\ &\equiv \exists_{x\in\mathbb{R}}\ (x^2 \leq 0). \end{aligned}$$

b. The negation of $\forall_{x\in\mathbb{R}}\ (x^2 > 1 \Rightarrow x \geq 2)$ is:

$$\begin{aligned} \neg\forall_{x\in\mathbb{R}}\ (x^2 > 1 \Rightarrow x \geq 2) &\equiv \exists_{x\in\mathbb{R}}\ \neg(x^2 > 1 \Rightarrow x \geq 2) \\ &\equiv \exists_{x\in\mathbb{R}}\ \neg(x \geq 2\ \vee\ x^2 \leq 1) \\ &\equiv \exists_{x\in\mathbb{R}}\ (x < 2\ \wedge\ x^2 > 1) \end{aligned}$$

Note that we used the fact that $\neg(P \Rightarrow Q) \equiv \neg(Q \vee \neg P) \equiv \neg Q \wedge P$.

Recall that if a and b are real numbers then:

$$\neg(a > b) \equiv (a \leq b), \quad \neg(a \geq b) \equiv (a < b),$$

$$\neg(a < b) \equiv (a \geq b), \quad \neg(a \leq b) \equiv (a > b).$$

4.7 Free and bound variables

Look at the formula

$$\exists_{x\in\mathbb{Z}}\ (x^2 < y). \tag{4}$$

In this formula, the quantifier is applied to x, but not to y. This means that the quantifier has no control whatsoever over y. We say that the variable x is *bound* by the quantifier $\exists$ whereas y is a *free variable*.

If the value of the variable y is not given, then the formula (4) is not a proposition. The true value of (4) depends on the value of y. For example, its truth value is **T** if $y = 1$ but it is **F** if $y = -1$. The formula (4) can be written in English as follows:

"There exists at least one element x of $\mathbb{Z}$ which satisfies $x^2 < y$".

We can also eliminate x and write it more simply as:

"There exists an integer whose square is less than y."

As you see, without knowing y, the truth value of the last sentence is ambiguous.

A formula without any free variables is called a *closed formula*.

Since there are no free variables in a closed formula, the truth value of a closed formula is well defined (i.e., it makes sense). Some references use the word *sentence* instead of *closed formula*.

4.8 Formulas with several quantifiers

The formula

$$\exists_{x\in\mathbb{N}}\exists_{y\in\mathbb{N}}\ (x > y) \tag{5}$$

is closed. Therefore its truth value is well defined. This formula can also be written as:

"There exists at least one $x \in \mathbb{N}$ for which the formula $\exists_{y\in\mathbb{N}}\ (x > y)$ is true".

Is this sentence true or false? It is true, because if (for example) you set $x = 3$ then $\exists_{y\in\mathbb{N}}\ (x > y)$ is true (for example, by setting $y = 2$). we can write (5) in different form like:

"There exists at least one $x \in \mathbb{N}$ and at least one $y \in \mathbb{N}$ such that $x > y$"

or

"There exist elements x, y of $\mathbb{N}$ for which $x > y$ holds".

Here are a few more examples:

Exmaple 4.8.1. The formula

$$\forall_{x\in\mathbb{R}}\forall_{y\in\mathbb{R}}\ (x^2 + y^2 \geq 4xy) \tag{6}$$

is closed. It can also be written as

"For every $x \in \mathbb{R}$, the formula $\forall y \in \mathbb{R}\ (x^2 + y^2 \geq 4xy)$ is true"

or

"For every $x \in \mathbb{R}$ and every $y \in \mathbb{R}$, the condition $x^2 + y^2 \geq 4xy$ holds"

or

"For every $x, y \in \mathbb{R}$, the condition $x^2 + y^2 \geq 4xy$ holds".

If you want to avoid variables completely, you can say:

"The sum of squares of two real numbers is at least as big as four times their product".

But comprehending the last form requires some thought!

What is the truth value of (6)? It takes some effort, but after some trial and errors, you realize that its truth value is **F**. (For example, we can set $x = 1$ and $y = 1$.)

Exercise.

Check that the formula $\forall_{a\in\mathbb{R}}\forall_{b\in\mathbb{R}}\ (a^2 + b^2 \geq 2ab)$ is closed. What is its truth value?

4.9 Order of quantifiers

Suppose that $\varphi = \varphi(x, y)$ is a predicate with two variables x and y. (For example, φ can be $x^2 + y^2 \geq 4xy$.) Then we have the following equivalences:

$$\exists_{x\in A}\exists_{y\in B}\ \varphi(x,y) \equiv \exists_{y\in B}\exists_{x\in A}\ \varphi(x,y)$$

and

$$\forall_{x\in A}\forall_{y\in B}\ \varphi(x,y) \equiv \forall_{y\in B}\forall_{x\in A}\ \varphi(x,y)$$

For instance, let us look at the formula studied in Example 4.8.1. If we change the order of quantifiers, we obtain:

$$\forall_{y\in\mathbb{R}}\forall_{x\in\mathbb{R}}\ (x^2 + y^2 \geq 4xy) \tag{7}$$

which can also be written as:

"For every $y \in \mathbb{R}$ and every $x \in \mathbb{R}$, the condition $x^2 + y^2 \geq 4xy$ holds"

from which we can understand that (6) and (7) have the same meaning.

Sometimes we write

$$\exists_{a,b\in A} \quad \text{instead of} \quad \exists_{a\in A}\exists_{b\in A}$$

and

$$\forall_{a,b\in A} \quad \text{instead of} \quad \forall_{a\in A}\forall_{b\in A}$$

For example, the formula

$$\forall_{x\in\mathbb{R}}\forall_{y\in\mathbb{R}}\ (x^2 + y^2 \geq 4xy)$$

may also be written as

$$\forall_{x,y\in\mathbb{R}}\ (x^2 + y^2 \geq 4xy)$$

which is slightly shorter. Similarly, the formula

$$\exists_{x\in\mathbb{N}}\exists_{y\in\mathbb{N}}\ (x > y)$$

may also be writen as

$$\exists_{x,y\in\mathbb{N}}\ (x > y)$$

However, when the two quantifiers are not the same, changing the order of quantifiers usually changes the meaning of the formula.

Exmaple 4.9.1. The formula

$$\forall_{x\in\mathbb{N}}\exists_{y\in\mathbb{R}}\ (y^2 = x) \tag{8}$$

is closed. It can also be written as

"For every $x \in \mathbb{N}$, the formula $\exists_{y\in\mathbb{R}}\ (y^2 = x)$ is correct"

or

"For every $x \in \mathbb{N}$, there exists a $y \in \mathbb{R}$ such that $y^2 = x$"

or

"For every natural number x, there exists a real number y such that $y^2 = x$".

Of course with a bit more thought, we can see that the variables can be avoided:

"Every natural number is the square of a real number".

From high school mathematics you know that the truth value of (8) is **T**.

Let us see what happens after changing the order of quantifiers. The formula (8) turns into:

$$\exists_{y\in\mathbb{R}}\forall_{x\in\mathbb{N}}\ (y^2 = x) \tag{9}$$

which can be written as:

"There exists an element $y \in \mathbb{R}$ such that for every $x \in \mathbb{N}$, the condition $y^2 = x$ holds".

This is clearly false! Therefore the truth value of (9) is **F**, i.e., (8) and (9) have different meanings.

Remark. If $\varphi = \varphi(x)$ is a predicate with one variable x then the formula

$$\exists!_{x\in A}\ \varphi$$

means the following equivalent sentences in English:

- *There exists a unique element a of A which satisfies the property φ.*
- *Exactly one of the elements of A satisfies the property φ.*

The symbol $\exists!$ helps us write formulas more conveniently, but from a mathematical point of view it is not really necessary! In fact one can write a longer formula which only uses the existential and universal quantifiers and means the same thing as $\exists!_{x\in A}\ \varphi(x)$. Here is that formula:

$$\exists_{x\in A}\ \varphi(x)\ \wedge\ \forall_{x_1,x_2\in A}\ \Big((\varphi(x_1)\wedge\varphi(x_2))\Rightarrow(x_1 = x_2)\Big). \tag{10}$$

4.10 Negation of a formula with several quantifiers

To negate a formula with several quantifier, we do it *one quantifier at a time*, using the formulas

$$\neg\forall_{x\in A}\,\varphi \equiv \exists_{x\in A}\,\neg\varphi \quad \text{and} \quad \neg\exists_{x\in A}\,\varphi \equiv \forall_{x\in A}\,\neg\varphi$$

which we have seen before. Here are some examples:

a. The negation of (6) is:

$$\begin{aligned}\neg\forall_{x\in\mathbb{R}}\forall_{y\in\mathbb{R}}\,(x^2+y^2\geq 4xy) &\equiv \exists_{x\in\mathbb{R}}\,\neg\big(\forall_{y\in\mathbb{R}}\,(x^2+y^2\geq 4xy)\big)\\ &\equiv \exists_{x\in\mathbb{R}}\exists_{y\in\mathbb{R}}\,\neg(x^2+y^2\geq 4xy)\\ &\equiv \exists_{x\in\mathbb{R}}\exists_{y\in\mathbb{R}}\,(x^2+y^2<4xy)\end{aligned}$$

b. The negation of (8) is:

$$\begin{aligned}\neg\forall_{x\in\mathbb{N}}\exists_{y\in\mathbb{R}}\,(y^2=x) &\equiv \exists_{x\in\mathbb{N}}\,\neg\big(\exists_{y\in\mathbb{R}}\,(y^2=x)\big)\\ &\equiv \exists_{x\in\mathbb{N}}\forall_{y\in\mathbb{R}}\,\neg(y^2=x)\big)\\ &\equiv \exists_{x\in\mathbb{N}}\forall_{y\in\mathbb{R}}\,(y^2\neq x)\big)\end{aligned}$$

c. What is the negation of "$\forall_{x,y\in\mathbb{N}}\,(x+y\leq xy+1)$"? We can write it as

$$\forall_{x\in\mathbb{N}}\forall_{y\in\mathbb{N}}\,(x+y\leq xy+1)$$

and continue as above, but there is also a shortcut:

$$\begin{aligned}\neg\forall_{x,y\in\mathbb{N}}\,(x+y\leq xy+1) &\equiv \exists_{x,y\in\mathbb{N}}\,\neg(x+y\leq xy+1)\\ &\equiv \exists_{x,y\in\mathbb{N}}\,(x+y>xy+1)\end{aligned}$$

4.11 An application: continuity

Suppose that $f:\mathbb{R}\to\mathbb{R}$ is a function. In your analysis (or calculus) courses, you have encountered the notion of continuity. For example, the precise meaning of the sentence

"*f is continuous at $x=2$*"

is the following sentence:

> "*For every $\varepsilon>0$, there exists a $\delta>0$ such that for every $x\in\mathbb{R}$, if $|x-2|<\delta$ then $|f(x)-f(2)|<\varepsilon$.* "

We can write this sentence in terms of quantifiers as follows:

$$\forall_{\varepsilon\in\mathbb{R}^+}\exists_{\delta\in\mathbb{R}^+}\forall_{x\in\mathbb{R}}\,(|x-2|<\delta\Rightarrow|f(x)-f(2)|<\varepsilon). \tag{11}$$

How can we write a formula which says "$f(x)$ *is discontinuous at* $x = 2$" ? We can simply find the negation of (11):

$$\begin{aligned}
&\neg\forall_{\varepsilon\in\mathbb{R}^+}\exists_{\delta\in\mathbb{R}^+}\forall_{x\in\mathbb{R}}\ (|x-2|<\delta \Rightarrow |f(x)-f(2)|<\varepsilon)\\
&\qquad\equiv \exists_{\varepsilon\in\mathbb{R}^+}\neg\exists_{\delta\in\mathbb{R}^+}\forall_{x\in\mathbb{R}}\ (|x-2|<\delta \Rightarrow |f(x)-f(2)|<\varepsilon)\\
&\qquad\equiv \exists_{\varepsilon\in\mathbb{R}^+}\forall_{\delta\in\mathbb{R}^+}\ \neg\forall_{x\in\mathbb{R}}(|x-2|<\delta \Rightarrow |f(x)-f(2)|<\varepsilon)\\
&\qquad\equiv \exists_{\varepsilon\in\mathbb{R}^+}\forall_{\delta\in\mathbb{R}^+}\exists_{x\in\mathbb{R}}\ \neg(|x-2|<\delta \Rightarrow |f(x)-f(2)|<\varepsilon)\\
&\qquad\qquad\equiv \exists_{\varepsilon\in\mathbb{R}^+}\forall_{\delta\in\mathbb{R}^+}\exists_{x\in\mathbb{R}}\ (|x-2|<\delta\ \wedge\ |f(x)-f(2)|\geq\varepsilon).
\end{aligned}$$

Exercise.

a. Determine the truth value of each of the following formulas.

   (a) $\forall_{x\in\mathbb{Z}}\exists_{y\in\mathbb{Z}}\ (x+y=0)$

   (b) $\exists_{y\in\mathbb{Z}}\forall_{x\in\mathbb{Z}}\ (x+y=0)$

b. Write the negation of the following formulas.

   (a) $\forall_{x\in\mathbb{Z}}\exists_{y\in\mathbb{Z}}\ (xy<y^2)$

   (b) $\exists_{y\in\mathbb{Z}}\forall_{x\in\mathbb{Z}}\ (xy\leq 1)$

   (c) $\exists_{\varepsilon\in\mathbb{R}^+}\forall_{n\in\mathbb{N}^+}\forall_{r\in\mathbb{R}^+}\ (\frac{r}{r^n+1}<\varepsilon)$

c. First write the following sentences as formulas using quantifiers. Then write the negation of each of your formulas. (Recall that $\mathbb{R}^+$ denotes the set of positive real numbers and $\mathbb{N}^+$ denotes the set of positive integers.)

   (a) For every two positive real numbers x and y, there exists a positive integer n such that $nx > y$.

   (b) For every positive real number x there exists a positive integer n such that $\dfrac{1}{n} < x$.

   (c) For every positive real number x there exists a positive integer n such that $\dfrac{1}{n} < x^n$.

d. Write the negation of formula (10) (the formula which defines $\exists!$ in terms of $\exists$ and $\forall$). .

4.12 Proving statements which contain quantifiers

4.13 Proving $\exists_{x\in A}\varphi(x)$

To prove $\exists_{x\in A}\varphi(x)$, you should show that at least one element x of A satisfies $\varphi(x)$.

But how?

The simplest way to do so is to give that element explicitly. This kind of proof is called a *constructive proof*. Let us prove the following proposition:

Proposition 4.13.1. $\exists_{n\in\mathbb{N}}\ (n^2+n=20)$*?*

Proof. Setting $n=4$ we obtain $n^2+n=4^2+4=16+4=20$. □

Here is another example:

Proposition 4.13.2. *There exists a* 2×2 *matrix* A *such that* $A^2=0$ *but* $A\neq 0$*.*

Proof. Let $A=\begin{bmatrix}0&0\\1&0\end{bmatrix}$. Clearly $A\neq 0$. However, $A^2=0$ because:

$$A^2=\begin{bmatrix}0&0\\1&0\end{bmatrix}\begin{bmatrix}0&0\\1&0\end{bmatrix}=\begin{bmatrix}0&0\\0&0\end{bmatrix}$$

□

Finding a constructive proof may not be easy, and may even be impossible. However, the beauty of mathematics is that sometimes you can prove that what you want exists without actually being able to give it explicitly! This kind of proof is called a *nonconstructive* proof. Here is an example:

Proposition 4.13.3. *There exists an irrational number* x *such that* $x^{\sqrt{2}}\in\mathbb{Q}$*.*

Proof. We know that $\sqrt{2}$ is rational. Unfortunately, we do not know whether $\sqrt{2}^{\sqrt{2}}$ is rational or not. However, we know that exactly one of the following two statements is true:
• $\sqrt{2}^{\sqrt{2}}$ is rational. Therefore $x=\sqrt{2}$ is an irrational sumber such that $x^{\sqrt{2}}\in\mathbb{Q}$.
• $\sqrt{2}^{\sqrt{2}}$ is irrational. Setting $x=\sqrt{2}^{\sqrt{2}}$, we see that x is irrational and

$$x^{\sqrt{2}}=\left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}}=\sqrt{2}^{(\sqrt{2}\times\sqrt{2})}=\sqrt{2}^{2}=2\in\mathbb{Q},$$

i.e., $x^{\sqrt{2}}\in\mathbb{Q}$

In the above argument we showed that definitely at least one of the two numbers $\sqrt{2}$ and $\sqrt{2}^{\sqrt{2}}$ satisfies the statement of the proposition, i.e., we have *proved* the proposition rigorously, although we still do not know exactly which of these numbers prove it. □

Exercise.

a. Prove that there exist two 2×2 matrices A and B such that $AB \neq BA$.

b. Prove the following : $\exists_{a,b \in \mathbb{Z}} \ (1.9b^2 < a^2 < 2b^2)$

4.14 Proving $\forall_{x \in A} \varphi(x)$

To prove $\forall_{x \in A} \varphi(x)$, you should show that every element x of A satisfies $\varphi(x)$.

The proof of $\forall_{x \in A} \varphi(x)$ usually starts with a phrase such as *"Let $x \in A$"*, *"Fix an element $x \in A$"*, or another phrase which basically means:

"Suppose that x is an arbitrary element of A".

Then this phrase is followed by some arguments which lead to the conclusion that x satisfies the property $\varphi(x)$. Here is how the proof looks like:

Proposition. $\forall_{x \in A} \ \varphi(x)$
Proof. Let $x \in A$. Then blahblah...
$\vdots$
...and this implies that x satisfies φ. □

Remark. It is very important to know that after the phrase "*Let $x \in A$*", the variable x becomes a *constant*, i.e., a fixed element of A.

Here are some examples:

Proposition 4.14.1. $\forall_{x \in \mathbb{R}} \ (x^2 > 2 \ \vee \ x < 2)$.

Proof. Let $x \in \mathbb{R}$. Then we know that exactly one of the following statements holds:
$\underline{x < 2}$: in this case clearly the truth value of $(x^2 > 2 \ \vee \ x < 2)$ is **T**.
$\underline{x \geq 2}$: in this case we have $x^2 \geq 4 > 2$, i.e., $x^2 > 2$. Again we see that the truth value of $(x^2 > 2 \ \vee \ x < 2)$ is **T**.

Therefore we have proved that x satisfies the condition $(x^2 > 2 \ \vee \ x < 2)$. □

Proposition 4.14.2. $\forall_{x \in \mathbb{R}^+} \exists_{n \in \mathbb{N}} \ (\frac{1}{x+n} < x)$.

Remark. The proposition asserts the following statement:

"For every positive real number x, there exists at least one natural number n which satisfies $\frac{1}{x+n} < x$."

Proof. Let $x \in \mathbb{R}^+$. We want to prove the following statement:

$$\exists_{n \in \mathbb{N}} \ (\frac{1}{x+n} < x). \tag{12}$$

Since $x > 0$ and $n \geq 0$, we have $x + n > 0$. Therefore the inequality $\frac{1}{x+n} < x$ is equivalent to

$$1 < x(x+n) \quad \text{or} \quad 1 < x^2 + nx \quad \text{or} \quad 1 - x^2 < nx \quad \text{or} \quad \frac{1-x^2}{x} < n.$$

Therefore in order to prove that (12) is true it suffices to prove that

$$\exists_{n \in \mathbb{N}} \ (\frac{1-x^2}{x} < n) \tag{13}$$

is true. But it is clear why (13) is true: it suffices to take an $n \in \mathbb{N}$ which is bigger than $\frac{1-x^2}{x}$.

□

Remark. Again I remind you that after the phrase "*Let $x \in \mathbb{R}^+$*" the value of x is fixed. The fact that throughout the proof x is fixed is exactly why we can claim that there exists an $n \in \mathbb{N}$ such that $\frac{1-x^2}{x} < n$.

Exercise.

Prove the following statements.

a. $\forall_{n \in \mathbb{N}} \ (\frac{n}{n+1} < \frac{n+1}{n+2})$

b. $\forall_{x \in \mathbb{R}} \exists_{n \in \mathbb{N}} \ \frac{x^2+1}{n+1} < \frac{1}{2}$

4.15 Disproving statements which contain quantifiers

Disproving an assertion means proving that it is false, i.e., proving that its negation is true.

For example, if you want to disprove $\forall_{x\in A}\ \varphi(x)$, then you should prove $\neg\forall_{x\in A}\ \varphi(x)$, which is the same as $\exists_{x\in A}\neg\varphi(x)$. Therefore:

> To disprove $\forall_{x\in A}\ \varphi(x)$, you should show that there exists at least one element of A which does not satisfy the property φ.

Exmaple 4.15.1. To disprove $\forall_{n\in\mathbb{N}}\ (n^2 \leq 2^n)$, it suffices to say that $n = 3$ is an element of $\mathbb{N}$ for which $n^2 > 2^n$.

> **Important Remark**
>
> - To disprove $\forall_{x\in A}\varphi(x)$ it is enough to give an example of an element of A which satisfies the condition φ.
>
> - To prove $\forall_{x\in A}\varphi(x)$ it is NOT enough to give an example of an element of A which satisfies the condition φ.
>
> It is a common mistake by many students to believe that to prove $\forall_{x\in A}\varphi(x)$ all they need to do is to given an example of such x. This is erroneous.

Exercise.

a. For each of the following assertions,

- prove it if you believe that it is true,
- disprove it if you believe that it is false.

(a) $\exists_{n\in\mathbb{N}}\ (\frac{3n+4}{4n+3} < \frac{95}{124})$

(b) $\forall_{x\in\mathbb{R}}\ (\frac{1}{2x^2+3} \leq \frac{3}{10})$

(c) $\forall_{x\in\mathbb{R}}\ (\frac{1}{x^2+1}) \leq 1$

b. Let A and B be the subsets of $\mathbb{R}^2$ defined by

$$A = \{\ (x, y) \in \mathbb{R}^2 \mid y \geq 0 \text{ and } x \leq y^2\ \} \text{ and } B = \{\ (x, y) \in \mathbb{R}^2 \mid x \leq y^2\ \}.$$

Consider the usual vector addition of $\mathbb{R}^2$:

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2).$$

Show that A is closed under vector addition, but B is not. In other words, show that

$$\forall_{v_1, v_2 \in A}\ (v_1 + v_2 \in A)$$

is true, but

$$\forall_{v_1, v_2 \in B}\ (v_1 + v_2 \in B)$$

is false.

c. For any two vectors (x_1, y_1, z_1) and (x_2, y_2, z_2) we define

$$(x_1, y_1, z_1) \star (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, x_1 + 2z_1 + z_2).$$

(a) Disprove the following assertion:

$$\forall_{v_1, v_2 \in \mathbb{R}^3}\ (v_1 \star v_2 = v_2 \star v_1).$$

(b) Consider the line in $\mathbb{R}^3$ given by

$$L = \{\ (x, y, z) \in \mathbb{R}^3 \mid 2y - z = 3 \text{ and } x + 2y = 8\ \}.$$

Prove or disprove the following assertion:

$$\forall_{v_1, v_2 \in L}\ (v_1 \star v_2 = v_2 \star v_1).$$

5 Naïve Set Theory

5.1 What is a set, really?

We begin by reviewing some concepts that we have used already. The main point is to make sure that we are on the same page.

A *set* is a collection of objects [1]. The objects in the set are called the *elements* (or *members*) of the set.

If a set has finitely many elements then it can be denoted as $\{x_1, \ldots, x_n\}$. Here $x_1, \ldots, x_n$ are the elements of the set.) A set with exactly one element is called a *singleton*. For instance, the set $\{5\}$ is a singleton. It is important to note that the set $\{5\}$ is *not* the same as the number 5: $5 \neq \{5\}$. Moreover, when we write elements of sets, we don't repeat them. For example, we don't write $\{1, 1\}$. In fact repetition should be ignored; for example $\{1, 1, 2\} = \{1, 2\}$.

Two sets are called *equal* if they have the same members. For example

$$\{4\} \neq \{-4\} \quad \text{and} \quad \{0, 1, 2\} \neq \{0, 2\},$$

but

$$\{1, -2, 3\} = \{-2, 1, 3\},$$

i.e., the order in which the elements of a set are written is not important. More precisely:

> Two sets A and B are equal iff
>
> $$\forall_{x \in A}\, (x \in B) \;\; \wedge \;\; \forall_{x \in B}\, (x \in A).$$

Another way to write $A = B$ is: $\forall_x\, (x \in A \Leftrightarrow x \in B)$.

What does $A \neq B$ mean? The negation of the formula expressing $A = B$ is:

$$\begin{aligned} \neg\big(\forall_{x \in A}\, (x \in B) \;\; \wedge \;\; \forall_{x \in B}\, (x \in A)\big) \;\; &\equiv \;\; (\neg \forall_{x \in A}\, (x \in B)) \;\vee\; \neg(\forall_{x \in B}\, (x \in A)) \\ &\equiv \;\; \exists_{x \in A}\, \neg(x \in B) \;\vee\; \exists_{x \in B}\, \neg(x \in A) \\ &\equiv \;\; \exists_{x \in A}\, (x \notin B) \;\; \vee \;\; \exists_{x \in B}\, (x \notin A) \end{aligned}$$

In other words:

$$A \neq B \;\Leftrightarrow\; \exists_{x \in A}\, (x \notin B) \;\; \vee \;\; \exists_{x \in B}\, (x \notin A)$$

[1] As I said before, we are mostly interested in mathematical objects, such as numbers, matrices, points in the xy-plane, etc. But we can extend the theory to real life sets using our common sense!

A set can be an element of another set. For example, $\{\{1\}\}$ is a set which is a singleton and whose only element is the set $\{1\}$. Note that $1 \notin \{\{1\}\}$. Also note that $\{1\} \neq \{\{1\}\}$.

Exercise.

How many elements does $\{1, \{1, \{-1, 2\}, -3\}, 5\}$ have?

A set is called *finite* if it has only finitely many elements. Otherwise, it is called an *infinite* set.

Exmaple 5.1.1.

- The sets $\mathbb{N}, \mathbb{Q}, \mathbb{R}, \mathbb{R}^+$ are infinite.
- If a and b are two numbers then the sets $\{a, b\}$ and $\{a\}$ are equal iff $a = b$.

The number of elements of a set X is denoted by $\lvert X \rvert$.

Exmaple 5.1.2.

- $|\{1, 5\}| = 2$.
- $|\mathbb{N}| = \infty$.
- $|\{2, \{2, \{2\}\}\}| = 2$.

A set A is said to be a *subset* of another set B if every element of A is an element of B. When A is a subset of B we write $A \subseteq B$. More precisely:

$$A \subseteq B \Leftrightarrow \forall_{x \in A} (x \in B)$$

If A is not a subset of B we write $A \nsubseteq B$.

Exmaple 5.1.3.

- $\{2, 3, 5\} \subseteq \{1, 2, 3, 4, 5\}$ but note that $\{1, 3, 5\} \notin \{1, 2, 3, 4, 5\}$.
- $\{1, 3\} \in \{-1, \{1, 3\}\}$ but $\{1, 3\} \nsubseteq \{-1, \{1, 3\}\}$.
- $\{1\} \subseteq \{1, \{1\}\}$ and $\{1\} \in \{1, \{1\}\}$.
- $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$.

Proposition 5.1.4. *Two sets A and B are equal iff $A \subseteq B$ and $B \subseteq A$. In other words:*

$$A = B \Leftrightarrow (A \subseteq B \ \wedge \ B \subseteq A).$$

Proof. Exercise. □

If $A \subseteq B$ but $A \neq B$ then we may clarify this by writing $A \subsetneq B$. For example $\mathbb{N} \subsetneq \mathbb{R}$.

Exercise.

Prove that if $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.

There is a set with no elements. It can be denoted by $\{\}$ or $\varnothing$. (In mathematics the symbol $\varnothing$ is more commonly used.) The set with no elements is called the *empty set*. Note that $\varnothing \neq \{\varnothing\}$. In fact $|\{\varnothing\}| = 1$ and $\varnothing \in \{\varnothing\}$.

Theorem 5.1.5. *The empty set is a subset of every set.*

Proof. We use the method of proof by contradiction. Assume that there exists a set A such that $\varnothing \nsubseteq A$. This means that there exists an element in $\varnothing$ which does not belong to A. In particular, there exists an element in A...! But hey...that is absurd...because $\varnothing$ was supposed to have no elements. Contradiction! □

Remark. Is the emptyset unique? Intuitively, the answer is yes. More formally, you can argue as follows: assume that $\varnothing_1$ and $\varnothing_2$ are two sets with no elements. By applying Theorem 5.1.5 twice we have $\varnothing_1 \subseteq \varnothing_2$ and $\varnothing_2 \subseteq \varnothing_1$. By Proposition 5.1.4 we have $\varnothing_1 = \varnothing_2$.

You may not believe that this formal argument is necesary. Hey, it's clear that there is a unique set with no elements, isn't it...? At this point, you are probably right. But in a few weeks you will see why I mentioned this argument!

5.2 Sets of numbers

We have already seen some sets of numbers:

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

If you add or multiply two elements of $\mathbb{N}$ you end up with an element of $\mathbb{N}$. This means that $\mathbb{N}$ is *closed* under addition and multiplication. However, if you subtract an element of $\mathbb{N}$ from another element, sometimes you get a number which is not in $\mathbb{N}$ anymore. We say that $\mathbb{N}$ is not closed under subtraction. How about other sets of numbers and other operations? The following table tells you which sets of numbers are closed with respect to which operations. (In the case of $\div$ we assume that we are not dividing by zero.)

	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	$\mathbb{R}^+$
$+$	Yes	Yes	Yes	Yes	Yes
$\times$	Yes	Yes	Yes	Yes	Yes
$-$	No	Yes	Yes	Yes	No
$\div$	No	No	Yes	Yes	Yes

Exercises.

a. Prove that if a and b are rational numbers such that $a < b$, then there exists a rational number c such that $a < c < b$.

b. Prove that there is no smallest positive rational number. In other words, prove that the set of positive rational numbers does not have a minimum.

c. Let x and y be two positive rational numbers. Prove that there exists a natural number n such that $nx > y$.

5.3 Subsets defined by a property

Suppose that P is a property and $P(x)$ means that x has property P. We can obtain a set

$$\{x \ : \ P(x)\}$$

which is the set of all x which have property P.

If X is a set and P is a property then

$$\{x \in X \ : \ P(x)\}$$

denotes the subset of X containing the elements of X which have property P. Another way to describe this subset is:

$$\{x \ : \ x \in X \ \wedge \ P(x)\}.$$

Exmaple 5.3.1.

- $\{x \in \mathbb{N} \ : \ x \text{ is even}\}$ is the set of even natural numbers. We can also describe this set by its elements: $\{0, 2, 4, 6, 8, ...\}$ but this description is somewhat vague because it does not explicitly tell us what comes after 8. Of course our common sense tells us it will be 10, but the language of mathematics is supposed to be understandable to Martians!

- $\{2x \ : \ x \in \mathbb{Z}\}$ is the set of even integers. We can also describe it by $\{..., -6, -4, -2, 0, 2, 4, 6, ..\}$ which, again, is somewhat vague and not recommended.

Exercise.

Show that the set $\{2x \ : \ x \in \mathbb{Q}\}$ is equal to $\mathbb{Q}$.

Notation. If r is a number then the sets $r\mathbb{N}, r\mathbb{Z}, r\mathbb{Q}, ...$ are defined as follows: for example, we define $r\mathbb{N}$ as

$$r\mathbb{N} = \{rn \ : \ n \in \mathbb{N}\}.$$

(The definitions of the other sets will be similar.) For example,

$$3\mathbb{N} = \{0, 3, 6, 9, 12, ...\} \quad \text{and} \quad \frac{1}{2}\mathbb{Z} = \{..., -\frac{3}{2}, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, ...\}$$

We can also consider sets like $a\mathbb{N} + b = \{an + b \ : \ n \in \mathbb{N}\}$, etc.

The following general principle is the main tool for constructing new sets:

> For any given property P, there exists a unique set whose elements are exactly the objects which satisfy P.

This obvious principle should be taken very seriously for the following two reasons:

1. Any set that we want to work with should be defined precisely by a property.
2. We will see that using the above principle carelessly could cause a lot of trouble! For example, consider the following set:

   $$U = \{A \ : \ A \text{ is a set}\}.$$

   In other words U is the set of all sets. In particular, $U \in U$. Can you write a set which belongs to itself explicitly (i.e., by listing its elements)?

   I hope you agree with me that the set U is a bit weird! But in fact it is dangerous! In fact it is a time bomb which will explode in a few weeks! But the good thing is that at that time we will also see how to use the above principle safely.

5.4 Intervals

If $a, b \in \mathbb{R}$, we define the following subsets of $\mathbb{R}$:

$$
\begin{aligned}
(a,b) &= \{x \in \mathbb{R} : a < x < b\} \\
(a,b] &= \{x \in \mathbb{R} : a < x \leq b\} \\
[a,b) &= \{x \in \mathbb{R} : a \leq x < b\} \\
[a,b] &= \{x \in \mathbb{R} : a \leq b \leq b\} \\
(a,\infty) &= \{x \in \mathbb{R} : x > a\} \\
[a,\infty) &= \{x \in \mathbb{R} : x \geq a\} \\
(-\infty,a) &= \{x \in \mathbb{R} : x < a\} \\
(-\infty,a] &= \{x \in \mathbb{R} : x \leq a\} \\
(-\infty,\infty) &= \mathbb{R}
\end{aligned}
$$

(Note that in the above relations we do not need to assume $a < b$.) These subsets of $\mathbb{R}$ are called *intervals*.

Exmaple 5.4.1.

- $(0,\infty) = \mathbb{R}^+$.
- $(1,-2) = \varnothing$
- $[2,2] = \{2\}$.
- *Some authors use the symbol* $\mathbb{R}^{\geq 0}$ *which means* $[0,\infty)$.

Exercise.

a. Show that if $a \geq b$ then $(a,b) = \varnothing$.

b. Describe the set $\{x \in \mathbb{R} \ : \ x^2 \in [0,1)\}$ by an interval.

c. Determine the elements of the set $\{x \in \mathbb{R} \ : \ x^2 \in (0,1)\}$. Is this set an interval?

5.5 Sets of sets

In many cases, sets whose elements are sets arise naturally. Here are a few examples of such sets:

Exmaple 5.5.1.

Let $S = \{\ (a,b) \ : \ a,b \in \mathbb{R}\ \}$. The elements of S are certain intervals, i.e., they are certain subsets of $\mathbb{R}$. For example, $(1,3) \in S$, but $(0,\infty) \notin S$. Is $\varnothing$ an element of S?

Let $X = \{A \subseteq \mathbb{R} \ : \ 1 \in A\}$. In other words, X is the set of subsets of $\mathbb{R}$ which contain 1. For example, $\{1,2\} \in X$ and $(0,1] \in X$ but $(1,2) \notin X$.

5.6 Power sets

Given any set A, the set containing all of the subsets of A is called the *power set* of A, and is denoted by $\wp(A)$.

Exmaple 5.6.1.

- $\wp(\{-1,5\}) = \{\varnothing, \{-1\}, \{5\}, \{-1,5\}\}$.
- $\wp(\varnothing) = \{\varnothing\}$
- $\wp(\{\varnothing\}) = \{\varnothing, \{\varnothing\}\}$

Theorem 5.6.2. *If A is a finite set with n elements (i.e., $|A| = n$) then $|\wp(A)| = 2^n$.*

Proof. To make a subset X of A, we have two choices for each element: either to put it in X, or to not put it in X. Since there are n elements in A, the total number of possibilities is $\underbrace{2 \times 2 \times \cdots \times 2 \times 2}_{n \text{ times}} = 2^n$. □

Exercise.

a. Let $A = \{2,3\}$ and $B = \{A\}$. How many elements does A have? How many elements does B have?

b. Let $X = \{0,1\}$. What are the elements of the set $\{\wp(A) \ : \ A \in \wp(X)\}$? (Write them down.)

c. How many elements does $\wp(\wp(\{\{\varnothing\}, \varnothing\}))$ have? Write them down.

d. Prove that $X \subseteq Y$ iff $\wp(X) \subseteq \wp(Y)$. Conclude that $X = Y$ iff $\wp(X) = \wp(Y)$.

e. Prove that for any set X we have $\{\varnothing, \{X\}\} \in \wp(\wp(\wp(X)))$.

f. Prove that for any set X we have $\{\wp(A) \ : \ A \subseteq X\} \in \wp(\wp(\wp(X)))$.

g. Let X be a set with n elements. How many elements are there in the set $\{A \in \wp(X) \ : \ |A| = n-1\}$?

h. Let X be a set with n elements. How many elements are there in the set $\{A \in \wp(X) \ : \ |A| \text{ is odd}\}$?

i. Let X be a set with n elements. How many elements are there in the set $\{A \in \wp(X) \ : \ |A| \text{ is divisible by } 3\}$?

j. Recall that $\mathbb{R}^+$ is the set of positive real numbers and $\mathbb{Q}^+$ is the set of positive rational numbers. Let the sets A, B, and C be defined as follows:

$$A = \{\ X \in \wp(\mathbb{R})\ :\ \exists_{\varepsilon\in\mathbb{R}^+}\ (\ (-\varepsilon,\varepsilon) \subseteq X\ )\ \}$$
$$A = \{\ X \in \wp(\mathbb{R})\ :\ \exists_{\varepsilon\in\mathbb{R}^+}\ (\ [-\varepsilon,\varepsilon] \subseteq X\ )\ \}$$
$$A = \{\ X \in \wp(\mathbb{R})\ :\ \exists_{\varepsilon\in\mathbb{Q}^+}\ (\ [-\varepsilon,\varepsilon] \subseteq X\ )\ \}$$

Prove that $A = B = C$.

5.7 Indexed sets

Let us consider the set

$$\{3n - 2\ :\ n \in \mathbb{N}\}.$$

This set is said to be *indexed* (or parametrized) by $\mathbb{N}$. The set $\mathbb{N}$ is called the *index set*.

As you saw above, sometimes the elements of one set can be indexed by elements of another set. If I denotes the index set, then any indexed set can be written as $\{x_i\ :\ i \in I\}$. Sometimes other notations are used, e.g., $(x_i)_{i\in I}$ or $\{x_i\}_{i\in I}$. When the index set is clear from the context, we may even write $(x_i)_i$.

5.8 Operations with sets

We are now going to see how one can generate new sets from the old!

5.9 Union

Let X and Y be two sets. The *union* of X and Y is the set denote by $X \cup Y$ and defined by:

$$X \cup Y = \{\ a\ :\ (a \in X)\ \vee\ (a \in Y)\ \}.$$

Remember that elements are not repeated, i.e., if an element belongs to both of X and Y, we do *not* consider it twice.

Exmaple 5.9.1.

- If $X = \{1, 3, 4, 5\}$ and $Y = \{3, 5, 6\}$ then $X \cup Y = \{1, 3, 4, 5, 6\}$.
- $[-1, 5) \cup [5, 8) = [-1, 8)$.
- For every set X, we have $X \cup X = X$.
- For every set X, we have $X \cup \varnothing = X$.
- For every two sets X and Y, we have $X \subseteq X \cup Y$. Moreover, $X \cup Y = X$ iff $Y \subseteq X$.
- For every two sets X and Y, we have $X \cup Y = Y \cup X$. (Commutativity.)
- For every three sets X, Y, and Z, we have $(X \cup Y) \cup Z = X \cup (Y \cup Z)$. (Associativity.)

Since the union is commutative and associative, it does not matter is which order you write the union of sets, and you do not need to put parentheses. For example, the union

$$A \cup B \cup C \cup D$$

is well defined, and is equal to

$$A \cup C \cup B \cup D$$

or

$$B \cup C \cup D \cup A$$

or any other permutation of these sets.

We can also take a union of an infinite number of sets. But infinity is a source of trouble in mathematics(!) and this should be done carefully. The formal way to do this is as follows:

> Let I be an index set and $\{A_i \ : \ i \in I\}$ be a *family of sets*, i.e., a set parametrized by I whose elements are sets. Then $\bigcup_{i\in I} A_i$ is defined as follows:
>
> $$\bigcup_{i\in I} A_i = \{x \ : \ \exists_{i\in I}\,(x \in A_i)\ \} = \{x \ : \ \text{for some } i \in I\,,\ x \in A_i\ \}$$

Sometimes, instead of writing $\bigcup_{i\in\mathbb{N}} A_i$ we write $\bigcup_{n=1}^{\infty} A_n$. Also, when X is a set of sets (i.e., its elements are sets), then instead of $\bigcup_{A\in X} A$ we write $\cup X$.

Exmaple 5.9.2. What is $\bigcup_{n=1}^{\infty}(0, 1 - \frac{1}{n})$ equal to? We will now prove that

$$\bigcup_{n=1}^{\infty}(0, 1 - \frac{1}{n}) = (0, 1).$$

According to Proposition 5.1.4, it suffices to prove that

$$\bigcup_{n=1}^{\infty}(0, 1 - \frac{1}{n}) \subseteq (0, 1) \quad \text{and} \quad (0, 1) \subseteq \bigcup_{n=1}^{\infty}(0, 1 - \frac{1}{n}).$$

First we prove that $\bigcup_{n=1}^{\infty}(0, 1 - \frac{1}{n}) \subseteq (0, 1)$. If $t \in \bigcup_{n=1}^{\infty}(0, 1 - \frac{1}{n}) \subseteq (0, 1)$, then for some $n \in \mathbb{N}^+$ we have $x \in (0, 1-\frac{1}{n})$. But clearly, $(0, 1-\frac{1}{n}) \subseteq (0, 1)$, and it follows that $x \in (0, 1)$.

Conversely, let us prove that $(0, 1) \subseteq \bigcup_{n=1}^{\infty}(0, 1 - \frac{1}{n})$. Let $x \in (0, 1)$. Therefore $x < 1$. We want to show that for some $n \in \mathbb{N}^+$, $x \in (0, 1 - \frac{1}{n})$, or equivalently,

$$x < 1 - \frac{1}{n} \quad \text{or} \quad \frac{1}{n} < 1 - x \quad \text{or} \quad 1 < (1 - x)n \quad \text{or} \quad \frac{1}{1-x} < n. \tag{14}$$

Looking at the last inequality in (14), we realize that if we take $n \in \mathbb{N}^+$ large enough such that $\frac{1}{1-x} < n$, then $x < 1 - \frac{1}{n}$ and therefore $x \in (0, 1 - \frac{1}{n})$.

Exercise.

Prove the following statements about unions of intervals:

a. $\bigcup_{n=1}^{\infty}(\frac{1}{n}, 2 - \frac{1}{n}) = (0, 2)$

b. $\bigcup_{n=1}^{\infty}(-\frac{1}{n},\frac{2}{n}) = (-1,2)$

c. $\bigcup_{r\in\mathbb{Q}}(-\infty,r) = \mathbb{R}$

d. *Bonus Problem.* Is it possible to express the open interval $(0,1)$ as a disjoint union of closed intervals? (Degenerate intervals, i.e., intervals of the form $[a,a]$, are not allowed.) Justify your answer.

5.10 Intersection

Let X and Y be two sets. The *intersection* of X and Y is the set denoted by $X \cap Y$ and defined by:

$$X \cap Y = \{\ a\ :\ (a \in X)\ \wedge\ (a \in Y)\ \}.$$

If $X \cap Y = \varnothing$, then X and Y are called *disjoint* sets.

Exmaple 5.10.1.

- If $X = \{1,2,4,5\}$ and $Y = \{0,2,5,7,9\}$ then $X \cap Y = \{2,5\}$.
- $(0,1) \cap [.5,1.5) = [.5,1)$.
- For every set X, we have $X \cap X = X$.
- For every set X, we have $X \cap \varnothing = \varnothing$.
- For every two sets X and Y, we have $X \cap Y \subseteq X$. Moreover, $X \cap Y = X$ iff $X \subseteq Y$.
- For every two sets X and Y, we have $X \cap Y = Y \cap X$. (Commutativity.)
- For every two sets X and Y, we have $(X \cap Y) \cap Z = X \cap (Y \cap Z)$. (Associativity.)

Since the intersection is commutative and associative, it does not matter is which order you write the intersection of sets, and you do not need to put parentheses. For example, the intersection

$$A \cap B \cap C \cap D$$

is well defined, and is equal to

$$B \cap D \cap A \cap C$$

or

$$A \cap D \cap B \cap C$$

or any other permutation of these sets.

Just like the case of unions, one can form the intersction of any (nonempty and possibly infinite) family of sets:

Let I be a nonempty index set and $\{A_i \,:\, i \in I\}$ be a family of sets. Then $\bigcap_{i\in I} A_i$ is defined as follows:

$$\bigcap_{i\in I} A_i = \{x \,:\, \forall_{i\in I}\,(x \in A_i)\,\} = \{x \,:\, \text{for every } i \in I\,,\, x \in A_i\,\}$$

Sometimes, instead of writing $\bigcap_{i\in\mathbb{N}} A_i$ we write $\bigcap_{n=1}^{\infty} A_n$. Also, when X is a set of sets (i.e., its elements are sets), then instead of $\bigcap_{A\in X} A$ we write $\cap X$.

Exercise.

Prove the following statements.

a. $\displaystyle\bigcap_{n=1}^{\infty}(n,\infty) = \varnothing$

b. $\displaystyle\bigcap_{r<-1 \text{ and } s>2} (r,s] = [-1,2]$

c. $\displaystyle\bigcap_{n=1}^{\infty}(-\frac{2}{n},1) = [0,1)$

d. If $X = \{\{-1, 1, 2\}, \mathbb{Q}, \mathbb{R}^+\}$ then $\cap X = \{1, 2\}$.

e. For every two sets X and Y, we have $X \cap Y = X$ iff $X \cup Y = Y$.

Union and intersection are distributive over each other. In other words:

For every three sets X, Y, and Z, we have

$$X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$$

and

$$X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z).$$

The above laws are sometimes called distributivity Laws. (Don't they resemble the distributivity laws of propositional logic?)

Exercise.
Prove the above distributivity Laws.

5.11 Difference

Let X and Y be two sets. The difference $X \setminus Y$ is read "X *minus* Y" and defined by:

$$X \setminus Y = \{\ a\ :\ (a \in X)\ \wedge\ (a \notin Y)\ \}.$$

Exmaple 5.11.1.

- If $X = \{1, 2, 4, 5\}$ and $Y = \{2, 5, 6\}$ then $X \setminus Y = \{1, 4\}$.
- $\mathbb{R} \setminus \mathbb{R}^+ = (-\infty, 0]$
- $\mathbb{Z} \setminus 2\mathbb{Z}$ is equal to the set of odd integers.
- For every set X, we have $X \setminus X = \varnothing$ and $X \setminus \varnothing = X$.
- For every two sets X and Y, we have $X \setminus Y \subseteq X$. Moreover, $X \setminus Y = \varnothing$ iff $X \subseteq Y$.

Note that usually $X \setminus Y \neq Y \setminus X$.

For example, if $X = \{1, 2\}$ and $Y = \{1, 3\}$ then $X \setminus Y = \{2\}$ but $Y \setminus X = \{3\}$.

Exercise.

Is the difference associative? In other words, is it true that $(X \setminus Y) \setminus Z = X \setminus (Y \setminus Z)$?

If X is a set and Y is a subset of X, then the set $X \setminus Y$ is sometimes called the *complement* of Y (in X), and denoted by Y^c. (Note that although there is no X in the notation Y^c, it is assumed that X is fixed.)

Exmaple 5.11.2.

- If $X = \{1, 2, 3, 4, 5\}$ and $Y = \{1, 3, 5\}$, then $Y^c = \{2, 4\}$.
- If $X = \mathbb{R}$ and $Y = (-\infty, 0]$, then $Y^c = \mathbb{R}^+$.

Here are some of the properties of differences:

For a fixed set X we have:

- $X^c = \varnothing$ and $\varnothing^c = X$.
- if $Y \subseteq X$ then $(Y^c)^c = Y$

5.12 Symmetric Difference

As you saw above, the difference of two sets is not symmetric, i.e., in general $X \setminus Y \neq Y \setminus X$. However, there is another difference operation between sets which is symmetric. If X and Y are two sets, their *symmetric difference* is denoted by $X \Delta Y$ and defined by:

$$X \Delta Y = (X \setminus Y) \cup (Y \setminus X).$$

In other words, $X \Delta Y$ is the set of elements which belong to one of X and Y but not to both of them.

Exmaple 5.12.1.

If $X = \{1, 2, 3, 4\}$ and $Y = \{2, 4, 5\}$ then $X \Delta Y = \{1, 3, 5\}$.

Here are some of the properties of the symmetric difference:

For every three sets X, Y, and Z, we have:

- $X \Delta Y = Y \Delta X$. (Commutativity.)
- $(X \Delta Y) \Delta Z = X \Delta (Y \Delta Z)$. (Associativity.)
- $X \Delta \varnothing = \varnothing \Delta X = X$.
- $X \Delta X = \varnothing$.

Exercise.

a. Given a set X, show that for every $A, B \subseteq X$ we have $A \setminus B = A \cap B^c$.

b. Prove that for every three sets A, B, and C, we have:

$$\begin{aligned} B \cap (A \setminus B) &= \varnothing \\ (A \cup B) \setminus C &= (A \setminus C) \cup (B \setminus C) \\ (A \cap B) \setminus C &= (A \setminus C) \cap (B \setminus C) \\ (A \setminus B) \setminus C &= A \setminus (B \cup C) \end{aligned}$$

c. Find $\bigcup_{n \in \mathbb{N}} [n, 2n]$.

d. Find $\bigcup_{n \in \mathbb{N}} (n, 2n]$.

e. Find $\bigcup_{n \in \mathbb{N}} (n, n^2)$.

f. Let I be a nonempty index set and $\{Y_i : i \in I\}$ be a family of subsets of a set X, i.e., $Y_i \subseteq X$ for every $i \in I$. Prove that

$$\Big(\bigcup_{i \in I} Y_i\Big)^c = \bigcap_{i \in I} Y_i^c$$

and

$$\Big(\bigcap_{i \in I} Y_i\Big)^c = \bigcup_{i \in I} Y_i^c$$

g. Suppose that X and Y are two nonempty sets of sets. Prove that

$$(\cup_{A \in X} A) \cap (\cup_{B \in Y} B) = \cup_{A \in X, B \in Y} (A \cap B).$$

and

$$(\cap_{A \in X} A) \cup (\cap_{B \in Y} B) = \cap_{A \in X, B \in Y} (A \cup B).$$

h. Let I be an index set and $\{A_i \,:\, i \in I\}$ be a family of sets indexed by I. Prove that for every set B we have:

$$\big(\bigcup_{i\in I} B_i\big) \setminus B = \bigcup_{i\in I}(B_i \setminus B)$$

5.13 Cartesian product

You have already seen the space $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$, whose points are identified by ordered pairs (a, b) of real numbers[2], a.k.a. the coordinates of points. We can generalize the concept of product to sets. What do we expect from an ordered pair (a, b)? We expect the following sentence to be true:

$$(a, b) = (a', b') \;\; \textit{iff} \;\; a = a' \textit{ and } b = b'.$$

Does $\{a, b\}$ have this property? No, because for example $(2, -1) \neq (-1, 2)$ while $\{2, -1\} = \{-1, 2\}$.

Definition 5.13.1. The *Cartesian product* of two sets X and Y is the set denoted by $X \times Y$ and defined as:

$$X \times Y = \{\, (x, y) \,:\, x \in X \;\wedge\; y \in Y \,\}.$$

Exmaple 5.13.2. If $X = \{1, 2\}$ and $Y = \{\mathrm{a}, \mathrm{b}\}$ then

$$X \times Y = \{\, (1, \mathrm{a}), (1, \mathrm{b}), (2, \mathrm{a}), (2, \mathrm{b}) \,\}$$

How can we define $\mathbb{R}^3$? We can use the same idea, i.e., instead of using ordered pair we can use ordered triples (a, b, c) where $a, b, c \in \mathbb{R}$. Similarly, the product $X \times Y \times Z$ of three sets is the set of all ordered triples (x, y, z) such that $x \in X$, $y \in Y$, and $z \in Z$. We can also generalize it to product of families of sets, using ordered n-tuples.

Exercise.

a. Let $|X| = m$ and $|Y| = n$. How many elements does $X \times Y$ have?

b. What is $\varnothing \times X$? What is $X \times \varnothing$?

c. Prove that

$$(X \cup Y) \times Z = (X \times Z) \cup (Y \times Z) \qquad , \qquad (X \cap Y) \times Z = (X \times Z) \cap (Y \times Z)$$

and

$$(X \setminus Y) \times Z = (X \times Z) \setminus (Y \times Z).$$

[2]The notation (a, b) should not be confused with an interval.

d. Let I be a nonempty index set and $\{X_i \ : \ i \in I\}$ be a family of sets indexed by I. Prove that

$$(\bigcup_{i\in I} X_i) \times Z = \bigcup_{i\in I}(X_i \times Z) \qquad , \qquad (\bigcap_{i\in I} X_i) \times Z = \bigcap_{i\in I}(X_i \times Z)$$

and

$$Z \times (\bigcup_{i\in I} X_i) = \bigcup_{i\in I}(Z \times X_i) \qquad , \qquad Z \times (\bigcap_{i\in I} X_i) = \bigcap_{i\in I}(Z \times X_i)$$

6 Functions

6.1 Definition of functions

You have already seen several functions in your calculus courses. For example, $f(x) = \sqrt{x-1}$ is a real-valued function with domain $[1, \infty)$ and range $[0, \infty)$. The concept of functions in set theory is very similar, only more general.

Intuitively, a *function* (or a *map* or a *transformation*) $f : X \to Y$ from a set X to a set Y is a rule that to each element $x \in X$ assigns a unique element $f(x) \in Y$. The set X is called the *domain* of f, and the set Y is called the *codomain* of f. The set of values that f attains, i.e., the set $f(X)$, is called the *image* or the *range* of f. More precisely:

> if $f : X \to Y$ is a function, then for every $S \subseteq X$ the *image* of S under f is denoted by $f(S)$ and defined by
>
> $$f(S) = \{\, y \in Y \ : \ \exists_{x \in X}\ (f(x) = y)\,\}.$$

Exmaple 6.1.1. Let us consider the function $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = x^2.$$

As you see, f is a rule which maps every real number x to its square (i.e., x^2). The domain and the codomain of f are both equal to $\mathbb{R}$, but the image (or range) of f is equal to $[0, \infty)$. Therefore the codomain and the image are not necessarily the same.

Remark. If $f : X \to Y$ is any given function, you can always set $Y_1 = f(X)$ and consider the function $f_1 : X \to Y_1$ which is defined by the same rule, i.e., by setting $f_1(x) = f(x)$ for every $x \in X$. Note that the codomain and image of f_1 are the same.

> if $f : X \to Y$ is a function, then for every $S \subseteq Y$ the *preimage* of S under f is denoted by $f^{-1}(S)$ and defined by
>
> $$f^{-1}(S) = \{\, x \in X \mid f(x) \in S \,\}.$$

Two functions $f_1 : X_1 \to Y_1$ and $f_2 : X_2 \to Y_2$ are said to be *equal* if $X_1 = X_2$, $Y_1 = Y_2$, and $f_1(x) = f_2(x)$ for every x. For example, the functions

$$f : \mathbb{R} \to \mathbb{R} \quad , \quad f(x) = \sqrt{x^2}$$

and

$$g : \mathbb{R} \to \mathbb{R} \quad , \quad g(x) = |x|$$

are equal. (Why?)

Exmaple 6.1.2.

- For every set X we have the *identity function* $\mathrm{id}_X : X \to X$, which is defined by

$$\mathrm{id}(x) = x \quad \text{for every } x \in X$$

- Let X and Y be two arbitrary sets and $y_\circ$ be a fixed element of Y. A function $g : X \to Y$ for which we have

$$g(x) = y_\circ \quad \text{for every } x \in X$$

  is called a *constant* function. The element $y_\circ$ is called the *value* of this constant function. (It is the only value that this function attains.) Note that the image of a constant function is a singleton.

- If $f : X \to Y$ is a function and $A \subseteq X$, the *restriction* of f to A is denoted by $f|_A$. It is the function $f|_A : A \to Y$ which is defined by

$$f|_A(a) = f(a) \quad \text{for every } a \in A.$$

- Let X and Y be two sets. The functions $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ defined by

$$\pi_1((x, y)) = x \quad \text{and} \quad \pi_2((x, y)) = y$$

  are called the *projections* onto the first and second components respectively.

6.2 Composition of functions

Let X, Y, and Z be three sets. If $f : X \to Y$ and $g : Y \to Z$ are functions, then their *composition* is a function $g \circ f : X \to Z$ which is defined by

$$(g \circ f)(x) = g(f(x)) \quad \text{for every } x \in X.$$

Note that although $g \circ f$ is defined, $f \circ g$ is not necessarily defined (unless $X = Z$). Even if $f \circ g$ is defined, the equality $f \circ g = g \circ f$ is usually false.

If $f : X \to X$ is a function then we can compose f with itself several times, i.e., we can form

$$f \circ f \ , \ f \circ f \circ f \ , \ f \circ f \circ f \ , \text{ etc.}$$

In general, we denote $\underbrace{f \circ \cdots \circ f}_{n \text{ times}}$ by f^n. It is easily seen that $f^m \circ f^n = f^{m+n}$.

The composition of functions has the following important properties:

Proposition 6.2.1. *Let $f : X \to Y$, $g : Y \to Z$ and $h : Z \to W$ be three functions. Then*

a. $h \circ (g \circ f) = (h \circ g) \circ f$,

b. $f \circ \mathrm{id}_X = f$ *and* $\mathrm{id}_Y \circ f = f$.

Proof. a. First observe that both $h \circ (g \circ f)$ and $(h \circ g) \circ f$ are functions from X to W. Next, for every $x \in X$ we have

$$(h \circ (g \circ f))(x) = h\big((g \circ f)(x)\big) = h(g(f(x))$$

and

$$((h \circ g) \circ f)(x) = (h \circ g)\big(f(x)\big) = h(g(f(x)).$$

It follows that $(h \circ (g \circ f))(x) = ((h \circ g) \circ f)(x)$ for every $x \in X$. Consequently, $h \circ (g \circ f) = (h \circ g) \circ f$.

b. We prove the equality $f \circ \mathrm{id}_X = f$ and leave the other equality as an exercise. Observe that $f \circ \mathrm{id}_X$ is a function from X to Y. Moreover, for every $x \in X$,

$$(f \circ \mathrm{id}_X)(x) = f(\mathrm{id}_X(x)) = f(x),$$

i.e., $f \circ \mathrm{id}_X = f$ for every $x \in X$. Consequently, $f \circ \mathrm{id}_X = f$.

□

6.3 The graph of a function

Let $f : X \to Y$ be a function. The *graph* of f, which is denoted by $\mathrm{Gph}(f)$, is defined by

$$\mathrm{Gph}(f) = \{\ (a, b) \in X \times Y \ \ :\ \ b = f(a)\ \}.$$

In other words, $\mathrm{Gph}(f)$ is the subset of $X \times Y$ whose elements are exactly the ordered pairs $(x, f(x))$ for every $x \in X$.

Every function $f : X \to Y$ is uniquely determined by its graph. The main property of the graph is the following:

For every $x \in X$ there exists a unique element $y \in Y$ such that $(x, y) \in \mathrm{Gph}(f)$.

Conversely, if $A \subseteq X \times Y$ satisfies the above property then it is the graph of a function from X to Y. This leads us the following more rigorous (but abstract) definition of a function in terms of sets:

Definition 6.3.1. *A function $f : X \to Y$ is a subset $A \subseteq X \times Y$ with the property that for every $x \in X$ there exists a unique element $y \in Y$ such that $(x, y) \in A$. In terms of quantifiers, this condition can be written as:*

$$\forall_{x \in X} \exists!_{y \in Y} \, (\, (x, y) \in A \,)$$

The set whose elements are functions from X to Y is denoted by $\mathrm{Func}(X, Y)$.

6.4 Further examples of functions

Exmaple 6.4.1.

- If $f : X_1 \to Y_1$ and $g : X_2 \to Y_2$ are two functions, then we can define a function $f \times g : X_1 \times X_2 \to Y_1 \times Y_2$ by $(f \times g)(x_1, x_2) = (f(x_1), g(x_2))$.
- Given a function $f : X \to Y$, we can define a new function

  $$f_{\wp} : \wp(X) \to \wp(Y)$$

  by

  $$f_{\wp}(A) = f(A), \quad \text{i.e.,} \quad f_{\wp}(A) = \{f(a) \, : \, a \in A\} \text{ for every } A \subseteq X.$$

  We can also define a function

  $$f^{\wp} : \wp(Y) \to \wp(X)$$

  by

  $$f^{\wp}(B) = f^{-1}(B), \quad \text{i.e.,} \quad f^{\wp}(B) = \{x \in X \, : \, f(x) \in B\} \text{ for every } B \subseteq Y.$$

- Let $f : X_1 \to Y_1$ and $g : X_2 \to Y_2$ be two functions. If $X_1 \cap X_2 = \varnothing$, then one can define a function $f \cup g : X_1 \cup X_2 \to Y_1 \cup Y_2$ by

  $$(f \cup g)(x) = \begin{cases} f(x) & \text{if } x \in X_1, \\ g(x) & \text{if } x \in X_2. \end{cases}$$

Exercise.

a. How many functions are there from a set of m elements to a set of n elements?

b. Let $f : X \to Y$ be a function and A and B be subsets of X. Are $f(A \setminus B)$ and $f(A) \setminus f(B)$ equal? If not, what inclusion relation between them is satisfied? Justify your answers.

c. Let $f : X \to Y$ be a function. Prove that $f(\varnothing) = \varnothing$.

d. Let $f\mathbb{N} \to \mathbb{N}$ be a function defined by $f(x) = 2x + 1$. For example, $f(1) = 3$ and $f(5) = 11$.

   (i) Find $f(\mathbb{N})$, $f(f(\mathbb{N}))$, and $f(f(f(\mathbb{N})))$.

   (ii) Find $f^n(\mathbb{N})$ for every positive integer n.

e. Let $f : X \to Y$ be a function.

   (i) Prove that if $\{B_i \ : \ i \in I\}$ is a family of subsets of Y then

$$f^{-1}(\bigcup_{i\in I} B_i) = \bigcup_{i\in I} f^{-1}(B_i)$$

   and

$$f^{-1}(\bigcap_{i\in I} B_i) = \bigcap_{i\in I} f^{-1}(B_i)$$

   (ii) Prove that if $B \subseteq Y$ then $f^{-1}(B^c) = \big(f^{-1}(B)\big)^c$.

   (iii) Prove that if $\{A_i \ : \ i \in I\}$ is a family of subsets of X then

$$f(\bigcup_{i\in I} A_i) = \bigcup_{i\in I} f(A_i)$$

   and

$$f(\bigcap_{i\in I} A_i) \subseteq \bigcap_{i\in I} f(A_i).$$

   Demonstrate with an example that it is possible to have

$$f(\bigcap_{i\in I} A_i) \neq \bigcap_{i\in I} f(A_i).$$

   (iv) If $A \subseteq X$, what inclusion relation between $f(A^c)$ and $f(A)^c$ is satisfied?

f. Give an example of a function $f : \wp(\mathbb{N}) \to \mathbb{N}$ such that $f(X) \in X$ for every $X \in \wp(\mathbb{N}) \setminus \{\varnothing\}$.

g. Give an example of a function $f : \wp(\mathbb{Z}) \to \mathbb{Z}$ such that $f(X) \in X$ for every $X \in \wp(\mathbb{Z}) \setminus \{\varnothing\}$.

h. *(Bonus problem)* Give an example of a function $f : \wp(\mathbb{Q}) \to \mathbb{Q}$ such that $f(X) \in X$ for every $X \in \wp(\mathbb{Q}) \setminus \{\varnothing\}$.

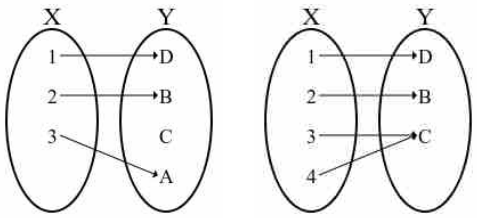
7 Injections, Surjections, Bijections

7.1 Injections

A function $f : X \to Y$ is said to be *injective* (or an *injection*) iff

$$\forall_{x_1,x_2 \in X}\big(f(x_1) = f(x_2) \Rightarrow (x_1 = x_2)\big).$$

The idea is that an injective function $f : X \to Y$ does not map two distinct elements of X to the same element of Y.



injective not injective

Exmaple 7.1.1.

- The function
$$f : \mathbb{R} \to \mathbb{R} \quad , \quad f(x) = x^2$$
is not injective, because for example $f(1) = 1 = f(-1)$. However, it is easily checked that the restriction $f\big|_{[0,\infty)} : [0, \infty) \to \mathbb{R}$ is injective.

- The functions
$$f : \mathbb{N} \to \mathbb{N} \quad , \quad f(n) = n + 5$$
and
$$g : \mathbb{N} \to \mathbb{Q} \quad , \quad f(n) = \frac{n}{2}$$
are injective.

- If X and Y are two sets and $x_\circ \in X$ then the function
$$f : Y \to X \times Y \quad , \quad f(y) = (x_\circ, y)$$
is injective.

- If $f : X_1 \to Y_1$ and $g : X_2 \to Y_2$ are injections then the function
$$f_1 \times f_2 : X_1 \times X_2 \to Y_1 \times Y_2 \quad , \quad (f_1 \times f_2)(x_1, x_2) = (f(x_1), f(x_2))$$
is also an injection.

Proposition 7.1.2. *Let $f : X \to Y$ and $g : Y \to Z$ be two functions.*

a. If f and g are injections then $g \circ f : X \to Z$ is also an injection.

b. If $g \circ f$ is an injection then f is also an injection.

Proof.

a. For every $x_1, x_2 \in X$,

$$\begin{array}{lcll} (g \circ f)(x_1) = (g \circ f)(x_2) & \Rightarrow & g(f(x_1)) = g(f(x_2) & \text{(by definition of composition)} \\ & \Rightarrow & f(x_1) = f(x_2) & \text{(by injectivity of } g\text{)} \\ & \Rightarrow & x_1 = x_2. & \text{(by injectivity of } f\text{)} \end{array}$$

Consequently, $g \circ f$ is injective.

b. For every $x_1, x_2 \in X$,

$$\begin{array}{lcll} f(x_1) = f(x_2) & \Rightarrow & g(f(x_1)) = g(f(x_2)) & \\ & \Rightarrow & (g \circ f)(x_1) = (g \circ f)(x_2) & \\ & \Rightarrow & x_1 = x_2 & \text{(by injectivity of } g \circ f\text{)} \end{array}$$

Consequently, f is injective.

□

Remark. Note that if $g \circ f$ is an injection it is not necessarily true that g is an injection. For example, consider the functions

$$f : \{0\} \to \{0, 1\} \quad , \quad f(0) = 1$$

and

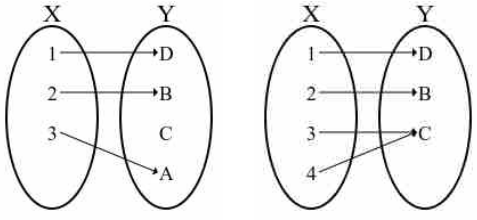
$$g : \{0, 1\} \to \{1\} \quad , \quad g(0) = g(1) = 1.$$

7.2 Surjections

A function $f : X \to Y$ is said to be *surjective* (or a *surjection*) iff $f(X) = Y$, i.e.,

$$\forall_{y \in Y} \exists_{x \in X} \big(y = f(x)\big).$$

The idea is that the image of a surjective function $f : X \to Y$ is equal to its codomain.



not sujrective surjective

Exmaple 7.2.1.

- The function
$$f : \mathbb{R} \to \mathbb{R} \quad , \quad f(x) = x^2$$
is not surjective, because for example $-1 \notin f(\mathbb{R})$. However, it is easily checked that the map
$$g : \mathbb{R} \to [0, \infty) \quad , \quad g(x) = x^2$$
is surjective.

- The functions
$$f : \mathbb{Z} \to \mathbb{N} \quad , \quad f(n) = |n|$$
and
$$g : \wp(\mathbb{Q}) \to \wp(\mathbb{Z}) \quad , \quad f(A) = A \cap \mathbb{Z}$$
are surjective, but not injective. (Why?)

- If X and Y are two nonempty sets then the projections $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ are surjections

- If $f : X_1 \to Y_1$ and $g : X_2 \to Y_2$ are surjections then the function
$$f_1 \times f_2 : X_1 \times X_2 \to Y_1 \times Y_2 \quad , \quad (f_1 \times f_2)(x_1, x_2) = (f(x_1), f(x_2))$$
is also a surjection.

Proposition 7.2.2. *Let $f : X \to Y$ and $g : Y \to Z$ be two functions.*

a. If f and g are surjections then $g \circ f : X \to Z$ is also a surjection.

b. If $g \circ f$ is a surjection then g is also a surjection.

Remark. Note that if $f : X \to Y$ is an arbitrary function then the function
$$f_1 : X \to f(X) \quad , \quad f_1(x) = f(x) \text{ for all } x \in X$$
is always a surjection.

Exercise.

a. Prove Proposition 7.2.2.

b. By an example, demonstrate that if $g \circ f$ is a surjection it does not necessarily follow that f is a surjection.

7.3 A characterisation of injections and surjections

Theorem 7.3.1. *Let X and Y be two sets and $f : X \to Y$ be a function. Then:*

a. f is a surjection iff for every $B \subseteq Y$, we have $f(f^{-1}(B)) = B$.

b. f is an injection iff for every $A \subseteq X$, we have $f^{-1}(f(A)) = A$.

Proof. We only prove (a). The proof of (b) is similar and left as an exercise.

Assume f is surjective. Let $B \subseteq Y$. to prove that $f(f^{-1}(B) = B$, it is enough to prove that $f(f^{-1}(B)) \subseteq B$ and $B \subseteq f(f^{-1}(B))$.

- **Proof of $f(f^{-1}(B)) \subseteq B$.** Let $b \in f(f^{-1}(B))$. Then $b = f(a)$ for some $a \in f^{-1}(B)$. From the definition of $f^{-1}(B)$ it follows that $f(a) \in B$. Since $b = f(a)$, it follows that $b \in B$.

- **Proof of $B \subseteq f(f^{-1}(B))$.** Let $b \in B$. Since f is surjective, there exists an $x \in X$ such that $f(x) = b$. By the definition of $f^{-1}(B)$, we have $x \in f^{-1}(B)$. This means that $f(x) \in f(f^{-1}(B))$. But we also know that $b = f(x)$. Consequently, $b \in f(f^{-1}(B))$.

We now prove the converse statement. Assume that for every $B \subseteq Y$, we have $f(f^{-1}(B)) = B$. We prove that f is surjective. Let $y \in Y$. Consider the set $B = \{y\}$. By our assumption, $f(f^{-1}(\{y\})) = \{y\}$. Therefore there exists an $x \in f^{-1}(\{y\})$ such that $f(x) = y$. Obviously $x \in X$ (because $f^{-1}(\{y\}) \subseteq X$).

□

Exercise.

Complete the proof of part (b) of Theorem 7.3.1.

7.4 Bijections

A function $f : X \to Y$ which is both injective and surjective is called *bijective* (or a *bijection*). If X and Y are two sets and there exists a bijective function $f : X \to Y$, then X and Y are said to be in *bijective correspondence*.

Proposition 7.4.1.

a. The composition of two bijections is a bijection.

b. If $g \circ f$ is a bijection then f is an injection and g is a surjection.

c. If $f : X \to Y$ is an injection then the function $f : X \to f(X)$ is a bijection.

Proof. This proposition follows from Proposition 7.1.2 and Proposition 7.2.2

□

Proposition 7.4.2. *If $f : X \to Y$ is a bijection, then there exists a unique bijection $g : Y \to X$ for which we have*

$$f \circ g = \mathrm{id}_Y \quad \textit{and} \quad g \circ f = \mathrm{id}_X.$$

Proof. The existence of g follows by "reversing the arrows", as the picture below shows.

The uniqueness of g is proved as follows. Suppose that $h : Y \to X$ is a function such that $f \circ h = \mathrm{id}_Y$ and $h \circ f = \mathrm{id}_X$. Then:

$$h = \mathrm{id}_X \circ h = (g \circ f) \circ h = g \circ (f \circ h) = g \circ \mathrm{id}_Y = g.$$

□

Definition 7.4.3. If $f : X \to Y$ is a bijection, the unique function $f^{-1} : Y \to X$ which satisfies

$$f \circ f^{-1} = \mathrm{id}_Y \quad \text{and} \quad f^{-1} \circ f = \mathrm{id}_X$$

is called the *inverse* of f.

Note that there is a difference between the notions inverse of f and preimage of a set under f:

Different meanings of f^{-1}

- If $f : X \to Y$ is a function, the inverse of f exists iff f is a bijection. It is a function
$$f^{-1} : Y \to X$$
which satisfies
$$\forall_{x \in X} [\, f^{-1}(f(x)) = x \,] \qquad \text{and} \qquad \forall_{y \in Y} [\, f(f^{-1}(y)) = y \,].$$
If the function f^{-1} exists, then it is unique.

- If $f : X \to Y$ is a function, and $S \subseteq Y$, the preimage of S under f always exists (i.e., the assumption that f is bijective is not needed anymore). It is a subset of X, denoted by $f^{-1}(S)$, which is defined as follows:
$$f^{-1}(S) = \{\, x \in X \;:\; f(x) \in S \,\}.$$

Proposition 7.4.4.

a. If $f : X \to Y$ is a bijection, then $(f^{-1})^{-1} = f$.

b. If $f : X \to Y$ and $g : Y \to Z$ are bijections, then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Exercise.

a. Prove Proposition 7.4.4.

b. Find all the bijections from $\{0, 1, 2\}$ onto $\{0, 1, 2\}$. Find all the bijections from $\{0, 1, 2, 3\}$ onto $\{1, 2, 3, 4\}$. Find all the surjections from $\{0, 1, 2, 3\}$ onto $\{0, 1, 2\}$.

c. Let X and Y be finite sets. Show that if $|X| \neq |Y|$ then there is no bijection from X onto Y. How many bijections are there from X onto Y when $|X| = |Y|$?

d. Let $f : X \to Y$ be a bijection. Prove that for every $A, B \subseteq X$,

(i) $f(A \cap B) = f(A) \cap f(B)$.

(ii) $f(A \cup B) = f(A) \cup f(B)$.

(iii) $f(A \setminus B) = f(A) \setminus f(B)$.

(iv) The functions $f_{\wp} : \wp(X) \to \wp(Y)$ and $f^{\wp} : \wp(Y) \to \wp(X)$ are also bijections.

e. Let X be an arbitrary set. Show that $\mathrm{Func}(\{0, 1\}, X)$ and $X \times X$ are in bijective correspondence.

f. Let X be an arbitrary set. Show that $\mathrm{Func}(\{0, 1, 2\}, X)$ and $X \times X \times X$ are in bijective correspondence.

8 Binary Operations

8.1 Binary operations on sets

We saw earlier that the sets $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, ...$ are closed under familiar operations such as addition or multiplication. These operations are special cases of binary operations, which are defined below.

Definition 8.1.1. Let X be a set. A *binary operation* on X is a function $X \times X \to X$.

Remark. A binary operation on a set X is usually denoted by symbolds like $\star, \cdot, +, \otimes, ...$, i.e., we write $x \star y$, $x \cdot y$, $x + y$, $x \otimes y$, or even xy, and call it the *product* of x and y. Also, to emphasize that a set X is endowed with a binary operation $\star$, we write $(X, \star)$.

Exmaple 8.1.2.

- The usual addition is a binary operation on $\mathbb{N}$ as well as on $\mathbb{Z}$. Subtraction is not a binary operation on $\mathbb{N}$ because for example $1 - 2 = -1 \notin \mathbb{N}$. Division is not a binary operation on $\mathbb{Q}$ but it is a binary operation on $\mathbb{Q} \setminus \{0\}$.
- If X is an arbitrary set, then $\cap, \cup, \setminus$ and Δ are binary operations on $\wp(X)$.

8.2 Binary operations with extra properties

Some (but not all) binary operations may have the following properties:

- **Associativity.** A binary operation on a set X which satisfies

  $$(x \star y) \star z = x \star (y \star z)$$

  for every $x, y, z \in X$ is called *associative*. Note that when $\star$ is associative, parenthesizing the expression $x \star y \star z$ (or more generally, any product of elements of X) is not necessary. However, if X is not associative, even an expression as simple as $x \star x \star x$ could be be ambiguous, because $(x \star x) \star x$ and $x \star (x \star x)$ can have different values.

- **Commutativity.** A binary operation on a set X which satisfies $x \star y = y \star x$ for every $x, y \in X$ is called *commutative*.

- **Identity element.** Let $\star$ be a binary operation on a set X. An element $e_L \in X$ such that

  $$e_L \star x = x \text{ for every } x \in X$$

is called a *left identity* for $\star$. Similarly, an element $e_R \in X$ such that

$$x \star e_R = x \text{ for every } x \in X$$

is called a *right identity* for $\star$. An element of X which is both a left identity and a right identity for $\star$ is called an *identity* element for $\star$.

Proposition 8.2.1. *Let $(X, \star)$ be a set endowed with a binary operation.*

a. *Suppose that $\star$ has a left identity e_L and a right identity e_R. Then $e_L = e_R$. (In particular, $\star$ has an identity element.)*

b. *The identity element for $\star$, if exists, is unique.*

Proof. a. To prove that $e_L = e_R$ note that $e_L \star e_R = e_R$ and $e_L \star e_R = e_R$. It follows immediately that $e_L = e_R$ is an identity element for $\star$.

b. To prove that the identity element is unique, assume that e and e' are two identity elements. Then

$$e \star e' = e \text{ and } e \star e' = e'$$

from which it follows that $e = e'$.

□

Exmaple 8.2.2.

- The identity element of $(\mathbb{Z}, +)$ is 0.
- $(\mathbb{Z}, -)$ does not have a left identity but 0 is its unique right identity.
- Let X be an arbitrary set. The identity element of $(\wp(X), \cup)$ is $\varnothing$. The identity element of $(\wp(X), \cap)$ is X.
- Left left (or right) identity element is not necessarily unique. For example, consider the binary operation $\star$ defined on $X = \{1, 2\}$ by

$$1 \star 1 = 1 \quad , \quad 1 \star 2 = 2 \quad , \quad 2 \star 1 = 1 \quad , \quad 2 \star 2 = 2.$$

Clearly both of 1 and 2 are left identity elements.

- **Inverse.** Let $(X, \star)$ be a set endowed with a binary operation, and assume that $\star$ has an identity element $e \in X$, i.e.,

$$x \star e = e \star x = x \text{ for every } x \in X.$$

Let $x \in X$. A *right inverse* of x is an element $y \in X$ such that $x \star y = e$. Similarly, a *left inverse* of x is an element $z \in X$ such that $z \star x = x$. Note that the left and right inverse do not necessarily exist.

An element $y \in X$ which is both a left inverse and a right inverse of x is called the *inverse* of x. When x has an inverse, it is also called *invertible*.

Proposition 8.2.3. *Suppose that $(X, \star)$ is a set endowed with a binary operation. Assume that $\star$ is associative and has an identity element e.*

a. If an element x has a left inverse y and a right inverse z, then $y = z$ (and consequently, the element $y = z$ is the inverse of x)

b. The inverse of an element $x \in X$, if exists, is unique.

Proof. a. We have

$$y = y \star e = y \star (x \star z) = (y \star x) \star z = e \star z = z.$$

It follows immediately that $y = z$ is an inverse of x.

b. Suppose that x has two inverses y and y'. Then

$$y = y \star e = y \star (x \star y') = (y \star x) \star y' = e \star y' = y',$$

i.e., $y = y'$.

□

Exercise.

Each of the following sets is endowed with the given binary operation. In each case determine whether the operation (i) is associative, (ii) is commutative, (iii) has a left identity, (iv) has a right identity. If there exists a left or a right identity, determine the elements which have a left or a right inverse.

a. $(\wp(A), \cup)$.

b. $(\wp(A), \setminus)$.

c. $(\wp(A), \Delta)$.

d. The sets $\mathbb{N}, \mathbb{Q}, \mathbb{R}$ with $+$ or $\times$.

e. The sets $\mathbb{N} \setminus \{0\}$, $\mathbb{Z} \setminus \{0\}$, $\mathbb{R} \setminus \{0\}$ with $\div$.

f. $(2\mathbb{Z}, +)$.

g. $(\mathbb{R}, \star)$ where for every $x, y \in \mathbb{R}$, $x \star y = x + y - a$ (here $a \in \mathbb{R}$ is fixed).

h. $(X, \star)$ where X is an arbitrary nonempty set and for every $x, y \in X$, $x \star y = e$ (here e is a fixed element of X).

i. $(X, \star)$ where X is an arbitrary nonempty set and for every $x, y \in X$, $x \star y = x$.

j. $(\mathbb{R}, \star)$ where for every $x, y \in \mathbb{R}$, $x \star y = \max\{x, y\}$.

k. $(\mathbb{R}, \star)$ where for every $x, y \in \mathbb{R}$, $x \star y = x - y$.

l. $(\mathbb{R}, \star)$ where for every $x, y \in \mathbb{R}$, $x \star y = |x - y|$.

8.3 Operations with functions

You have already seen that composition is a binary operation on $\mathrm{Func}(X, X)$. We will now see that there are other interesting binary operations on functions.

Exmaple 8.3.1.

- Let X be an arbitrary set. The *pointwise addition* of two functions $f, g \in \mathrm{Func}(X, \mathbb{R})$ is the function from X to $\mathbb{R}$ which is usually denoted by $f + g$ and is defined by

$$(f + g)(x) = f(x) + g(x) \quad \text{for every } x \in X.$$

  The *pointwise multiplication* of two functions $f, g \in \mathrm{Func}(X, \mathbb{R})$ is the function from X to $\mathbb{R}$ which is usually denoted by $f \cdot g$ or fg and is defined by

$$(fg)(x) = f(x)g(x) \quad \text{for every } x \in X.$$

  More generally:

> More generally, when X is an arbitrary set and Y is a set with a binary operation $\star$, the set $\mathrm{Func}(X, Y)$ can be endowed with a binary operation $\bigstar$ which is defined as follows:
>
> For every $f, g \in \mathrm{Func}(X, Y)$ and every $x \in X$, $(f \bigstar g)(x) = f(x) \star g(x)$

Remark. certain properties of $(Y, \star)$ extend to $(\mathrm{Func}(X, Y), \bigstar)$. For example, if $(Y, \star)$ is commutative, then $(\mathrm{Func}(X, Y), \bigstar)$ is commutative as well. To see this, note that for every $x \in X$:

$$(f \bigstar g)(x) = f(x) \star g(x) = g(x) \star f(x) = (g \bigstar f)(x).$$

Consequently, $f \bigstar g = g \bigstar f$.

Exmaple 8.3.2.

- For every set A, we can consider $(\wp(A), \cap)$. We have seen that $\cap$ is commutaive and associative. The (unique) identity element for $\cap$ is A. The only element of $\wp(A)$ which has an inverse is A. (You should check all of these statements.)

- Let us consider $(\mathbb{Z}, +)$, where $+$ means the usual addition. It is easily checked that $+$ is associative and commutative, the (unique) identity element for $+$ is 0, and every element $x \in \mathbb{Z}$ has a (unique) inverse, namely $-x$.

- Let us consider $(\mathbb{Z}, \times)$, where $\times$ means the usual multiplication. It is easily checked that $\times$ is associative and commutative, the (unique) identity element for $\times$ is 1, and the only elements $x \in \mathbb{Z}$ which have an inverse are 1 and -1.

- Let us consider $(\mathbb{Q} \setminus \{0\}, \div)$ where $\div$ is the usual division. This binary operation is neither associative nor commutative. For example:

$$1 \div 2 \neq 2 \div 1 \text{ and } (8 \div 4) \div 2 \neq 8 \div (4 \div 2).$$

  There is no left identity, but 1 is a right identity, as $x \div 1 = x$ for every $x \in \mathbb{Q} \setminus \{0\}$. (Consequently, there is no identity element for $\div$.)

8.4 An alternative characterization of injections and surjections

The next proposition suggests an alternative characterization of the injections and surjections from a set X to itself.

Let X be a nonempty set, and $(\mathrm{Func}(X, X), \circ)$ be the set of functions from X into X, endowed with composition as a binary operation. Recall that composition of functions is associative and has an identity element, namely id_X.

Proposition 8.4.1. *Let $X \neq \varnothing$ and $(\mathrm{Func}(X, X), \circ)$ be as above.*

a. An element $f \in \mathrm{Func}(X, X)$ is injective iff it has a left inverse.

b. An element $f \in \mathrm{Func}(X, X)$ is surjective iff it has a right inverse.

Proof. We only prove part (a). (The proof of part (b) is very similar.)

• If f has a left inverse, then there exists a function $g : X \to X$ such that $g \circ f = \mathrm{id}_X$. In particular, $g \circ f$ is injective. Therefore Proposition 7.1.2 implies that f is injective.

• Conversely, let $f : X \to X$ be injective. Choose an arbitrary element $x_\circ \in X$ and define the function $g : X \to X$ by

$$g(x) = \begin{cases} x' & \text{if } f(x') = x \\ x_\circ & \text{if } x \notin f(X). \end{cases}$$

Injectivity of f implies that g is well defined. It is easy to check that $g \circ f = \mathrm{id}_X$.

□

Exercise.

a. Show that if $(Y, \star)$ is associative, then $(\mathrm{Func}(X, Y), \star)$ is associative too.

b. Show that if $(Y, \star)$ has a left (resp. right) identity, then $(\mathrm{Func}(X, Y), \star)$ has a left (resp. right) identity as well.

c. Show that if $(Y, \star)$ has an identity element and every element of Y has an inverse, then $(\mathrm{Func}(X, Y), \star)$ satisfies the same properties.

d. Let $f : X_1 \to Y_1$ and $g : X_2 \to Y_2$ be two functions. In general (i.e., when $X_1 \cap X_2 \neq \varnothing$) when and how can we modify the definition of $f \cup g$ to obtain a function from $X_1 \cup X_2$ to $Y_1 \cup Y_2$?

e. Let X be an arbitrary set and $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be two functions.

   (i) Prove that $f + g = s \circ (f \times g)$ where $s : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ is defined by $s(x, y) = x + y$.

   (ii) Prove that $fg = p \circ (f \times g)$ where $p : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ is defined by $p(x, y) = xy$.

9 Relations

Let X be an arbitrary set. A *binary relation* (or simply a *relation*) on X (or between elements of X) is a subset of $X \times X$.

> Suppose that $R \subseteq X \times X$ is a relation on X. For every $x, y \in X$, we write xRy iff $(x, y) \in R$, and $x \not R y$ iff $(x, y) \notin X \times X$.

Usually, when we know the meaning of a relation, instead of the letter R we use various symbols. You are already familiar with many relations, e.g., $<, \leq, \subseteq, \equiv, ...$ Other symbols like $\simeq, \approx, \prec, \lhd, ...$ are also used.

Exmaple 9.0.1.

a. Let $X = \{1, 2, 3, 4\}$. We can define a relation $<$ on X by

$$R \ = \ \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}.$$

Observe that xRy really means "x *is less than* y".

b. More genreally, the relation

$$R \ = \ \{(x, y) \ : \ x, y \in \mathbb{R} \text{ and } x < y\}$$

is the usual order relation on $\mathbb{R}$.

c. Two obvious relations in every set X are $\varnothing$ and $X \times X$.

d. Another obvious relation on every set is *equality*, which is defined as follows:

$$R = \{(x, y) \in X \times X \ : \ x = y\} = \{(x, x) \ : \ x \in X\}.$$

e. We can define a relation on $\mathbb{Z}$:

$$R = \{(m, n) \ : \ m, n \in \mathbb{Z} \text{ and } n \text{ is divisible by } m\}.$$

This relation is called *divisibility*, and usually instead of aRb we write $a|b$. For example, $3|6$ and $(-2)|4$ but $3 \not| 5$.

f. Let X be an arbitrary set. Then we can define the following relations on $\wp(A)$:

(i) $R = \{(A, B) \ : \ A, B \subseteq X \text{ and } A \subseteq B\}$.

(ii) $R = \{(A, B) \ : \ A, B \subseteq X \text{ and } A \text{ and } B \text{ are in bijective correspondence }\}$.

(iii) $R = \{(A, B) \ : \ A, B \subseteq X \text{ and } A \cap B = \varnothing\}$.

(iv) Last but certainly not least, every function $f : X \to X$ corresponds to a subset of $X \times X$ (its graph) and hence is a relation.

9.1 Basic properties of relations

Let X be a set and $R \subseteq X \times X$ be a binary relation on X. Then:

a. R is called *reflexive* if xRx for every $x \in X$,

b. R is called *irreflexive* if $x \not R x$ for every $x \in X$,

c. R is called *symmetric* if $\forall_{x,y \in X}\ (xRy \Rightarrow yRx)$,

d. R is called *antisymmetric* if $\forall_{x,y \in X}\ \big((xRy \wedge yRx) \Rightarrow (x = y)\big)$

e. R is called *transitive* if $\forall_{x,y,z \in R}\ \big((xRy \wedge yRz) \Rightarrow (xRz)\big)$

Exmaple 9.1.1. :

- The relation "$=$" (equality) on $\mathbb{Z}$ is reflexive, symmetric, antisymmetric, and transitive, but not irreflexive.
- The relation "$\leq$" on $\mathbb{R}$ is reflexive, antisymmetric, and transitive, but not irreflexive or symmetric.
- The relation $<$ is irreflexive, antisymmetric, and transitive, but not reflexive or symmetric.
- For any arbitrary nonempty set X, the relation $\subseteq$ on $\wp(X)$ is reflexive, antisymmetric, and transitive, but not irreflexive or symmetric.

Remark. Let R be a relation on a set X. Then there is a "smallest" reflexive relation S containing R; it is defined by:

$$S = R \cup \{(x, x)\ :\ x \in X\}.$$

S is called the *reflexive closure* of R. Clearly, if R' is a reflexive relation on X containing R, then $S \subseteq R'$. Which is why we call S smallest.

We can also define the *symmetric closure* and the *transitive closure* of any relation R in a similar fashion. (See exercises below.)

Exercise.

a. Determine which of the relations of Example 9.0.1 are reflexive, irreflexive, symmetric, antisymmetric, or transitive.

b. Let R be a relation on an arbitrary set X. Show that there exists a smallest symmetric relation containing R. (It is called the *symmetric closure* of R.)

c. Let R be a relation on an arbitrary set X. Show that there exists a smallest transitive relation containing R. (It is called the *transitive closure* of R.)

9.2 Equivalence relations

Let X be an arbitrary set. A binary relation R on X which is reflexive, symmetric, and transitive is called an *equivalence relation*. In other words, an equivalence relation satisfies the following properties:

a. For every $x \in X$, xRx.

b. For every $x, y \in X$, if xRy then yRx.

c. For every $x, y, z \in X$, if xRy and yRz then xRz.

Sometimes equivalence relations are denoted by the symbol $\equiv$ or $\simeq$.

Exmaple 9.2.1.

- Equality is an equivalence relation on any set X.
- However, the relations $<$ and $\leq$ on $\mathbb{R}$ are not equivalence relations.
- The relation $R = X \times X$ on X is an equivalence relation (but the relation $R = \varnothing$ is not.)
- Let X be a subset of $\mathbb{R}$ or $\mathbb{C}$. Then the relation defined by

  $$aRb \text{ if and only if } |x| = |y|$$

  is an equivalence relation on X.
- The relation

  $$R = \{(x, y) \ : \ x, y \in \mathbb{R} \text{ and } x - y \in \mathbb{Z}\}$$

  is an equivalence relation on $\mathbb{R}$. (Check this.)

- Let $f \in \mathrm{Func}(\mathbb{R}, \mathbb{R})$. Then the relation $\equiv_f$ on $\mathbb{R}$ defined by

$$x \equiv_f y \ \Leftrightarrow \ f(x) = f(y)$$

is an equivalence relation on $\mathbb{R}$.

- Let X and Y be two sets and $A \subseteq X$ be fixed. Then the relation $\equiv_A$ on $\mathrm{Func}(X, Y)$ defined by

$$f \equiv_A g \ \Leftrightarrow \ f(A) = g(A)$$

is an equivalence relation.

- The next example, which is stated in Proposition 9.2.2 below, is of great importance in number theory.

Proposition 9.2.2. *Let n be a positive integer. Then the relation $\equiv_n$ on $\mathbb{Z}$ defined by*

$$x \equiv_n y \ \Leftrightarrow \ (x - y) \text{ is divisible by } n$$

is an equivalence relation.

Proof. We should prove that $\equiv_n$ is reflexive, symmetric and transitive.

Proof of reflexivity: For every $x \in Z$ we have $x - x = 0$ which is divisible by n. Therefore $x \equiv_n x$.

Proof of symmetricity: If $x, y \in \mathbb{Z}$ and $x - y$ is divisible by n then $(x - y) = nk$ for some integer k. But then $y - x = n(-k)$, i.e., $y - x$ is divisible by n as well.

Proof of transitivity: Suppose that $x, y, z \in \mathbb{Z}$ such that $x \equiv_n y$ and $y \equiv_n z$. Then $(x - y) = nk_1$ and $(y - z) = nk_2$ where k_1 and k_2 are integers. Therefore $x - z = (x - y) + (y - z) = nk_1 + nk_2 = n(k_1 + k_2)$. It follows that $x - z$ is also divisible by n, i.e., $x \equiv_n z$. □

9.3 Equivalence classes and partitions

Suppose that $\equiv$ is an equivalence relation on a set X. For every $a \in X$, the *equivalence class* of a, which is denoted by $\overline{a}$ or $[a]$, is the set

$$[a] \ = \ \{x \in X \ : \ x \equiv a\}.$$

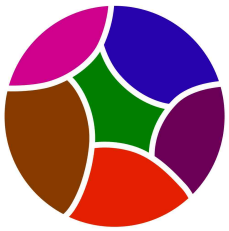
The equivelance classes of an equivalence relation on X yield a special family of subsets of X which is called a *partition* of X:

Let X be an arbitrary set. A *partition* of X is a family $\mathscr{S}$ of subsets of X which satisfies the following properties:

- $\varnothing \notin \mathscr{S}$,
- $\cup \mathscr{S} = X$,
- For every $A, B \in \mathscr{S}$, if $A \neq B$ then $A \cap B = \varnothing$.

Exmaple 9.3.1.

- $\mathscr{S} = \{\{1\}, \{2,3\}, \{4,5\}\}$ and $\mathscr{S} = \{\{1,2\}, \{3,4,5\}\}$ are partitions of $\{1,2,3,4,5\}$.
- $\mathscr{S} = \{(n, n+1] \ : \ n \in \mathbb{Z}\}$ and $\mathscr{S} = \{(2n, 2n+2] \ : \ n \in \mathbb{Z}\}$ are partitions of $\mathbb{R}$.



Visualizing a partition

Proposition 9.3.2. *If X is a set and $\equiv$ is an equivalence relation on X then the set*

$$\{ [a] \ : \ a \in X \}$$

is a partition of X. Conversely, if $\mathscr{S}$ is a partition of X then the binary relation $\equiv$ on X defined by

$$x \equiv y \ \Leftrightarrow \ \exists_{A \in \mathscr{S}} \, (x \in A \ \wedge \ y \in A)$$

is indeed an equivalence relation whose equivalence classes are the elements of $\mathscr{S}$.

Proof. Let $\equiv$ be an equivalence relation. First note that for every $x \in X$ we have $x \in [x]$. This implies that

$$\varnothing \notin \{ [a] \ : \ a \in X \}$$

and

$$\bigcup \{ [a] \ : \ a \in X \} = X.$$

Next we prove that if $[a_1], [a_2] \in \{ [a] \ : \ a \in X \}$ and $[a_1] \neq [a_2]$ then $[a_1] \cap [a_2] = \varnothing$. We use the method of proof by transposition. Assume that $[a_1] \cap [a_2] \neq \varnothing$. Choose an

element $x \in [a_1] \cap [a_2]$. Since $x \in [a_1]$, we have $x \equiv a_1$. Since $x \in [a_2]$, we have $x \equiv a_2$, and by symmetry of $\equiv$ we have $a_2 \equiv x$. Now observe that

$$(a_2 \equiv x \ \wedge \ x \equiv a_1) \ \Rightarrow \ (a_2 \equiv a_1 \ \wedge \ a_1 \equiv a_2).$$

Next we prove that $[a_1] \subseteq [a_2]$ and $[a_2] \subseteq [a_1]$ (from which it immediately follows that $[a_1] = [a_2]$).

▶ Proof of $[a_1] \subseteq [a_2]$: for every $s \in [a_1]$ we have $s \equiv a_1$, and

$$(s \equiv a_1 \ \wedge \ a_1 \equiv a_2) \Rightarrow (s \equiv a_2) \qquad \text{(by transitivity of } \equiv)$$

which means that $s \in [a_2]$. This completes the proof of $[a_1] \subseteq [a_2]$.

▶ Proof of $[a_2] \subseteq [a_1]$: for every $s \in [a_2]$ we have $s \equiv a_2$, and

$$(s \equiv a_2 \ \wedge \ a_2 \equiv a_1) \Rightarrow (s \equiv a_1) \qquad \text{(by transitivity of } \equiv)$$

which means that $s \in [a_1]$. This completes the proof of $[a_2] \subseteq [a_1]$.

The proof of the converse (i.e., that the relation obtained from a partition $\mathscr{S}$ is an equivalence relation) is left as an exercise.

□

Proposition 9.3.2 can be paraphrased as the following important fact:

> For every set X, there exists a bijective correspondence between the set of equivalence relations on X and the set of partitions of X.

Let X be a set and $\equiv$ be an equivalence relation on X. The set of equivalence classes of X, i.e., the set

$$\{ \, [x] \ : \ x \in X \, \}$$

is called the *quotient* of X by $\equiv$. The quotient of X by $\equiv$ is denoted by $X/\equiv$.

Exmaple 9.3.3. Let us determine the equivalence classes of some examples of equivalence relations and the corresponding quotients.

- Equality ("=") is an equivalence relation in every set. The equivalence classes of "=" are singletons $\{x\}$ where $x \in X$. Therefore the partition of X corresponding to "=" is $\{ \, \{x\} \ : \ x \in X \, \}$. Clearly $X/=$ is in bijective correspondence with X.

- Let us consider the equivalence relation $\equiv_4$ on $\mathbb{Z}$, which is given by

$$x \equiv_4 y \quad \Leftrightarrow \quad (x-y) \text{ is divisible by } 4.$$

It can be seen easily that the partition of $\mathbb{Z}$ corresponding to $\equiv_4$ is

$$\{A_0, A_1, A_2, A_3\}$$

where

$$A_i = \{4n + i \ : \ n \in \mathbb{Z}\} \text{ for } i = 0, 1, 2, 3.$$

In other words, $A_i = 4\mathbb{Z} + i$. Of course A_0, A_1, A_2, and A_3 are the equivalence classes of this equivalence relation. The quotient $\mathbb{Z}/\equiv_4$ is a set with four elements. Sometimes this quotient is written as $\mathbb{Z}/4\mathbb{Z}$ or $\mathbb{Z}/4$ or $\{\ \overline{0}, \overline{1}, \overline{2}, \overline{3}\ \}$.

- More generally, if n is a positive integer, we can consider the equivalence relation $\equiv_n$ on $\mathbb{Z}$, defined by

$$x \equiv_n y \quad \Leftrightarrow \quad (x-y) \text{ is divisible by } n.$$

The quotient of $\mathbb{Z}$ by $\equiv_n$ is also denoted by

$$\mathbb{Z}/2 \text{ or } \mathbb{Z}/n\mathbb{Z} \text{ or } \{\overline{0}, \overline{1}, \ldots, \overline{n-1}\}.$$

- Let us consider the equivalence relation $\equiv_{\{-1\}}$ on $X = \mathrm{Func}(\mathbb{R}, \mathbb{R})$ defined by

$$f \equiv_{\{-1\}} g \quad \Leftrightarrow \quad f(-1) = g(-1).$$

The partition of $\mathrm{Func}(\mathbb{R}, \mathbb{R})$ corresponding to this equivalence relation is $\{A_t \ : \ t \in \mathbb{R}\}$ where

$$A_t = \{\ f : \mathbb{R} \to \mathbb{R} \ : \ f(-1) = t\ \} \quad \text{for } t \in \mathbb{R}.$$

The quotient of $\mathrm{Func}(\mathbb{R}, \mathbb{R})$ by $\equiv_{\{-1\}}$ is in bijective correspondence to $\mathbb{R}$. The correspondence can be concretely described as follows: if $[f]$ is an equivalence class (where f is an arbitrary element in this class), the correspondence is given by

$$[f] \ \mapsto \ f(-1)$$

- Consider the equivalence relation $\sim$ on $\mathbb{R}$ defined by

$$x \sim y \ \Leftrightarrow \ (x-y) \in \mathbb{Z}$$

The equivalence class of an element $r \in \mathbb{R}$ is the set $r + \mathbb{Z}$. How can we describe the quotient $\mathbb{R}/\sim$ concretely? It is easy to see that there is a *unique* element $r' \in r + \mathbb{Z}$ such that $0 \leq r' < 1$. Therefore the set $\mathbb{R}/\sim$ is in bijective correspondence with the interval $[0, 1)$.

- Consider the equivalence relation $\equiv$ on $\mathbb{R}$ defined by

$$x \equiv y \;\Leftrightarrow\; (x-y) \in \mathbb{Q}$$

The equivalence class of an element $x \in \mathbb{R}$ is $x + \mathbb{Q}$. What is the quotient $\mathbb{R}/\equiv$? In this case, it is not possible to find a concrete description of this quotient. But certainly this quotient does exist. (More on this issue later!)

- Consider the equivalence relation $\equiv$ on $\mathbb{R}^2$ defined by

$$(x_1, y_1) \equiv (x_2, y_2) \;\Leftrightarrow\; x_1^2 + y_1^2 = x_2^2 + y_2^2.$$

The partition of $\mathbb{R}^2$ that corresponds to $\equiv$ is $\{A_t \;:\; t \in [0, \infty)\}$ where

$$A_t = \{(x, y) \in \mathbb{R}^2 \;:\; x^2 + y^2 = t\}.$$

Clearly the A_t , $t \in [0, \infty)$, are the equivalence classes of $\equiv$. Each equivalence class A_t is a circle of radius t centered at the origin. The quotient $\mathbb{R}^2/\equiv$ is in bijective correspondence with the interval $[0, \infty)$, and the bijection between $\mathbb{R}^2/\equiv$ and $[0, \infty)$ is given by

$$[(x, y)] \;\mapsto\; x^2 + y^2$$

Exercise.

a. Prove the unproven converse of Proposition 9.3.2.

b. Prove that it is impossible to find an equivalence relation $\equiv$ on $\mathbb{R}$ for which the following statement is true:

$$x \equiv y \;\Leftrightarrow\; (x-y) \notin \mathbb{Q}.$$

c. *(Bonus question)*Prove that there exists an equivalence relation $\equiv$ on $\mathbb{R}$ with the following two properties:

(i) if $x \equiv y$ then $x - y \notin \mathbb{Q}$.

(ii) every equivalence class of $\equiv$ is an infinite set.

9.4 Canonical projection and induced map

Let X be a set and $\sim$ be an equivalence relation on X. Then:

a. the function

$$\pi : X \to X/\sim \quad , \quad \pi(x) = [x]$$

is a surjection. It is called the *canonical projection* from X onto $X/\sim$.

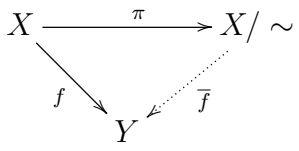
b. Let Y be another set and $f : X \to Y$ be a function. Assume that:

$$\text{if } x \sim y \text{ then } f(x) = f(y).$$

Then we can find a function

$$\overline{f} : X/\sim \to Y$$

defined by $\overline{f}([x]) = f(x)$. The function $\overline{f}$ is called the map *induced* from f.



Exmaple 9.4.1. Let $\equiv$ be the equivalence relation on $\mathbb{R} \times \mathbb{R}$ defined by

$$(x, y) \equiv (x', y') \;\Leftrightarrow\; x = x'.$$

(Observe that the elements of $\mathbb{R}^2/\equiv$ are in correspondence with vertical lines in $\mathbb{R}^2$.)

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function $f\big((x, y)\big) = -2x^2$. Then

$$(x, y) \equiv (x', y') \;\Rightarrow\; x = x' \;\Rightarrow\; f\big((x, y)\big) = -2x^2 = -2x'^2 = f\big((x', y')\big).$$

Therefore f induces a function $\overline{f} : \mathbb{R}^2/\equiv \;\to\; \mathbb{R}$ defined by

$$\overline{f}\big([(x, y)]\big) = -2x^2.$$

However, the function $g : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined by $g(x, y) = x + y$ does not define an induced map $\overline{g} : \mathbb{R} \times \mathbb{R}/\equiv \;\to\; \mathbb{R}$. The reason is that g is not constant on equivalence classes: for example, $(1, 0) \equiv (1, 1)$ but $g(1, 0) = 1 + 0 = 1$ and $g(1, 1) = 1 + 1 = 2$, and therefore $g(1, 0) \neq g(1, 1)$.

Exercise.

a. Let $X = \mathbb{R}$ or $\mathbb{C}$. Show that the relation $\equiv$ defined by

$$x \equiv y \;\Leftrightarrow\; \exists_{r \in X \setminus \{0\}}(x = ry)$$

is an equivalence relation. How many equivalence classes does $\equiv$ have?

b. Let X be an arbitrary set.

(i) Suppose that $S \subseteq \wp(X \times X)$ is a set such that every element of S is an equivalence relation on X. Prove that $\cap S$ is also an equivalence relation on S.

(ii) Conclude that for every relation R on X, there exists a unique "smallest" equivalence relation containing R. (This equivalence relation is sometimes said to be *generated* by R.) *Hint* : To start the proof, consider the set of all equivalence relations that contain R.

(iii) Let $X = \bigcup_{n=1}^{\infty}\{(x, y) \;:\; 2n + 1 \leq x^2 + y^2 \leq 2n + 2\}$. Define the relation R on X by:

$$(x, y)R(x', y') \;\Leftrightarrow\; \text{the line segment connecting } (x, y) \text{ and } (x', y') \text{ lies in } X.$$

Show that R is not an equivalence relation. Determine the equivalence relation generated by R by means of identifying its equivalence classes.

c. Consider the relation $\sim$ on $\mathrm{Func}(\mathbb{R}, \mathbb{R})$ defined by

$$f \sim g \;\Leftrightarrow\; \exists_{\varepsilon \in \mathbb{R}^+} \forall_{x \in (-\varepsilon, \varepsilon)} f(x) = g(x).$$

- Prove that $\sim$ is an equivalence relation.
- Prove that if f and g are two distinct polynomials then $[f] \neq [g]$.

d. Let X be an arbitrary set. On $\wp(X)$, consider the relation $\equiv$ defined by

$$A \equiv B \;\Leftrightarrow\; A \Delta B \text{ is finite.}$$

(i) Prove that $\equiv$ is an equivalence relation over $\wp(X)$.

(ii) Prove that if X is finite then $\equiv$ has exactly one equivalence class.

(iii) Conversely, prove that if $\wp(X)/\equiv$ has exactly one element then X is finite.

(iv) Prove that the quotient set $\wp(\mathbb{N})/\equiv$ is infinite.

Suppose that $A, B, A_1, B_1 \subseteq X$ such that $A \equiv A_1$ and $B \equiv B_1$. show that

(v) $A \cap B \equiv A_1 \cap B_1$.

(vi) $A \cup B \equiv A_1 \cup B_1$.

(vii) $A^c \equiv A_1^c$.

(viii) $A \setminus B \equiv A_1 \setminus B_1$.

(ix) $A \Delta B \equiv A_1 \Delta B_1$.

9.5 Partial orders

Let X be an arbitrary set. A *partial order* $\preceq$ on X is a relation on X which is

(a) **reflexive**, i.e., $\forall_{x\in X}\ (x \preceq x)$,

(b) **transitive**, i.e, $\forall_{x,y,z\in X}\ \big((x \preceq y \wedge y \preceq z) \Rightarrow (x \preceq z)\big)$,

(c) **antisymmetric**, i.e., $\forall_{x,y\in X}\ ((x \preceq y\ \wedge\ y \preceq x) \Rightarrow (x = y))$.

A partial order may also be represented by other symbols, e.g., $\leq$, $\trianglelefteq$.
A *strict partial order* $\prec$ on X is a relation which is

(a) **transitive**, i.e, $\forall_{x,y,z\in X}\ \big((x \prec y \wedge y \prec z) \Rightarrow (x \prec z)\big)$,

(b) **irreflexive**, i.e., $\forall_{x\in X}\ (\ x \not R x\ )$.

A strict partial order may also be represented by other symbols, e.g., $<$, $\lhd$.

A set which is endowed with a partial order is called a *partially ordered set* (or sometimes simply an *ordered set*). To emphasize that there is a partial order $\leq$ on a set X, we may write $(X, \leq)$. Similarly, a set endowed with a strict partial order is called a *strictly ordered set*.

It is important to note that if "$<$" is a strict partial order on a set X, then

$$\nexists_{x,y\in X}\ (x < y\ \wedge y < x).$$

In fact, if there exist $x, y \in X$ such that $x < y$ and $y < x$, then by transitivity we have:

$$x < y\ \wedge\ y < x\ \Rightarrow\ x < x$$

which contradicts irreflexivity.

Consequently, every strict partial order is antisymmetric.

Exmaple 9.5.1.

- The relation "$<$" (which is read as "*strictly less than*") is a strict partial order on $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$.

- The relation "$\leq$" (which is read as "*less than or equal to*") is a partial order on $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$.
- For an arbitrary set X, inclusion (i.e., $\subseteq$) is a partial order on $\wp(X)$, and strict inclusion (i.e., $\subsetneq$) is a strict partial order on $\wp(X)$.
- On $\mathbb{Z}$ define the relation $\lhd$ by

  $$x \lhd y \;\Leftrightarrow\; y < x.$$

  It is easy to check that $\lhd$ is a strict partial order. In general, the inverse of a (strict) partial order is a (strict) partial order.
- On $\mathbb{R}$, the relation

  $$x \prec y \;\Leftrightarrow\; |x| < |y|$$

  is a strict partial order. Note that $1 \not\prec -1$ and $-1 \not\prec 1$.
- On $\mathbb{N}$, the relation

  $$x \preceq y \;\Leftrightarrow\; y \text{ is divisible by } x, \text{ i.e., } \exists m \in \mathbb{Z}\ (y = mx)$$

  is a partial order. Note that $\preceq$ is not a partial order on $\mathbb{Z}$, as $-2 \preceq 2$ and $2 \preceq -2$ but $2 \neq -2$..
- The empty relation $\varnothing \subseteq X \times X$ is a strict partial order on any set X.

Partial orders and strict partial orders are related by the following proposition.

Proposition 9.5.2. *Let X be an arbitary set.*

a. If $\preceq$ is a partial order on X then the relation $\prec$ defined by

$$a \prec b \;\Leftrightarrow\; a \preceq b \;\wedge\; a \neq b$$

is a strict partial order on X.

b. Conversely, if $\prec$ is a strict partial order on X then the relation $\preceq$ defined by

$$a \preceq b \;\Leftrightarrow\; a \prec b \;\vee\; a = b$$

is a strict partial order on X.

Proof. (a) To prove that $\prec$ is a strict partial order, we need to check the following properties:

- *Transitive.* Let $x, y, z \in X$ such that $x \prec y$ and $y \prec z$. Then $x \preceq y$ and $y \preceq z$, and therefore $x \preceq z$. Moreover, $x \neq y$ and $y \neq z$.

  To conclude that $x \prec z$, we need to make sure that $x \neq z$. But if $x = z$ then we have

  $$x \leq y \ \wedge \ y \leq x \ \Rightarrow x = y \quad \text{because } \preceq \text{ is antisymmetric}$$

  which contradicts the fact that $x \neq y$.

- *Irreflexive.* That $x \not\prec x$ for every $x \in X$ is clear by the definition of $\prec$.

(b) To prove that $\preceq$ is a partial order, we need to check the following properties:

- *Reflexive.* That $x \preceq x$ for every $x \in X$ is clear by the definition of $\preceq$.

- *Transitive.* Let $x, y, z \in X$ such that $x \preceq y$ and $y \preceq z$. There are four cases to consider:

  (i) $x \prec y$ and $y \prec z$. Then since $\prec$ is transitive, we have $x \prec z$, which implies that $x \preceq z$.

  (ii) $x \prec y$ and $y = z$. Then clearly $x \prec z$, hence $x \preceq z$.

  (iii) $x = y$ and $y \prec z$. Then clearly $x \prec z$, hence $x \preceq z$.

  (iv) $x = y$ and $y = z$. Then $x = z$, hence clearly $x \preceq z$ by the definition of $\preceq$.

- *Antisymmetric.* We use the method of proof by contradiction. Suppose that there exist $x, y \in X$ such that $x \preceq y$, $y \preceq x$, and $x \neq y$. From the definition of $\preceq$ it follows that $x \prec y$ and $y \prec x$. But since $\prec$ is transitive, we should have $x \prec x$, which is impossible because $\prec$ is irreflexive.

□

Let $(X, \leq)$ be a partially ordered set.

a. An element $x_\circ \in X$ which satisfies $\forall_{x \in X} \ (x \leq x_\circ)$ is called a *maximum* element of $(X, \leq)$.

b. An element $x_\circ \in X$ which satisfies $\nexists_{x \in X} \ (x_\circ \leq x \ \wedge \ x_\circ \neq x)$ is called a *maximal* element of $(X, \leq)$.

c. An element $x_\circ \in X$ which satisfies $\forall_{x \in X} \ (x_\circ \leq x)$ is called a *minimum* element of $(X, \leq)$.

d. An element $x_\circ \in X$ which satisfies $\nexists_{x \in X} \ (x \leq x_\circ \ \wedge \ x_\circ \neq x)$ is called a *minimal* element of $(X, \leq)$.

Exmaple 9.5.3.

- Let $X = \{1, 2, 3\}$. Suppose $X_1 = \wp(X)$. Then $(\wp(X), \subseteq)$ is a partially ordered set, and has a (unique) maximum element M, where $M = \{1, 2, 3\}$. Now set

  $$X_2 = \{\ \{1\}, \{1, 3\}, \{2, 3\}, \varnothing\ \}.$$

  Then $(X_2, \subseteq)$ is also a partially ordered set. Observe that $(X_2, \subseteq)$ does not have a maximum element, and it has two maximal elements: $\{1, 3\}$ and $\{2, 3\}$.

- Let $X = \mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ and coincide the strict partial order $\prec$ on X defined by

  $$(x, y) \prec (x', y') \ \Leftrightarrow\ x < x'.$$

  By Proposition 9.5.2, the relation $\preceq$ defined by

  $$(x, y) \preceq (x', y') \ \Leftrightarrow\ x < x' \ \vee\ (x, y) = (x', y')$$

  is a partial order on X. Observe that $(X, \preceq)$ does not have maximum, maximal, minimum, or minimal elements. Consider the subset Y of X given by

  $$Y = \{\ (1, 0), (1, 2), (0, 0), (1, -3)\ \}.$$

  Then Y is a partially ordered set with the ordering induced from X. It is easily checked that $(0, 0)$ is both a minimum and a minimal element. Y does not have a maximum element but it has three maximal elements: $(1, 0), (1, 2), (1, -3)$.

A partially ordered set may have zero, one, or several maximal (or minimal) elements.

Proposition 9.5.4. *A partially ordered set $(X, \leq)$ has at most one maximum (resp. minimum) element. If $(X, \leq)$ has a maximum (resp.) minimum element, then this element is the unique maximal (resp. minimal) element of $(X, \leq)$.*

Proof. We only give the proof for maximum/maximal elements. (The case of minimum/minimal elements is very similar.)

Suppose x and x' are two maximum elements of X. Then we should have $x \leq x'$ and $x' \leq x$, from which it follows that $x = x'$ (since $\leq$ is antisymmetric).

Next suppose that there exists a maximum element $x_\circ \in X$. First we prove that $x_\circ$ is maximal. The proof is by contradiction: if $x_\circ$ is not maximal, then by definition there exists a $y \in X$ such that $y \neq x_\circ$ and $x_\circ \leq y$. But since $x_\circ$ is maximum, we have $y \leq x_\circ$, and we conclude that $x_\circ = y$ (because $\leq$ is antisymmetric). But this is impossible (because we also have $x_\circ \neq y$).

Finally we show that the maximum element $x_\circ$ is the unique maximal element of $(X, \leq)$. Suppose x' is a maximal element of X and $x' \neq x_\circ$. Since $x_\circ$ is maximum, we have $x' \leq x_\circ$. But this contradicts the fact that x' is maximal.

□

We conclude that the maximum element of $(X, \leq)$, if exists, is also maximal. However, a maximal element of $(X, \leq)$ is not necessarily maximum. The same statement holds for minimum/minimal elements.

Two elements x, y of a partially ordered set $(X, \leq)$ are called *comparable* iff $x \leq y$ or $y \leq x$.

Exmaple 9.5.5. Let $X = \{\{1,2\},\{2,3\},\{1,3\},\{1,3,4\}\}$ partially ordered by inclusion "$\subseteq$". Then $\{1,3\}$ and $\{1,3,4\}$ are comparable, but $\{1,3\}$ and $\{2,3\}$ are not comparable.

A partially ordered set $(X, \preceq)$ is called *totally ordered* (or *linearly ordered*) iff every pair of elements of X are comparable, i.e.,

$$\forall_{x,y \in X}\ (x \preceq y\ \vee y \preceq x).$$

The partial order of a totally ordered set is sometimes called a *total order*.

9.6 Transfinite induction

Let $(X, \preceq)$ be a well-ordered set. For every $x \in X$, we define the interval $(-\infty, x)$ as follows:

$$(-\infty, x) = \{y \in X\ :\ y \prec x\} = \{y \in X\ :\ y \preceq x\ \wedge\ y \neq x\ \}.$$

Proposition 9.6.1. *Let $(X, \preceq)$ be a well-ordered set. Suppose that S is a subset of X which satisfies*

$$\forall_{a \in X}[(-\infty, a) \subseteq S\ \Rightarrow a \in X)].$$

Then $S = X$.

Proof. The proof is by contradiction. Suppose that $S \neq X$. Then $X \setminus S$ is a nonempty subset of X. Since X is well-ordered, $X \setminus S$ has a minimum element. Let $x_\circ \in X \setminus S$ be the minimum element of $X \setminus S$. Clearly $(-\infty, x_\circ) \subseteq S$, and by the hypothesis of the theorem, $x_\circ \in S$, which contradicts the assumption that $x_\circ \in X \setminus S$.

□

Proposition 9.6.1 is sometimes called the *principle of transfinite induction*. It is an induction principle which is valid for any well-ordered set $(X, \preceq)$. It has the following immediate corollary:

Corollary 9.6.2. *Let $(X, \preceq)$ be a well-ordered set and $P(x)$ be a condition which depends on $x \in X$. Suppose that for every $a \in X$, the following implication is correct:*

$$\text{If } P(x) \text{ is true for every } x \in (-\infty, a), \text{ then } P(a) \text{ is also true.} \tag{15}$$

Then $\forall_{x \in X} P(x)$.

Proof. Consider the subset S of X defined by

$$S = \{x \in X \ : \ P(x)\}.$$

It is easily checked that S satisfies the hypothesis of Proposition 9.6.1. Consequently, $S = X$.

□

Remark. 1. Observe that since $(\mathbb{N}, \leq)$ is well-ordered, Proposition 9.6.1 can be applied to $(X, \preceq) = (\mathbb{N}, \leq)$. It implies that the usual induction on $\mathbb{N}$ is a valid method of proof.

2. Let $X \neq \varnothing$. Suppose that you want to use Corollary 9.6.2 to prove that $\forall_{x \in X} P(x)$. Then you should check the condition (15) for *every* $a \in X$. In particular, since X is well-ordered, it has a minimum element $a_\circ$. In this case, $(-\infty, a_\circ) = \varnothing$ and therefore checking condition (15) for $a = a_\circ$ amounts to checking that $P(a_\circ)$ is true. The case $P(a_\circ)$ is called the base of induction.

> If you do not check that the base of induction is true, your proof is not complete.

In Example 1.1.1, we also checked $P(0)$.

Exmaple 9.6.3.

- $(\mathbb{R}, \leq)$ is totally ordered (where $\leq$ is the usual "less than or equal to" ordering).
- If $X = \{\ [1, a) \ : \ a \in \mathbb{R} \ \wedge \ a > 1\ \}$ then $(X, \subseteq)$ is totally ordered.

Exercise.

Let $(X, \preceq)$ be a totally ordered set. Prove that if $x_\circ \in X$ is a minimal element, then it is also a minimum element.

> A totally ordered set $(X, \preceq)$ is said to be *well-ordered* iff every nonempty subset of X has a minimum element, i.e.,
>
> $$\forall_{A \in \wp(X)} \left(A \neq \varnothing \ \Rightarrow \ \exists_{a_\circ \in A} \forall_{a \in A} (a_\circ \leq a)\right).$$
>
> The relation $\preceq$ on a well-ordered set $(X, \preceq)$ is called a *well-ordering* on X.

Exmaple 9.6.4.

- $(\mathbb{N}, \leq)$ is well-ordered.
- $(\mathbb{R}, \leq)$ is not because for example the subset $(0, \infty)$ of $\mathbb{R}$ does not have a minimum element.
- The set $(\mathbb{Z}, \leq)$ is not well-ordered because for example $\mathbb{Z} \subseteq \mathbb{Z}$ does not have a minimum element.
- The set $(\mathbb{Z}, \preceq)$, where $\preceq$ is defined by

$$x \preceq y \;\Leftrightarrow\; (|x| < |y|) \;\vee\; (\; |x| = |y| \;\wedge\; x < y\;)$$

  is well-ordered. (Exercise: Prove this fact.)

The above examples show that there exist totally ordered sets which are not well-ordered.

Exercise.

(Bonus Problem.) Does there exist a well-ordering on $\mathbb{Q}$? Does there exist a well-ordering on $\mathbb{R}$?

Let $(X, \preceq)$ be a partially ordered set, and let $A \subseteq X$.

a. An element $x \in X$ is called an *upper bound* for A if $\forall_{a \in A}(a \preceq x)$.

b. An element $x \in X$ is called a *lower bound* for A if $\forall_{a \in A}(x \preceq a)$.

Note that lower bound and upper bound of A are not necessarily unique.

Let $(X, \preceq)$ be a totally ordered set, and let $A \subseteq X$.

a. An element $x \in X$ is called the *supremum* (or the *least upper bound*, or *l.u.b.*) of A if x is an upper bound of A and for every $x' \in X$, if x' is an upper bound of A then $x \preceq x'$; in other words:

$$\forall_{x' \in X} \left([\forall_{a \in A}(a \leq x')] \Rightarrow x \leq x' \right).$$

b. An element $x \in X$ is called the *infimum* (or the *greatest lower bound*, or *g.l.b.*) of A if x is a lower bound of A and for every $x' \in X$, if x' is a lower bound of A then $x' \preceq x$; in other words:

$$\forall_{x' \in X} \left([\forall_{a \in A}(x' \leq a)] \Rightarrow x' \leq x \right).$$

Note that the g.u.b. and l.u.b. of a subset, if exist, are unique. The supremum and infimum of A are denoted by $\sup(A)$ and $\inf(A)$ respectively.

Exmaple 9.6.5. Let us consider $(\mathbb{R}, \leq)$ where $\leq$ is the usual "less than or equal to" ordering. Then:

- $\sup\big([0,1)\big) = 1$ and $\inf\big([0,1)\big) = 0$. If $A = \{\frac{1}{n} \ : \ n \in \mathbb{N} \ \wedge \ n > 0\}$ then $\sup(A) = 1$ and $\inf(A) = 0$.
- $\sup(\mathbb{Z})$ and $\inf(\mathbb{Z})$ do not exist.
- $\sup(\mathbb{N})$ does not exist but $\inf(\mathbb{N}) = 0$.

Exercise.

a. Let $(X, \preceq)$ be a totally ordered set, and $A \subseteq X$. Prove that $\sup(A)$ and $\inf(A)$, if exist, are unique.

b. Let $(X, \preceq_X)$ and $(Y, \preceq_Y)$ be two partially ordered sets. Define a new relation on $X \times Y$ as follows. For every $(x, y) \in X \times Y$ and $(x', y') \in X \times Y$:

$$(x, y) \preceq (x', y') \ \Leftrightarrow \ (x \preceq_X x' \ \wedge \ x \neq x') \ \vee \ (x = x' \ \wedge \ y \preceq_Y y').$$

(This relation is called the *lexicographic order* on $X \times Y$.)

(i) Prove that $(X \times Y, \prec)$ is a partially ordered set.

(ii) Prove that if $(X, \preceq_X)$ and $(Y, \preceq_Y)$ are totally ordered, then $(X \times Y, \preceq)$ is also totally ordered.

(iii) Prove that if $(X, \preceq_X)$ and $(Y, \preceq_Y)$ are well-ordered, then $(X \times Y, \preceq)$ is also well-ordered.

c. Let $\prec_Y$ be a stict partial order on a set Y, and $f : X \to Y$ be a function. Define a relation $\prec_X$ on X as follows. For every $x_1, x_2 \in X$,

$$x_1 \prec_X x_2 \;\Leftrightarrow\; f(x_1) \prec_Y f(x_2).$$

Prove that $\prec_X$ is a strict partial order on X.

d. Let $\prec_Y$ be a strict partial order on a set Y. Let X be an arbitrary set and $A \subseteq X$. Define a relation $\prec$ on $\mathrm{Func}(X, Y)$ by

$$f \prec g \;\Leftrightarrow\; \forall_{a \in A}\, (f(a) \prec_Y g(a)).$$

Prove that $\prec$ is a strict partial order on $\mathrm{Func}(X, Y)$.

e. Find the supremum and infimum (if exist) of the following subsets of $\mathbb{R}$:

(i) $\{x \in \mathbb{R} \;:\; x^2 + x - 1 < 0\}$.

(ii) $\{x \in \mathbb{R} \;:\; x^2 + x - 1 > 0\}$.

(iii) $\{\frac{1}{n} + (-1)^n \;:\; n \in \mathbb{N}^+\}$.

(iv) $\{\frac{1}{1-x} \;:\; x \in \mathbb{R} \text{ and } x > 1\}$.

(iv) $\{\frac{1}{1-x} \;:\; x \in \mathbb{R} \text{ and } x < 1\}$.

f. Let $(\mathbb{R}, \leq)$ denote the real numbers with its usual "les than or equal to" ordering. A subset A of $\mathbb{R}$ is said to be a *well-ordered subset* iff it is well-ordered with the induced ordering, i.e., iff every nonempty subset of A has a minimum element. Consider the set

$$\mathscr{W}(\mathbb{R}) = \{\, A \in \wp(\mathbb{R}) \;:\; A \text{ is a well-ordered subset of } \mathbb{R} \,\}.$$

(i) Prove that if $A \in \mathscr{W}(\mathbb{R})$ and $B \in \mathscr{W}(\mathbb{R})$, then $A \cup B \in \mathscr{W}(\mathbb{R})$.

(ii) If $\{A_i \;:\; i \in I\}$ is a family of well-ordered subsets of $\mathbb{R}$, is it necessarily true that $\bigcup_{i \in I} A_i \in \mathscr{W}(\mathbb{R})$? Justify your answer (i.e., either prove or disprove it.)

(iii) Prove that if $A \in \mathscr{W}(\mathbb{R})$ and $B \subseteq A$ then $B \in \mathscr{W}(\mathbb{R})$.

(iv) Prove that if $A \in \mathscr{W}(\mathbb{R})$ and B is a subset of $\mathbb{R}$ such that $B \setminus A$ is finite, then $B \in \mathscr{W}(\mathbb{R})$.

(v) Give an example of subsets A and B of $\mathbb{R}$ such that $A \in \mathscr{W}(\mathbb{R})$, $A \setminus B$ is finite, and $B \notin \mathscr{W}(\mathbb{R})$.

10 Cardinality

Which one has more elements : $\mathbb{Q}$ or $\mathbb{N}$? Of course the intuitive answer to this question is :*"well, clearly $\mathbb{Q}$ has more elements"*. This answer is not mathematically valid. In fact since both $\mathbb{N}$ and $\mathbb{Q}$ are infinite sets, one should first address the following question: how can we *count* the number of elements of $\mathbb{N}$ or $\mathbb{Q}$? In other words, if X is an infinite set, then what does $|X|$ mean?

The above question also occurred to Georg Cantor, and he found a mathematically rigorous way to approach it. Cantor's idea was to use functions between two sets to measure their sizes. In this fashion, he could compare any two sets, without explicitly counting the number of their elements. The purpose of this chapter is to introduce this idea in more detail.

10.1 Sets of equal cardinality

Suppose you are given two (possibly infinite) sets A and B. Before answering the question *"which one has more elements"* let us try to answer a slightly simpler question: *Do they have the same number of elements?*

> Let A and B be given sets. We write $A \sim B$ iff there exists a bijective function $f : A \to B$.
>
> When $A \sim B$, we say that A and B are *equinumerous* or have the same *cardinality*.

Exmaple 10.1.1.

- Let $A = \{1, 3, 5\}$ and $B = \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\}$. Then $A \sim B$.
- Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 3, 4\}$. Then clearly $A \not\sim B$.
- Let $A = \mathbb{N}$ and $B = \mathbb{N}^+$. Then $A \sim B$ because for example the map

  $$f : \mathbb{N} \to \mathbb{N}^+$$

  defined by

  $$f(n) = n + 1 \text{ for every } n \in \mathbb{N},$$

  is a bijection.

- Let $A = \mathbb{N}$ and $B = \{0, 2, 4, 6, ...\} = \{2n \ : \ n \in \mathbb{N}\}$. Then $A \sim B$. To see this, observe that the function $f : A \to B$ defined by

  $$f(n) = 2n \text{ for every } n \in \mathbb{N}$$

  is a bijection.

- If $A = \{0, 1, 4, 9, 16, 25, ...\} = \{n^2 \ : \ n \in \mathbb{N}\}$ then $A \sim \mathbb{N}$. (For example, we can consider the bijection $f : \mathbb{N} \to A$ defined by $f(n) = n^2$.)

The proof of the following important proposition is simple and left as an exercise.

Proposition 10.1.2.

a. For every set A, $A \sim A$.

b. For every two sets A and B, if $A \sim B$ then $B \sim A$.

c. For every three sets A, B, C, if $A \sim B$ and $B \sim C$ then $A \sim C$.

Exercise.

Prove Proposition 10.1.2.

Proposition 10.1.3. $\mathbb{Z} \sim \mathbb{N}$.

Proof. For example, the function $f : \mathbb{Z} \to \mathbb{N}$ given by

$$f(n) = \begin{cases} 2n & \text{if } n \geq 0 \\ -2n - 1 & \text{if } n < 0 \end{cases}$$

is a bijection. (The idea is that $f(\mathbb{Z}^{\geq 0}) = 2\mathbb{N}$ and $f(\mathbb{Z}^{<0}) = 2\mathbb{N} + 1$.) □

Remark. Note that when $A \sim B$, in general we may have several different bijections $f : A \to B$. For example, here is another bijection $g : \mathbb{Z} \to \mathbb{N}$:

$$g(n) = \begin{cases} 2n + 1 & \text{if } n \geq 0 \\ -2n - 2 & \text{if } n < 0 \end{cases}$$

In this case, $g(\mathbb{Z}^{\geq 0}) = 2\mathbb{N} + 1$ and $f(\mathbb{Z}^{<0}) = 2\mathbb{N}$.

Proposition 10.1.4. *If $A \sim A'$ and $B \sim B'$ then $A \times B \sim A' \times B'$.*

Proof. Since $A \sim A'$, there exists a bijection $f : A \to A'$. Similarly, since $B \sim B'$, there exists a bijection $g : B \to B'$. Now the map

$$f \times g : A \times B \to A' \times B'$$

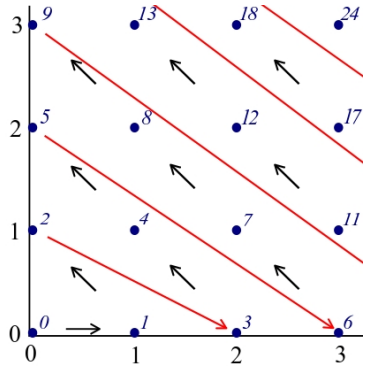
defined by $f(a, b) = (f(a), f(b))$ is a bijection. (Prove the latter fact!) Therefore $A \times B \sim A' \times B'$. □

Exercise.
Prove the following statements by means of giving explicit bijections:

$$2\mathbb{Z} \sim \mathbb{N} \quad , \quad 3\mathbb{Z} \sim \mathbb{N} \quad , \quad 3\mathbb{Z} \sim 5\mathbb{Z}$$
$$\mathbb{Z} \times \mathbb{Z} \sim \mathbb{N} \times \mathbb{N} \quad , \quad 2\mathbb{Z} \times \{0, 1\} \sim \mathbb{Z}$$

Proposition 10.1.5. $\mathbb{N} \times \mathbb{N} \sim \mathbb{N}$.

Proof. We give a geometric proof. The following picture explains how the bijective correspondence is obtained: we enumerate the elements of $\mathbb{N} \times \mathbb{N}$ as we "walk" along the path that is shown by the arrows. In other words, the bijection $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ is obtained as follows : $f(a, b) = n$ if and only if (a, b) is the $n+1$-th point that we visit along the path.



For instance, we have $f(0, 0) = 0$, $f(1, 0) = 1$, $f(0, 1) = 2$, $f(2, 0) = 3$, etc.

It is also possible to give an explicit formula for this function f.

$$f(m, n) = n + \frac{1}{2}(m + n)(m + n + 1).$$

It is also possible to prove algebraically that this formula defines a bijection. (This proof is rather complicated and therefore it is not given here.) This function is sometimes called *Cantor's pairing function*.

□

Recall that for a positive integere n, $\mathbb{Z}^n = \underbrace{\mathbb{Z} \times \cdots \times \mathbb{Z}}_{n \text{ times}}$.

Corollary 10.1.6. *For every positive integer n, we have $\mathbb{Z}^n \sim \mathbb{N}$.*

Proof. The proof is by induction on n. If $n = 1$, then the statement follows from Proposition 10.1.3. Next suppose that for some n we have $\mathbb{Z}^n \sim \mathbb{N}$. Then Proposition 10.1.4 and Proposition 10.1.3 imply that

$$\begin{aligned} \mathbb{Z}^{n+1} &= \mathbb{Z}^n \times \mathbb{Z} && \text{(by definition)} \\ &\sim \mathbb{N} \times \mathbb{Z} && \text{(by Proposition 10.1.4) and induction hypothesis} \\ &\sim \mathbb{N} \times \mathbb{N} && \text{(by Proposition 10.1.3)} \\ &\sim \mathbb{N} && \text{(by Proposition 10.1.5)} \end{aligned}$$

□

10.2 Comparing cardinalities

Let A and B be two sets. We write $A \preceq B$ iff there exists an injection $f : A \to B$.

Remark. Note that when $A \preceq B$, the injection $f : A \to B$ is in general not unique.

Exmaple 10.2.1.

- If $A = \{1, 2, 8\}$ and $B = \{1, 4, 5, 7\}$ then $A \preceq B$ but $B \not\preceq A$.
- $\varnothing \preceq A$ for every set A.

The proof of the next proposition is simple and left as an exercise.

Proposition 10.2.2.

a. For every set A, we have $A \preceq A$.

b. For every three sets A, B, and C,

$$A \preceq B \ \wedge B \preceq C \ \Rightarrow A \preceq C.$$

c. For every two sets A and B, if $A \subseteq B$ then $A \preceq B$.

d. For every two sets A and B, if $A \sim B$ then $A \preceq B$ and $B \preceq A$.

There is also an analogue of Proposition 10.1.4 for $\preceq$:

Proposition 10.2.3. *Prove that if $A \preceq A'$ and $B \preceq B'$ then $A \times B \preceq B \times B'$.*

Proof. The proof of this proposition is similar to that of Proposition 10.1.4 and is left as an exercise. □

One can now pose the following natural problems:

- **Problem A.** If $A \preceq B$ and $B \preceq A$, does it follow that $A \sim B$?
- **Problem B.** If A and B are arbitrary sets, is it true that at least one of

$$A \preceq B \text{ and } B \preceq A$$

hold?

These problems are very natural and easy to state, but difficult to solve. Why are they important? The following example clarifies this:

Exmaple 10.2.4. Let $A = [0, 1)$ and $B = [0, 1]$ be two intervals. Clearly, $A \preceq B$: for example consider the injective function $f : [0, 1) \to [0, 1]$ given by

$$f(a) = a \text{ for every } a \in A.$$

But, at the same time, we have $B \preceq A$: for example, consider the injective function $g : B \to A$ defined by

$$g(b) = \frac{b}{2} \text{ for every } b \in B.$$

Is $[0, 1) \sim [0, 1]$? It is a good exercise to try to prove this statement with the tools which are currently available to you. (Well, basically, you have almost nothing!) The solutions of Problems A and B above are quite vital for answering such questions.

Schröder- Bernstein Theorem. For every two sets A and B, if $A \preceq B$ and $B \preceq A$ then $A \sim B$.

Principle of Trichotomy For every two sets A and B, either $A \preceq B$ or $B \preceq A$.

Remark. The proofs of the two statements above are quite difficult, and beyond the scope of this course. You can find these proofs in various books on set theory (e.g., Halmos's book).

Proposition 10.2.5. *Let $A \preceq B$ be two sets. Prove that the following statements are equivalent:*

a. There exists an injective function $f : A \to B$.

b. There exists a surjective function $g : B \to A$.

Proof. We only prove that $(a) \Rightarrow (b)$. (The proof of $(b) \Rightarrow (a)$ is similar and left as an exercise.)

Suppose that $f : A \to B$ is an injection. We can define the surjection $g : B \to A$ by "reversing the arrows". More precisely, fix an arbitrary element $a_\circ \in A$ and set

$$g(b) = \begin{cases} a & \text{if } \exists_{a \in A}(f(a) = b), \\ a_\circ & \text{otherwise.} \end{cases}$$

Note that:

- g is well defined because f is injective.
- $g(B) = f^{-1}(B) = A$, i.e., g is surjective.

□

Remark. Proposition 10.2.5 essentially means that we can give an alternative definition of $A \preceq B$:

$A \preceq B$ iff there exists a surjection $g : B \to A$.

Proposition 10.2.5 ensures that the latter definition of $A \preceq B$ is equivalent to the original definition. When we want to prove $A \preceq B$, sometimes it is easier to use the new definition.

Exercise.

a. In Proposition 10.2.5 prove that $(b) \Rightarrow (a)$.

b. Find an explicit surjection map $f : [0, 1] \to [0, 1)$.

c. Find an explicit surjection map $f : \mathbb{Z} \to \mathbb{N}^+$.

d. Find an explicit surjection map $f : \mathbb{N} \to \mathbb{Z}$.

e. Find an explicit surjection $f : \mathbb{R} \to (0, 1)$.

Remark. We will see below that there exists a surjection $f : \mathbb{N} \to \mathbb{Q}$ but there is no surjection $g : \mathbb{N} \to \mathbb{R}$.

11 Countable Sets

Countable sets are the smallest infinite sets in a precise sense. Without further ado, we define these sets.

11.1 Definition of a countable set

A set A is called *countable* iff $A \preceq \mathbb{N}$.

Exmaple 11.1.1.

- Every finite set is countable.
- Corollary 10.1.6 implies that $\mathbb{Z}, \mathbb{Z} \times \mathbb{Z}, \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}, ...$ are all countable.
- If A is a countable set then every subset of A is also countable.

Intuitively, countable sets are the "smallest" infinite sets. The following proposition is our first attempt in the direction of making this intuition more precise:

Proposition 11.1.2. *If $A \subseteq \mathbb{N}$ and A is infinite, then $A \sim \mathbb{N}$.*

Proof. The idea of the proof is to use the fact that $\mathbb{N}$ is well-ordered. Consider the function $f : \mathbb{N} \to A$ given by

$$f(n) = \text{“}(n+1)^{\text{th}} \text{ lowest element of } A\text{”}.$$

For example,

$$\begin{aligned} & f(1) = \min(A) \\ \text{and } & f(2) = \min(A \setminus \{\min(A)\}) = \min\big(A \setminus \{f(1)\}\big) \\ \text{and } & f(3) = \min\big(A \setminus \{f(1), f(2)\}\big) \\ \text{and } & ... \end{aligned}$$

Since A is infinite, the above definition is valid and is a bijection between the two sets. □

More generally, we have:

Theorem 11.1.3. *If $A \preceq \mathbb{N}$ and A is infinite, then $A \sim \mathbb{N}$.*

Proof. Since $A \preceq \mathbb{N}$, there exists a function $f : A \to \mathbb{N}$. Set $B = f(A)$. Then $A \sim B$. Moreover, since A is infinite, B is also an infinite subset of $\mathbb{N}$. Proposition 11.1.2 implies that $B \sim \mathbb{N}$. Therefore:

$$A \sim B \sim \mathbb{N}$$

which completes the proof. □

Theorem 11.1.3 says the following:

If A is an infinite countable set, then $A \sim \mathbb{N}$.

Remark. If A is an infinite set, it is intuitively clear that $\mathbb{N} \preceq A$. However, the rigorous proof of this statement requires the Axiom of Choice.

Proposition 11.1.4. $\mathbb{Q} \sim \mathbb{N}$.

Proof. Since $\mathbb{Q}$ is infinite, from Theorem 11.1.3 it follows that it is enough to prove that $\mathbb{Q} \preceq \mathbb{N}$. Indeed it suffices to prove that $\mathbb{Q} \preceq \mathbb{Z}^2$, because:

$$\mathbb{Q} \preceq \mathbb{Z}^2 \ \wedge \ \underbrace{\mathbb{Z}^2 \preceq \mathbb{N}}_{\text{by Corollary 10.1.6}} \ \Rightarrow \ \mathbb{Q} \preceq \mathbb{N}.$$

We now prove that $\mathbb{Q} \preceq \mathbb{Z}^2$. It is well known that every rational number $r \in \mathbb{Q}$ can be written in a unique way in the form $r = \frac{a}{b}$ with the following properties:

- $a \in \mathbb{Z}$,
- $b \in \mathbb{N}^+$,
- the fraction $\frac{a}{b}$ cannot be reduced any further.

We call this representation of r its *reduced* form. Now define the function

$$f : \mathbb{Q} \to \mathbb{Z}^2$$

by

$$f(r) = (a, b) \qquad \text{if } \frac{a}{b} \text{ is the reduced form of } r.$$

It is easy to check that the function f is injective. Therefore we have proved that $\mathbb{Q} \preceq \mathbb{Z}^2$. □

Exercise.

Using induction, prove that for every $n \in \mathbb{N}^+$, we have $\mathbb{Q}^n \sim \mathbb{N}$.

Proposition 11.1.5. *If A and B are countable then $A \times B$ is also countable.*

Proof. We have $A \preceq \mathbb{N}$ and $B \preceq \mathbb{N}$. Therefore,

$$\underbrace{A \times B \preceq \mathbb{N} \times \mathbb{N}}_{\text{by Proposition 10.2.3}} \ \wedge \ \underbrace{\mathbb{N} \times \mathbb{N} \preceq \mathbb{N}}_{\text{by Proposition 10.1.5}} \ \Rightarrow \ A \times B \preceq \mathbb{N}.$$

□

Proposition 11.1.6. *Let $\{A_i \ : \ i \in I\}$ be a family of sets such that $I \subseteq \mathbb{N}$ and every A_i is countable. Then $\bigcup_{i \in I} A_i$ is also countable.*

Proof. For every $i \in I$, let $f_i : A_i \to \mathbb{N}$ be an injection. For every $x \in \bigcup_{i \in I} A_i$ define

$$s(x) = \min\{ \ j \ : \ x \in A_j \ \}.$$

Define the function $g : \bigcup_{i \in I} A_i \to \mathbb{N} \times \mathbb{N}$ by

$$g(x) = (f_{s(x)}(x), s(x)) \text{ for every } x \in \bigcup_{i \in I} A_i.$$

It is easy to check that g is an injection. It follows that $\bigcup_{i \in I} A_i \preceq \mathbb{N} \times \mathbb{N}$. Finally,

$$\bigcup_{i \in I} A_i \preceq \mathbb{N} \times \mathbb{N} \ \wedge \ \underbrace{\mathbb{N} \times \mathbb{N} \preceq \mathbb{N}}_{\text{by Proposition 10.1.5}} \ \Rightarrow \ \bigcup_{i \in I} A_i \preceq \mathbb{N}$$

□

With a minor modification of the above argument, we can prove the following important statement:

Theorem 11.1.7. *If I is a countable index set and $\{ \ A_i \ : \ i \in I \ \}$ is a family of sets such that each A_i is also countable, then $\bigcup_{i \in I} A_i$ is countable as well.*

The above fact is a very strong tool for proving a set is countable. It will be useful in homework and exam problems.

Exmaple 11.1.8. Let us show that the set $S = \{x \in \mathbb{R} \ : \ \cos(x) \in \mathbb{Q}\}$ is countable.

For every $c \in \mathbb{R}$, it is easy to see that the set

$$S_c = \{x \in \mathbb{R} \ : \ \cos(x) = c\}$$

is countable. In fact by high school trigonometry you know that

$$S_c = \{2\pi k \pm \alpha \ : \ k \in \mathbb{Z}\} \quad \text{for some } \alpha \in \mathbb{R}.$$

Now note that

$$S = \bigcup_{c \in \mathbb{Q}} S_c$$

and the index set, which is $\mathbb{Q}$, is also countable (see Proposition 11.1.4). Countability of S now follows immediately from Theorem 11.1.7.

Exercise.
Prove that the set $S = \{x \ : \ \sin(x) \in \mathbb{Q}\}$ is countable.

11.2 Adjoining countably many elements

Lemma 11.2.1. *Let A and B be disjoint infinite sets. Assume that B is a countable set. Then $A \cup B \sim A$.*

Proof. Clearly $A \preceq A \cup B$, and by the Schröder-Bernstein Theorem it is enough to prove that $A \cup B \preceq A$. Since B is countable, there exists an injection $f : B \to \mathbb{N}$, and since A is infinite, there exists an injection $g : \mathbb{N} \to A$ (see Remark 11.1). The idea is to use the Hilbert Hotel trick! Define the map $h : A \cup B \to A$ as follows:

$$h(x) = \begin{cases} x & \text{if } x \in A \setminus g(\mathbb{N}), \\ g(2m+1) & \text{if } \exists_{m \in \mathbb{N}}(x = g(m)), \\ g(2f(x)) & \text{if } x \in B. \end{cases}$$

It is easy to check that h is an injection. □

Corollary 11.2.2. *Let A and B be infinite sets. Assume that B is a countable set. Then $A \cup B \sim A$. (In other words, the disjointness assumption of Lemma 11.2.1 is redundant.*

Proof. Clearly $A \preceq A \cup B$. If we prove that $A \cup B \preceq A$, then the Schröder-Bernstein Theorem now implies that $A \cup B \sim A$.

We now prove that $A \cup B \preceq A$. Let $A_1 = A \times \{0\}$ and $B_1 = B \times \{1\}$. Then:

$$A_1 \cap B_1 = \varnothing \quad , \quad A \sim A_1 \quad , \quad B \sim B_1.$$

It is easy to see that $A \cup B \preceq A_1 \cup B_1$: for example, we can consider the injection $f : A \cup B \to A_1 \cup B_1$ defined by

$$f(x) = \begin{cases} (x, 0) & \text{if } x \in A \setminus B, \\ (x, 1) & \text{if } x \in B. \end{cases}$$

We now use Lemma 11.2.1 for A_1 and B_1. We have:

$$A \cup B \preceq A_1 \cup B_1 \preceq A_1 \preceq A.$$

Consequently, □

Let us state the above corollary again because of its importance:

If A is infinite and B is countable, then $A \cup B \sim A$.

Exercise.

Prove that the map $h : A \cup B \to A$ of Lemma 11.2.1 is an injection.

a. Suppose that $f : A \to \mathbb{Q}$ is a function such that for every $x \in \mathbb{Q}$, $f^{-1}(\{x\})$ is a countable set. Prove that A is a countable set.

b. Prove that the set $\bigcup_{n \in \mathbb{N}^+} \mathbb{Z}^n$ is countable.

c. Let $A = \{\texttt{A}, \texttt{B}, \texttt{C}, \ldots, \texttt{Z}, \square, \texttt{.}\}$. Prove that $\bigcup_{n \in \mathbb{N}^+} A^n$ is countable. (Note that this set contains all of the meaningful English sentences!)

12 Axiomatic Set Theory

12.1 Cantor's set theory and Russell's paradox

Georg Cantor (1845-1918) is considered to be the inventor of set theory. Cantor's set theory relied on two natural principles:

a. A set is a mathematical gadget which is identified by its elements; i.e., two sets are equal iff their elements are the same.

b. Sets are determined by conditions. In other words, for a given condition P, there is a unique set whose elements are exactly the (mathematical) objects with property P.

12.2 Russell's paradox

In 1901, Bertrand Russell discovered the following paradox, which jeopardized the second principle of Cantor's set theory:

Russell's Paradox

Consider the set

$$R = \{\ X\ :\ X \text{ is a set and } X \notin X\ \}.$$

In other words, R is the set containing all sets which are not elements of themselves. Is $R \in R$?

Let us try to answer Russell's question! Assume that $R \in R$. Then this means that R is a set which is not a member of itself. In other words, $R \notin R$!

On the other hand, if we assume that $R \notin R$, then from the definition of R it follows that $R \in R$!

Therefore we have shown that none of $R \in R$ and $R \notin R$ are possible!!!

Note that this paradox implies that Cantor's set theory is not consistent. In other words, it is not a good mathematical theory. At the same time, this means that all of the efforts to build mathematics on Cantor's set theory would be in vain. Therefore, this simple paradox is detrimental not only to Cantor's set theory, but also to all of mathematics.

The solution to this paradox is to rebuild set theory properly, i.e., to introduce *axioms* which identify sets more carefully and avoid Russell's paradox (and other paradoxes). This was done by Ernst Zermelo in 1908. Around 1922 Zermelo's list of axioms was enhanced by Abraham Fraenkel (and independently by Thoraif Skolem). The system that was obtained is nowadays called *ZF set theory*.

Later on, another axiom was added to ZF set theory. The addition of the latter axiom, called the *axiom of choice*, faced resentment by some of the greatest mathematicians of the twentieth century. However, Kurt Gödel and Paul Cohen proved that the axiom of choice is independent of ZF set theory. In other words they showed that although the axiom of choice can be stated within the world of ZF set theory, it cannot be proved or disproved in that world! Therefore its addition, which seemed quite necessary, would not lead to contradictions. The new system is nowadays called *ZFC set theory*.

12.3 The ZFC universe

The ZFC set theory is an axiomatic system which talks about three abstract concepts:

- *Objects ($a, b, x, y, ...$) :* Objects of ZFC are sets. They form what is called the *ZFC universe*.

- *Equality ($=$) :* for every two objects of the ZFC universe, we know that either $x = y$ or $x \neq y$.

- *Membership ($\in$) :* given any pair of objects x and y of the ZFC universe, we know that either $x \in y$ or $x \notin y$.

I should point out again that the objects of ZFC are [3] sets. There is nothing but sets. Therefore elements of objects of ZFC are sets themselves, and the elements of these elements are also sets, and so on. For example, the empty set $\varnothing$, the set $\{\varnothing, \{\varnothing\}\}$, or the set $\wp(\{\varnothing, \{\varnothing\}\})$ are objects of ZFC. But the set $\{1\}$ is not an object of ZFC, because the number 1, as we understand it, is not a set. ZFC simply does not understand it!

12.4 Some notation

A formula such as $\forall_x(...)$ means:

"for every object x of the ZFC universe, ...".

A formula such as $\exists_x(...)$ means:

"there exists an object x of the ZFC universe such that ...".

[3] Defining a mathematical system with a series of axioms means describing the relations between its abstract concepts using sentences that involve quantifiers. Therefore, we should forget about our intuitive understanding of the meaning of those concepts. When we want to prove or disprove a statement in that system, we are only allowed to use the axioms and the logically acceptable rules of deduction. From this point of view, it would be more rigorous to say that the objects of ZFC are abstract entities which are *meant to be* sets. This is a somewhat philosophical statement!

For example, the sentence

$$\forall_x(x = x \vee x \neq x)$$

means :

"for every object x in the ZFC universe, either $x = x$ or $x \neq x$."

which, by the above discussion, is **T**. On the other hand, the sentence

$$\forall_x \exists_y (y \in x)$$

means:

"for every object x of the ZFC universe, there exists an object y of the ZFC universe which belongs to x."

and we know that this sentence is **F** because $x = \varnothing$ is a counterexample to it.

12.5 The first three axioms of ZFC

The first axiom of ZFC is identical to Cantor's first principle. It describes how the concepts of equality and membership are related.

ZFC Axiom 1 (Extensionality). Two sets are equal iff they have the same elements. In other words:

$$\forall_{x,y}[x = y \;\Leftrightarrow\; \forall_z\big((z \in x) \Leftrightarrow (z \in y)\big)]$$

Exmaple 12.5.1. It is easy to come up with "artificial" examples of universes which do not satisfy Axiom 1. Consider a universe which consists of exactly three distinct objects a, b_1, b_2 and the membership relation $\in$ is described as follows:

$$\begin{array}{lll} a \in b_1, & a \in b_2, & a \notin a \\ b_1 \notin b_1, & b_1 \notin b_2, & b_1 \notin a \\ b_2 \notin b_1, & b_2 \notin b_2, & b_2 \notin a. \end{array}$$

By assumption $b_1 \neq b_2$, but you can see that in this universe b_1 and b_2 have the same elements. Therefore Axiom 1 does not hold for this universe.

The second axiom of ZFC is a weakened version of Cantor's second principle:

ZFC Axiom 2 (Specification). Given a set a and a condition $P(x)$, there exists a unique set b whose elements are exactly those $x \in a$ which satisfy $P(x)$. In other words:

$$\forall_a \exists_b \left(\forall x \left[(x \in b) \Leftrightarrow (x \in a \ \wedge \ P(x)) \right] \right)$$

Remark.

1. What does a "condition $P(x)$" mean mathematically? It means that $P(x)$ is a predicate with exactly one free variable x. For example, $P(x)$ can be

$$\forall_y (x \in y \ \wedge y \notin x)$$

or

$$\exists_y \forall_z \big((x \notin y \ \vee \ x \notin x) \Rightarrow y \in z \big)$$

but it cannot be

$$\exists_y \big((x \in z) \vee (y \in x) \big).$$

2. Note that when I stated Axiom 2 in English I said that b is unique. However, when I wrote is as a formula and using quantifiers, I did not use the symbol $\exists!$. This is because indeed the uniqueness of b follows from Axiom 1.

3. Axiom 2 is basically meant to allow us to define sets of the form

$$b = \{x \in a \ : \ P(x)\}.$$

In particular, if a is an object of the ZFC univerise (i.e., a set), then we can consider the set

$$b = \{x \in a \ : \ x \notin x\}$$

You may now feel embarrassed and expect that you are going to encounter Russell's paradox again! However, Axiom 2 is formulated carefully and protects us from Russell's attacks. In fact it can be shown that "the set of all sets" is not an object of ZFC. (See Proposition 12.5.4 below.)

Exmaple 12.5.2. Consider a universe with exactly three distinct objects α, β, γ such that membership is described by

$$\alpha \in \gamma, \ \beta \in \gamma, \ \beta \in \beta, \text{ and } x \notin y \text{ for any other pair of elements } x, y \text{ of the universe.}$$

You can check that α has no elements, the only element of β is β, and the elements of γ are α and β. Therefore Axiom 1 of ZFC holds for this universe. However, Axiom 2 fails: for example let us set

$$P(x) = \text{“}x \notin x\text{”}.$$

Then Axiom 2 says that there should exist an element of the universe which is defined by

$$\{x \in \gamma \ : \ P(x)\}.$$

In other words, there should exist an element of the universe which is defined by

$$\{x \in \gamma \ : \ x \notin x\}.$$

But you can see that the only $x \in \gamma$ which satisfies $x \notin x$ is $x = \alpha$. Consequently, if Axiom 2 holds then there should exist an element of the universe *which has only one element, that element being* α. But by investigating the objects of the universe, you can see that none of them satisfies this property.

Remark. Axiom 2 allows us to define intersection and difference of two sets: if a, b are two objects in the ZFC universe, then we set:

$$a \cap b = \{x \in a \ : \ x \in b\}$$

and

$$a \setminus b = \{x \in a \ : \ x \notin b\}.$$

However, the union of a and b *cannot* be defined by Axiom 2. Indeed the next example shows that one can cook up a universe which satisfies Axiom 1 and Axiom 2, but does not contain the union of two elements.

Exmaple 12.5.3. Consider the universe with objects a_1, a_2, a_3 such that

$$a_1 \in a_2, \ a_2 \in a_3, \ \text{ and for every other pair of elements } x, y, \text{ we have } x \notin y.$$

You can check that $a_2 \cup a_3$ does not exist. In other words, there does not exist an element b of this universe which satisfies

$$\forall_x (x \in b \ \Leftrightarrow \ x \in a_2 \vee x \in a_3).$$

Proposition 12.5.4. *There does not exist an element a of the ZFC universe which satisfies $\forall_x (x \in a)$. (In other words, in the ZFC universe the set of all sets does not exist.)*

Proof. The proof is by contradiction. Suppose that such an a exists. Then by Axiom 2, the set

$$b = \{x \in a \ : \ x \notin x\}.$$

exists in the ZFC universe. Now at least one of the following two cases should happen:

a. $b \in b$. In this case the condition that defines b implies that $b \notin b$. Contradiction!

b. $b \notin b$. In this case, since by the definition of a we have $b \in a$, it follows that the condition "$b \notin b$" should be false. In words, $b \in b$. Again, we reached a contradiction!

Since both of the cases lead to a contradiction, it follows that the set a cannot exist. $\square$

Remark. As you have noticed, the above proof is inspired by Russell's paradox.

The third axiom of the ZFC universe is somewhat philosophical! It is only meant to guarantee that the ZFC universe is not void.

ZFC Axiom 3 (Existence) There exists at least one set. In other words,

$$\exists_x (x = x).$$

Remark. Note that the condition $x = x$ really imposes no restriction on x. It says nothing about x!

We are now going to see that in ZFC, the empty set exists and is unique:

Proposition 12.5.5. *There exists a unique set in ZFC universe which has no elements.*

Proof.

Proof of existence: By Axiom 3, there exists a set x in ZFC. By Axiom 2, there exists a set a defined by

$$a = \{t \in x \ : \ t \neq t\}.$$

The set a does not have any elements, because if $b \in a$ then $b \neq b$ which is absurd.

Proof of uniqueness: The proof is by contradiction. Let a_1 and a_2 be two objects of ZFC with no elements such that $a_1 \neq a_2$. Therefore

$$\neg \forall_z \big((z \in a_1) \Leftrightarrow (z \in a_2) \big) \qquad \text{(see Axiom 1).}$$

After negation, we conclude that

$$\exists_z \big(z \in a_1 \ \wedge \ z \notin a_2) \vee (z \in a_2 \ \wedge \ z \notin a_1)$$

is true, which is absurd because both of $z \in a_1 \ \wedge \ z \notin a_2$ and $z \in a_2 \ \wedge \ z \notin a_1$ are false for every z. (Recall that a_1 and a_2 have no elements.) $\square$

As before, we denote the empty set of ZFC by $\varnothing$.

The next proposition basically states that the intersection of a family of sets is also a set (i.e., still an element of the ZFC universe).

Proposition 12.5.6. *If $a \neq \varnothing$ is a set, then there exists a unique set b with the following property:*

$$\forall_x \left[(x \in b) \Leftrightarrow (\forall_y (y \in a \Rightarrow x \in y)\right].$$

Proof. Since $a \neq \varnothing$, there exists an $a_\circ \in a$. By Axiom 2, there exists a set

$$b = \{x \in a_\circ \,:\, \forall_y (y \in a \Rightarrow x \in y)\}.$$

It is now clear that b is the desired set. □

The main point of Proposition 12.5.6 is that in ZFC, the definition

$$b = \{x \,:\, \forall_y (y \in a \Rightarrow x \in y)\}$$

is not valid, and therefore we have to find a workaround. (As you saw, it was very easy to do so.)

We now need more axioms to be able to construct more sets. (For example. as you saw above, still we cannot obtain unions of sets.)

ZFC Axiom 4 (Parity). For every two sets x, y, there exists a set z whose members are just x and y. In other words:

$$\forall_x \forall_y \exists_z \forall_t (t \in z \Leftrightarrow t = x \vee t = y).$$

Remark. Somewhat informally, Axiom 4 basically says that if a, b are sets then $\{a, b\}$ is also a set. Setting $a = b$, it follows that if a is a set then $\{a\}$ is also a set.

Exercise.

Prove, using the axioms of ZFC, that $\{\varnothing\} \neq \{\{\varnothing\}\}$ and $\{\{\varnothing\}\} \neq \{\{\{\varnothing\}\}\}$.

ZFC Axiom 5 (Union). If a is a set then $\cup a$ is also a set. In other words, for every set a there exists a set b such that every element of an element of a is in b. More formally:

$$\forall_a \exists_b \left(\forall_x [\exists_y (x \in y \,\wedge\, y \in a) \Leftrightarrow (x \in b)]\right).$$

Exercise.
Using induction, prove that if $a_1, ..., a_n$ are n arbitrary sets (where $n \in \mathbb{N}^+$) then $a_1 \cup \ldots \cup a_n$ is also a set.

From now on, for simplicity we use the following notation:

The meaning of inlcusion in ZFC

Given two objects x, y of the ZFC universe (i.e., two sets), we write $x \subseteq y$ iff the following statement holds:

$$\forall_a (a \in x \Rightarrow a \in y).$$

We are ready to state the sixth axiom:

ZFC Axiom 6 (Power set). The power set of any set is also a set. In other words:

$$\forall_a \exists_b \forall_x (x \in b \Leftrightarrow x \subseteq a)$$

Exercise.

a. Prove that $\{\{a\}, \{a, b\}\} = \{\{a'\}, \{a', b'\}\}$ iff $a = a'$ and $b = b'$.

b. In Naïve set theory we introduced ordered pairs and Cartesian products of sets. The axioms introduced so far do not explicitly mention the direct products. We have also seen that an ordered pair (a, b) is not the same as $\{a, b\}$. However, it is possible to define (a, b) in terms of sets. This was first done by Kazimierz Kuratowski (1896-1980). Use the above exercise to define the Cartesian product of two sets A and B.

12.6 Axiom of infinity

Axiom 3 of ZFC guarantees that there exists at least one set. From the five axioms we have discussed so far, one can *prove* the existence of several nonempty sets. However, from these axioms one cannot obtain infinite sets. To achieve infinity, yet another axiom is needed. But, how do we say in the language of ZFC that a set is infinite? (remember that the ZFC vocabulary we have developed so far is very limited.) The conventional way to introduce the concept of infinity is subtle and somewhat indirect.

If y is a set, then from Axioms 1–6 we can conclude that $y \cup \{y\}$ is also a set. (Prove this as an exercise!)

> From now on, for every set y, we denote $y \cup \{y\}$ by y^+ and we call it the *successor* of y.

For example, we have

$$\varnothing^+ = \{\varnothing\} \quad \text{and} \quad \{\varnothing\}^+ = \{\varnothing\} \cup \{\{\varnothing\}\} = \{\varnothing, \{\varnothing\}\}.$$

Exercise.

Prove (using Axioms 1–6 of ZFC) that for every set y we have $y \in y^+$ and $y \subseteq y^+$.

> A set a is called *inductive* iff it satisfies the following two conditions:
>
> a. $\varnothing \in a$,
>
> b. for every $x \in a$, we have $x^+ \in a$.

> **ZFC Axiom 7 (Infinity)** There exists at least one inductive set.

12.7 Natural numbers

Using Axiom 7, we can introduce the set of natural numbers in set theory. The customary way to do so is to let $\varnothing$ correspond to zero, and every $n \in \mathbb{N}^+$ is really $(n-1)^+$. Therefore we have:

$$\begin{aligned}
&0 = \varnothing \\
&1 = 0^+ = 0 \cup \{0\} = \varnothing \cup \{\varnothing\} = \{\varnothing\} \\
&2 = 1^+ = 1 \cup \{1\} = \{\varnothing\} \cup \{\{\varnothing\}\} = \{\varnothing, \{\varnothing\}\} \\
&3 = 2^+ = 2 \cup \{2\} = \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\} \\
&...
\end{aligned}$$

and $\mathbb{N} = \{0, 1, 2, 3, ...\} = \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}, \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\}, ...\}$.

Exercise.

Using induction, prove that in the above definition of natual numbers, we have $n = \{0, 1, ..., n-1\}$.

12.8 Axiom of foundation

Imagine you receive a big box as a birthday present from a mathematician friend of yours. You open the box and find another box in it. You open the latter box, and find yet another box in it,... finally, you realize that the birthday gift is nothing but an infinite number of boxes! Do you like this present?!!

There are reasons why mathematicians don't like sets which look like your birthday present. (Discussing these reasons is beyond the scope of this course.) The next axiom of ZFC, the Axiom of Foundation, basically ensures that sets like $\{\{\{\cdots\}\}\}$ are avoided. This is not so clear from the standard formulation of this axiom:

> **ZFC Axiom 8 (Foundation)** If x is a nonempty set, then there exists an element $y \in x$ such that $y \cap x = \varnothing$.

However, we have the following proposition:

Proposition 12.8.1. *No set is an element of itself. In otherwords,* $\nexists_a\, (a \in a)$.

Proof. The proof is by contradiction. Assume that there exists a set a such that $a \in a$. The remark after Axiom 4 implies that $\{a\}$ exists. By Axiom 8, there exists an element $b \in \{a\}$ such that $b \cap \{a\} = \varnothing$. However, a is the only element of $\{a\}$. Therefore we should have $a \cap \{a\} = \varnothing$. However, we also know that

$$a \in a \text{ and } a \in \{a\} \;\Rightarrow\; a \in a \cap \{a\} \;\Rightarrow\; a \cap \{a\} \neq \varnothing.$$

which is a contradiction. □

Exercise.

Using Axiom 8, prove that there do not exist sets a, b such that $a \in b$ and $b \in a$.

The last two axioms of ZFC are difficult, but very deep and important. Unfortunately, a detailed study of these axioms is beyond the scope of this course. We will only discuss them briefly.

> **ZFC Axiom 9 (Replacement).** Let a be a given set and $Q(x, y)$ be a condition with two free variables x, y, such that
>
> $$\forall_x (x \in a \Rightarrow \exists!_y Q(x, y)). \tag{16}$$
>
> Then there exists a set b such that:
>
> $$\forall_y [(y \in b) \;\Leftrightarrow\; \exists_x (x \in a \;\wedge Q(x, y))].$$

What is Axiom 9 good for? It is a tool for construction of new sets. The next two examples show how it is used.

Exmaple 12.8.2. Let a be any set. We want to prove that there exists a set b whose elements are the power sets of elements of a, i.e.,

$$\forall_x[x \in b \Leftrightarrow \exists_y(y \in a \ \wedge \ x = \wp(y))].$$

We can't simply write $b = \{\wp(x) \ : x \in a\}A$ because Axiom 2 does not support this kind of definition. (Remember Russell's Paradox?!!) However, we can use Axiom 9 for the following formula:

$$Q(x, y) = \text{“}x = \wp(y)\text{”}.$$

Since the power set of any set is unique, this $Q(x, y)$ satisfies the uniqueness condition (16). Therefore Axiom 9 implies that the set b exists. (As an exercise, try to prove the existence of b without Axiom 9.)

Exmaple 12.8.3. Does the set

$$s = \wp(\varnothing) \cup \wp(\wp(\varnothing)) \cup \wp(\wp(\wp(\varnothing))) \cup \ldots$$

exist? You can try to argue that s is equal to $\cup t$ where

$$t = \{\wp(\varnothing), \wp(\wp(\varnothing)), \wp(\wp(\wp(\varnothing))), \ldots\},$$

but then you have to prove that t itself indeed exists! To prove that t exists, you can probably try to argue that $t = \cup u$ where

$$u = \{\{\wp(\varnothing)\}, \{\wp(\wp(\varnothing))\}, \{\wp(\wp(\wp(\varnothing)))\}, \ldots\}.$$

But again, you need to justify that u also exists! After trying a few more minutes, you will realize that proving the existence of s is not as easy as you initially thought! In fact without Axiom 9, you cannot prove the existence of s and t.

But how can we use Axiom 9 to prove the existence of s? We need a condition $Q(x, y)$ that says "y is equal to $\underbrace{\wp(\wp(\cdots(\wp}_{x \text{ times}}(\varnothing))\cdots))$". Here is one way to do it: consider the formula $Q(x, y)$, where $x \in \mathbb{N}$ and y is an arbitrary set, given by

$$\begin{aligned} Q(x, y) = \text{“}\exists_b\exists_f \big[&(f : x^+ \to b \text{ is a function}) \wedge (f(0) = \varnothing) \\ &\wedge [\,\forall_m \ (m \in x) \Rightarrow \big(f(m^+) = \wp(f(m))\big)\,] \wedge (y = f(x))\big]\text{.”} \end{aligned}$$

If we take $a = \mathbb{N}$, then the uniqueness condition (16) of Axiom 9 holds. (It can be proved by induction on x.) Therefore Axiom 9 implies the existence of a set b which is indeed equal to the set t mentioned above. Existence of s now follows immediately.

ZFC Axiom 10 (Choice) Given a set x, there exists a function $\varphi : x \to \cup x$ such that $\forall_a \, (a \in x \Rightarrow \varphi(a) \in a)$.

Remark. Informally, the Axiom of Choice says the following:

Given a family $\{A_i \, : \, i \in I\}$ of sets, we can always pick one element from each A_i.

The latter statement seems trivial. But there is a subtle issue: how do we choose an element from each A_i? Do we have a definite rule? For example, if the A_i are subsets of $\mathbb{N}$, then we know what to do: for example we can choose $\min(A_i)$ from A_i. But for a general set, it may not be clear what to do.

Bertrand Russell explained this subtle point by the following example:

> Suppose you have infinitely many pairs of shoes. To choose one shoe from each pair is easy : from each pair you can just take the left shoe.
>
> Now suppose you have infinitely many pairs of socks. To choose one sock from each pair you need the axiom of choice (because you can't distinguish the left sock from the right sock).

A number of great mathematicians of the nineteenth and early twentieth century believed that the Axiom of Choice was nontrivial, or in other words too strong to be taken as an axiom. In fact the Axiom of Choice has striking consequences. The first importent consequence of the Axiom of Choice is the following:

Well-ordering principle

Every set can be well-ordered. In other words, for every set X, there exists a well-ordering on X.

This is striking because for example $(\mathbb{R}, \leq)$ (the usual ordering) is not well-ordered. Indeed it is impossible to obtain a concrete well-ordering of $\mathbb{R}$. However, the existence of such an ordering is guaranteed by the Axiom of Choice.

The second important consequence of the Axiom of Choice is the following:

Zorn's Lemma

Let $(X, \preceq)$ be a partially ordered set. Assume that every totally ordered subset of X has an upper bound. Then X has a maximal element.

Zorn's Lemma is a powerful tool to prove results in many areas of mathematics.

There is also a much more paradoxial consequence of the axiom of choice. It is called the *Banach-Tarski paradox* (although it is not really a paradox - it is just hard to believe!). It basically states that you can break a sphere into finitely many pieces and glue them together such that you obtain two new spheres *identical* to the original sphere...Yup!

13 Cantor's Diagonal Argument

13.1 Intervals and cardinality

Let us now study $\preceq$ and $\sim$ for intervals of $\mathbb{R}$. We have seen that $\mathbb{N} \sim \mathbb{Z}$. Is there a similar result for $\mathbb{R}$ as well? In other words, is it true that $\mathbb{R} \sim \mathbb{R}^+$? We begin with some concrete examples.

Proposition 13.1.1. $\mathbb{R}^+ \sim \mathbb{R}$.

Proof. From calculus we know that the function $f(x) = e^x$ is strictly increasing, $f(x) > 0$ for every x, and $f(x) = y \Leftrightarrow x = \ln(y)$. Therefore f is a bijection. □

How about subintervals of the form (a, b)? Are they equinumerous to $\mathbb{R}$? The next result partially answers this question. (See exercises at the end of this section for the general case.)

Proposition 13.1.2. $(0, 1) \sim \mathbb{R}$.

Proof. We saw in Proposition 13.1.1 that $\mathbb{R} \sim \mathbb{R}^+$. Therefore by transitivity of $\sim$ it is enough to prove that $(0, 1) \sim \mathbb{R}^+$. Again we use calculus! Consider the function

$$f : (0, 1) \sim \mathbb{R}^+ \quad , \quad f(x) = \tan\big(\frac{\pi x}{2}\big).$$

By drawing the graph of this function we can see that it is a bijection from $(0, 1)$ to $\mathbb{R}^+$. □

How about $[0, 1)$ vs. $(0, 1)$? With all the surprising facts you have seen so far, it would be quite plausible to expect that $[0, 1) \sim (0, 1)$. Still, this statement requires a rigorous proof.

Proposition 13.1.3. $[0, 1) \sim (0, 1)$.

Proof. We give a simple proof using the Schröder-Bernstein Theorem, and leave a proof independent of this theorem to the exercises.

Clearly $(0, 1) \preceq [0, 1)$. To show that $[0, 1) \preceq (0, 1)$, note that

$$[0, 1) \preceq \mathbb{R} \quad \text{and} \quad \underbrace{\mathbb{R} \sim \mathbb{R}^+}_{\text{by Proposition 13.1.1}} \quad \text{and} \quad \underbrace{\mathbb{R}^+ \sim (0, 1)}_{\text{by Proposition 13.1.2}} \quad \Rightarrow \quad [0, 1) \preceq (0, 1).$$

The Schröder-Bernstein Theorem now implies that $[0, 1) \sim (0, 1)$. □

Remark. From Proposition 13.1.3 and Proposition 13.1.2 it follows that

$$[0, 1) \sim \mathbb{R}.$$

Exercise.

a. Give an explicit bijection $f : (0,1) \to [0,1)$. (Hint: You want to do a Hilbert Hotel trick on the set $S = \{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, ...\}$. More precisely, prove that the map $f : [0,1) \to (0,1)$ defined by

$$f(x) = \begin{cases} x & \text{if } x \notin S, \\ \frac{1}{n+1} & \text{if } x = \frac{1}{n} \text{ for some } n \in \{2,3,4,5,6,...\}, \\ \frac{1}{2} & \text{if } x = 0. \end{cases}$$

is a bijection.)

b. Prove that $\mathbb{R} \times \mathbb{Q} \sim \mathbb{R}$.

c. Prove that for every two real numbers $a < b$ we have $[a,b] \sim \mathbb{R}$, $[a,b) \sim \mathbb{R}$, $(a,b] \sim \mathbb{R}$, and $(a,b) \sim \mathbb{R}$. (Hint: use the Schröder-Bernstein Theorem.)

d. Find an explicit bijective function $f : \mathbb{R} \to \mathbb{R} \setminus \mathbb{Z}$. (Note that this implies that $\mathbb{R} \setminus \mathbb{Z} \sim \mathbb{R}$.) (Hint: Use the idea of Hilbert's Hotel!)

13.2 Sets of strictly bigger cardinality than $\mathbb{N}$

So far we have seen that many sets which intuitively look quite bigger than $\mathbb{N}$ (like $\mathbb{Q}$ or $\mathbb{Z}^n$) are actually equinumerous to $\mathbb{N}$. It is natural ask if there are any infinite sets which are *strictly* bigger than $\mathbb{N}$? In other words, can we find a set A such that $A \not\preceq \mathbb{N}$?

From now on, to indicate that $A \not\preceq B$, we write $B \prec A$.

Informally, $B \prec A$ means that A is a "bigger" infinite set than B. A set A is called *uncountable* iff $\mathbb{N} \prec A$. (we will see examples of uncountable sets shortly!)

About the year 1873, Cantor studied this question and solved it by means of a method which he invented himself. This method is called *the diagonal argument*, and is one of the most powerful methods in set theory, logic, and computer sceience. In fact Russell once said that when he was thinking about his paradox he was inspired by Cantor's diagonal argument. Gödel also used Cantor's method in his proof of incompleteness theorem (which roughly says that there are mathematical statements that are correct but cannot be proven).

We are now going to use Cantor's diagonal argument to prove the following important theorem:

Theorem 13.2.1. *The interval $[0,1)$ is uncountable. In other words, $\mathbb{N} \prec [0,1)$.*

Proof. Let us first remind ourselves of general properties of decimal representations of real numbers.

Every real number $x \in [0, 1)$ has a decimal representation

$$x = .x_1x_2x_3\cdots$$

where each x_i is a decimal digit, i.e, $x_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. For example we have

$$\frac{1}{3} = .333333\cdots \qquad \text{or} \qquad \frac{1}{\sqrt{2}} = .707106781\cdots$$

The decimal representation of a number is unique, with one exception: when there are repreating 9's. For example:

$$.1399999999999\cdots = .14000000000000\cdots$$

> If we do not allow decimal representations with repeating 9's, then the decimal representation of every $x \in [0, 1)$ will be unique.

By Proposition 10.2.5, in order to show that $[0, 1) \not\preceq \mathbb{N}$ it is enough to prove the following statement:

There is no surjection $g : \mathbb{N} \to [0, 1)$.

We now prove the above statement. Let $g : \mathbb{N} \to [0, 1)$ be an arbitrary function. Let us write the values of g as follows:

$$\begin{aligned} g(0) &= c_{01}c_{02}c_{03}c_{04}\cdots \\ g(1) &= c_{11}c_{12}c_{13}c_{14}\cdots \\ g(2) &= c_{21}c_{22}c_{23}c_{24}\cdots \\ g(3) &= c_{31}c_{32}c_{33}c_{34}\cdots \\ &\cdots \end{aligned}$$

Consider the real number $c = .c_1c_2c_3\cdots$ where for each $i \in \mathbb{N}^+$,

$$c_i = \begin{cases} 1 & \text{if } c_{ii} \neq 1, \\ 2 & \text{if } c_{ii} = 1. \end{cases}$$

It is easy to see that $c \in [0, 1)$. Since g is surjective, this means that $c = g(m)$ for some $m \in \mathbb{N}$. However, this is absurd because the m-th digit of c is different from the m-th digit of $g(m)$. Therefore g is not surjective. □

We are now going to compare $\mathbb{N}$ and $\mathbb{R}$. It is clear that $\mathbb{N} \preceq \mathbb{R}$. We will now prove that this inequality is strict:

Theorem 13.2.2. $\mathbb{R}$ *is uncountable. In other words,* $\mathbb{N} \prec \mathbb{R}$.

Proof. The proof is by contradiction. Suppose, on the contrary, that $\mathbb{R} \preceq \mathbb{N}$. Clearly, $[0,1) \preceq \mathbb{R}$. Therefore we have

$$[0,1) \preceq \mathbb{R} \text{ and } \mathbb{N} \preceq \mathbb{R} \preceq \mathbb{N} \;\Rightarrow\; [0,1) \preceq \mathbb{N}$$

which contradicts the conclusion of Theorem 13.2.1. □

The idea of using the unique decimal representation of a real number can be useful in proving other statements:

Proposition 13.2.3. $\mathbb{R} \times \mathbb{R} \sim \mathbb{R}$.

Proof. Since $[0,1) \sim \mathbb{R}$ (see Remark 13.1) it is enough to prove that

$$[0,1) \times [0,1) \sim [0,1).$$

We define a function $f : [0,1) \times [0,1) \to [0,1)$ as follows : for every two real numbers $x, y \in [0,1)$, we write their unique decimal representations

$$x = .x_1x_2x_3\cdots \text{ and } y = .y_1y_2y_3\cdots$$

(we do not allow repeating 9's) and set

$$f(x,y) = .x_1y_1x_2y_2x_3y_3\cdots$$

It is easy to see that $f(x,y) \in [0,1)$.

The uniqueness of the decimal representations of x and y implies that f is injective. Therefore $[0,1) \times [0,1) \preceq [0,1)$. Since obviously $[0,1) \preceq [0,1) \times [0,1)$, by the Schröder-Bernstein Theorem we have

$$[0,1) \times [0,1) \sim [0,1).$$

□

Remark. Note that the function f given in the above proof is not surjective. For exmaple, there are no $x, y \in [0,1)$ such that $f(x,y) = .9090909090\cdots$ (Why?)

Exercise.

a. Prove, using induction, that for every $n \in \mathbb{N}^+$ we have $\mathbb{R}^n \sim \mathbb{R}$.

b. Prove that for every point $(x,y) \in \mathbb{R} \times \mathbb{R}$ there exists a line passing through (x,y) that does not pass through any point with rational coordinates (i.e., for every point on this line, at least one of its coordinates is an irrational number.)

c. You are having a nightmare about being lost in a strange forest: the terrain is flat (it is the xy-plane, i.e., $\mathbb{R} \times \mathbb{R}$) and the trees are very dense : a tree has grown at any spot with coordinates (x, y) where $x, y \in \mathbb{Q}$. You have a map and you know you are at a point (x_1, y_1) (at least one of x_1, y_1 is irrational). You are supposed to get to the campground which is located at a point (x_2, y_2) (again, at least one of (x_2, y_2) is irrational). Prove that you can wade through this dense forest and get to your campground. (More precisely, prove that you can move on a path without hitting the trees, i.e., without passing through points with rational coordinates, to get from (x_1, y_1) to (x_2, y_2).)
(Hint: Try walking on straight line segments!)

d. *(Bonus problem)* Find an explicit injective function $f : \mathbb{R} \to \mathbb{R} \setminus \mathbb{Q}$. (Note that this implies that $\mathbb{R} \setminus \mathbb{Q} \sim \mathbb{R}$.)

e. *(Bonus problem)* Suppose that $\mathscr{S}$ is a family of disjoint open intervals in $\mathbb{R}$. In other words, $\mathscr{S}$ satisfies the following properties:

(i) $\mathscr{S} \subseteq \{ (a, b) \ : \ a, b \in \mathbb{R}\}$,

(ii) for every $A \in \mathscr{S}$ and $B \in \mathscr{S}$, if $A \neq B$ then $A \cap B = \varnothing$.

Prove that S is countable.

f. *(Bonus problem)* Let $f : \mathbb{R} \to \mathbb{R}$ be an increasing function. (f is not necessarily continuous.) Let

$$\mathrm{Disc}(f) = \{x \ : \ f \text{ is discontinuous at } x \ \}.$$

Prove that $\mathrm{Disc}(f)$ is a countable subset of $\mathbb{R}$. (Since $\mathbb{R}$ is uncountable, this implies that any increasing function is continuous at uncountably many points!)

13.3 Cantor's diagonal argument and power sets

Cantor's diagonal argument has applications inside and outside set theory. In this section we mention an applications in set theory.

Proposition 13.3.1. *Let A be an arbitary set. Then $A \prec \wp(A)$.*

Proof. It is enough to show that there is no surjection $f : A \to \wp(A)$. The proof is by contradiction. Assume that there exists such a surjection. Consider the set

$$B = \{x \in A \ : \ x \notin f(x)\}.$$

Since $B \in \wp(A)$ and f is a surjection, there exists an $a \in A$ such that $B = f(a)$. We may now ask if $a \in B$ or $a \notin B$? We observe that:

- If $a \in B$ then by the definition of B we have $a \notin f(a) = B$.

- Similarly, if $a \notin B$ then $a \in f(a) = B$.

Both of the cases lead to absurdity! Therefore f cannot be surjective. □

Remark.
1. Note that it is clear that $A \preceq \wp(A)$. For example, the map $f : A \to \wp(A)$ defined by $f(a) = \{a\}$ is an injection.

2. From Proposition 13.3.1 it follows that

$$\mathbb{N} \prec \wp(\mathbb{N}) \prec \wp(\wp(\mathbb{N})) \prec \wp(\wp(\wp(\mathbb{N}))) \prec \cdots$$

Therefore by repeatedly taking power sets, we can get bigger and bigger infinities.

The above remark says in particular that $\mathbb{N} \prec \wp(\mathbb{N})$. You also saw in Theorem 13.2.2 that $\mathbb{N} \prec \mathbb{R}$. It is now natural to ask the following question:

which one is bigger? $\mathbb{R}$ *or* $\wp(\mathbb{N})$*?*

We are now going to answer this question. First we need two general lemmas.

Lemma 13.3.2. *Let A and B be two sets and $f : A \to B$ be a function. Then:*

a. f is a surjection iff for every $Y \subseteq B$, we have $f(f^{-1}(Y)) = Y$.

b. f is an injection iff for every $X \subseteq A$, we have $f^{-1}(f(X)) = X$.

Proof. The proof of this lemma is easy and left as an exercise. □

Lemma 13.3.3. *If $A \sim B$ then $\wp(A) \sim \wp(B)$.*

Proof. Suppose $f : A \to B$ is a bijection. Consider the map $f_\wp : \wp(A) \to \wp(B)$ as follows:

$$f_\wp(X) = f(X) \text{ for every } X \subseteq A.$$

It is enough to prove that $f_\wp$ is a bijection.

- **Proof of surjectivity of $f_\wp$.** Let $Y \subseteq B$. Set $X = f^{-1}(Y)$. Since f is a bijection, it is also a surjection and therefore we have

$$f(X) = f(f^{-1}(Y)) = Y.$$

Therefore $f_\wp(X) = Y$.

- **Proof of injectivity of $f_\wp$.** Let X_1, X_2 be subsets of A such that

$$f_\wp(X_1) = f_\wp(X_2).$$

Therefore $f(X_1) = f(X_2)$, and consequently,

$$f^{-1}(f(X_1)) = f^{-1}(f(X_2)) \tag{17}$$

Since f is a bijection, it is also an injection and therefore

$$X_1 = f^{-1}(f(X_1)) \text{ and } X_2 = f^{-1}(f(X_2)).$$

From this, and equation (17) above, it follows that $X_1 = X_2$.

□

Theorem 13.3.4. $\wp(\mathbb{N}) \sim \mathbb{R}$.

Proof. By the Schröder-Bernstein Theorem, to prove $\wp(\mathbb{N}) \sim \mathbb{R}$ we need to prove two things: $\wp(\mathbb{N}) \preceq \mathbb{R}$ and $\mathbb{R} \preceq \wp(\mathbb{N})$. In the proof we are going to use the decimal representation of real numbers. Let us first recall the main properties of decimal representations:

Every real number $x \in [0, 1)$ has a decimal representation

$$x = .x_1x_2x_3\cdots$$

where each x_i is a decimal digit, i.e,

$$x_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

For example we have

$$\frac{1}{3} = .333333\cdots \qquad \text{or} \qquad \frac{1}{\sqrt{2}} = .707106781\cdots$$

If we do not allow repeating 9's, then the decimal representation of x is unique.
Without excluding repeating 9's, some numbers may have two decimal representations. For example:

$$.1399999999999\cdots = .14000000000000\cdots$$

- **Proof of $\wp(\mathbb{N}) \preceq \mathbb{R}$.** Consider the function $f : \wp(\mathbb{N}) \to \mathbb{R}$ which is defined as follows: for every $X \subseteq \mathbb{N}$, set

$$f(X) = .x_0x_1x_2x_3x_4\cdots$$

where

$$x_i = \begin{cases} 1 & \text{if } i \in X, \\ 2 & \text{if } i \notin X. \end{cases}$$

We now prove that the function f is injective. Let $X, Y \subseteq \mathbb{N}$, and suppose that

$$f(X) = .x_0x_1x_2x_3\cdots \quad \text{and} \quad f(Y) = .y_0y_1y_2y_3\cdots$$

If $f(X) = f(Y)$, then for every $i \in \mathbb{N}$ we have $x_i = y_i$. This implies that

$$\forall_{i\in\mathbb{N}}\ (i \in X \Leftrightarrow i \in Y),$$

i.e., $X = Y$.

- **Proof of $\mathbb{R} \preceq \wp(\mathbb{N})$.** By Proposition 10.1.5 we have $\mathbb{N} \sim \mathbb{N} \times \mathbb{N}$, and therefore by Lemma 13.3.3 we have $\wp(\mathbb{N}) \sim \wp(\mathbb{N} \times \mathbb{N})$. Also, by Remark 13.1 we have $[0, 1) \sim \mathbb{R}$. Consequently, it is enough to prove that

$$[0, 1) \preceq \wp(\mathbb{N} \times \mathbb{N}).$$

To prove the latter statement, we construct an injection

$$g : [0, 1) \to \wp(\mathbb{N} \times \mathbb{N}).$$

This function is defined as follows: for every $x \in [0, 1)$, we write its unique decimal representation

$$x = .x_0x_1x_2x_3x_4\cdots \text{ where } \ x_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

and set

$$g(x) = \{\ (i, x_i)\ :\ i \in \mathbb{N}\ \}. \tag{18}$$

It is easy to see why g is an injection: suppose that $x, y \in [0, 1)$ such that $g(x) = g(y)$. If we write

$$x = .x_0x_1x_2\cdots \text{ and } y = .y_0y_1y_2\cdots$$

then from the definition of g in equation (18) it follows that $x_i = y_i$ for every $i \in \mathbb{N}$; i.e., $x = y$. □

Exercise.

a. Let $\wp_{\text{fin}}(\mathbb{N})$ denote the set of finite subsets of $\mathbb{N}$, i.e.,

$$\wp_{\text{fin}}(\mathbb{N}) = \{\ A \subseteq \mathbb{N}\ :\ |A| < \infty\ \}.$$

Prove that $\wp_{\text{fin}}(\mathbb{N}) \sim \mathbb{N}$.

b. Prove that $\text{Func}(\{0, 1\}, \mathbb{N}) \sim \mathbb{R}$. (Hint: find a bijection between $\text{Func}(\{0, 1\}, \mathbb{N})$ and $\wp(\mathbb{N})$.)

c. Prove that $\text{Func}(\mathbb{N}, \mathbb{N}) \sim \mathbb{R}$. (Hint: use the graph of f, i.e., $\text{Grp}(f) \subseteq \mathbb{N} \times \mathbb{N}$, to find an injection from $\text{Func}(\mathbb{N}, \mathbb{N})$ into $\wp(\mathbb{N} \times \mathbb{N})$.)

13.4 A Property of Infinite Sets

You saw in Propisition 10.1.5 and Proposition 13.2.3 that $\mathbb{N} \times \mathbb{N} \sim \mathbb{N}$ and $\mathbb{R} \times \mathbb{R} \sim \mathbb{R}$. It is now natural to ask the following question:

Is it true that $A \times A \sim A$ for every infinite set A?

The answer to this question is positive, but the proof, which is based on the Axiom of Choice (and Zorn's Lemma) is not easy. This proof is not given in these notes. Nevertheless, we give a list of results about the cardinality of infinite sets, and prove one of them to show how Zorn's Lemma can be used as a tool.

Proposition 13.4.1. *If A is an infinite set then $A \times \{0,1\} \sim A$.*

Proof. Recall that

$$A \times \{0,1\} = \{\ (a,b)\ :\ a \in A\ \wedge\ b \in \{0,1\}\ \}.$$

Clearly $A \preceq A \times \{0,1\}$, and by the Schröder-Bernstein Theorem it is enough to prove that

$$A \times \{0,1\} \preceq A. \tag{19}$$

Consider the partially ordered set $(\Omega, \lhd)$ where

$$\Omega = \{\ f : X \times \{0,1\} \to X\ :\ X \subseteq A \text{ and } f \text{ is injective }\}$$

and the partial order $\lhd$ is defined as follows : for every two elements $f, g \in \Omega$,

$$f \lhd g\ \Leftrightarrow\ f \text{ is a restriction of } g.$$

Note that $\Omega \neq \varnothing$: Since A is infinite, there exists an injection $\psi : \mathbb{N} \to A$, and we have

$$\psi(\mathbb{N}) \times \{0,1\} \preceq \mathbb{N} \times \mathbb{N} \sim \mathbb{N} \preceq \psi(\mathbb{N})$$

which means that there exists an injection $g_\psi : \psi(\mathbb{N}) \times \{0,1\} \to \psi(\mathbb{N})$.

It is also easily checked that every totally ordered subset of $(\Omega, \lhd)$ has an upper bound in Ω. Therefore by Zorn's Lemma, $(\Omega, \lhd)$ has maximal elements. Let

$$f_\circ : X_\circ \times \{0,1\} \to X_\circ$$

be a maximal element of $(\Omega, \lhd)$. We now prove the following statement:

$$A \setminus X_\circ \textit{ is finite.} \tag{20}$$

The proof of statement (20) is by contradiction. Assume, on the contrary, that $A \setminus X_\circ$ is infinite. Therefore there exists an injection $\beta : \mathbb{N} \to A \setminus X_\circ$. We now have $\beta(\mathbb{N}) \sim \mathbb{N}$ and therefore

$$\beta(\mathbb{N}) \times \{0,1\} \sim \mathbb{N} \times \{0,1\} \sim \mathbb{N} \sim \beta(\mathbb{N}).$$

Consequently, there exists a bijection

$$g_\beta : \beta(\mathbb{N}) \times \{0,1\} \to \beta(\mathbb{N}).$$

Now the function

$$f_\circ \cup g_\beta : (X_\circ \times \{0,1\}) \cup (\beta(\mathbb{N}) \times \{0,1\}) \to X_\circ \cup \beta(\mathbb{N})$$

is an element of Ω, and we have

$$f_\circ \lhd f_\circ \cup \psi_1$$

which contradicts maximality of $f_\circ$. The proof of statement (20) is complete.

We now finish the proof of Proposition 13.4.1. Recall that it is enough to prove statement (19). Statement (20) implies that $X_\circ \times \{0,1\}$ is infinite and

$$(A \setminus X_\circ) \times \{0,1\} \preceq \mathbb{N}.$$

Observe that

$$A \times \{0,1\} = (X_\circ \times \{0,1\}) \cup ((A \setminus X_\circ) \times \{0,1\})$$

and by Corollary 11.2.2 we have

$$A \times \{0,1\} \sim X_\circ \times \{0,1\} \preceq X_\circ \preceq A.$$

□

Theorem 13.4.2.

a. If A is an infinite set then $A \times A \sim A$.

b. If A and B are sets such that $A \preceq B$ and B is infinite then $A \cup B \sim B$.

c. If A and B are sets such that $A \prec B$ and B is infinite then $B \setminus A \sim B$.

d. If M is an infinite set and $S = \{A_i \,:\, i \in I\}$ is a family of sets such that

$$I \preceq M \text{ and } \forall_{i \in I}(A_i \preceq M)$$

then $\cup S \preceq M$.

Exercise.

a. Let A be an infinite set. Use Theorem 13.4.2 and induction to prove that for every $n \in \mathbb{N}^+$, we have $A^n \sim A$. (Here $A^n = \underbrace{A \times \cdots \times A}_{n \text{ times}}$.)

b. Let A be an infinite set and $\wp_{\text{fin}}(A)$ be the set of finite subsets of A, i.e.,

$$\wp_{\text{fin}}(A) = \{\ X \subseteq A\ :\ X \text{ is finite. }\}$$

Prove that $\wp_{\text{fin}}(A) \sim A$.

c. Let A be an infinite set, and $\text{Func}_{\text{c}}(\mathbb{N}, A)$ be the set of functions which are ultimately constant. In other words,

$$\text{Func}_{\text{c}}(\mathbb{N}, A) = \{\ f : \mathbb{N} \to A\ :\ \exists_{m \in \mathbb{N}} \exists_{a \in A} \forall n \in \mathbb{N}\ (n > m \Rightarrow f(n) = a)\ \}$$

Prove that $\text{Func}_{\text{c}}(\mathbb{N}, A) \sim A$.

14 The Universal Debugger

If you have ever done any programming with a computer, you know how annoying it can be. Sometimes you write a computer program, you check the code several times to make sure there are no errors, and then you run it on your computer: you leave the computer for a few hours, then come back to check if it has finished the job. You realize that it is still running your program ... you leave again for a few hours, then come back to see if it is done ... still running the program. You let the program run the computer overnight, while closing all other applications, hoping that the job will be done by tomorrow morning. Tomorrow morning : first thing you do after you get out of bed is check your computer ... still running the program ... Now you are starting to wonder : is it really working properly or is the program trapped in an infinite loop?

The situation becomes more complicated when a program terminates on test input data but gets trapped in an infinite loop on the *real* input data. For many computer programmers, having a tool to determine if a program will get trapped in an infinite loop or will terminate is a dream. But is it really possible to realize this dream? Is it possible to develop a software that does this?

In this section we are going to see a *mathematically rigorous* proof that such a software *cannot* exist. This astonishing fact was first discovered and proved by Alan Turing, who is nowadays considered as one of the pioneers of computer science. The ingenious proof of Turing uses Cantor's idea of diagonalization.

14.1 The problem statement

Let us first make the problem statement more precise.

- **What is a computer?** By a computer we just mean an ordinary computer! Your desktop or laptop would be good enough! [4]
- **What is a computer language?** A computer language is a series of instructions that the computer understands. For example, the command

  ```
  OUTPUT{``Hello world!''}
  ```

  means that the computer should output the phrase `Hello world!`. The command

  $$x := 1$$

[4] However, mathematicians (and theoretical computer scientists) usually like to use a theoretical model called the *Turing machine*. It is a very basic contraption: an infinitely long tape which consists of a sequence of cells to record information (this tape plays the role of the memory), and a *head* to read and write on the tape (this head plays the role of the microprocessor). Initially, on every cell a "0" or a "1" is written. The head can change a "0" to a "1" or vice versa.

means that the computer should set the value of a variable x equal to 1. The command

$$\texttt{IF } a \leq b \texttt{ THEN } b := a + 1$$

does the following: if $a \leq b$ then this command sets the value of the variable b equal to $a + 1$, and if $a \not\leq b$, then this command does nothing.

As you see, a computer program is a language, somewhat close to human language, which allows us to communicate with a computer.

- **What is a computer programme?** It is a list of commands in the computer language which are stored in a text file (i.e., a `.txt` file). Each line of the text file contains exactly one command. To run the program, the computer starts from the first line of the file, and follows the instructions line by line. (But note that a command such as

  `GO TO LINE 15`

  means that the computer should skip to line 15 of the file.) The command `END` indicates that a computer program has terminated, and the computer should stop.

- **What is a program input?** Every program receives an input as a text file. (Again, by a text file I simply mean a `.txt` file.) For example, consider the following program:

```
01  INPUT{p}
02  q := p * p
03  OUTPUT{q}
04  END
```

Let us give a name to the above program : from now on we will call it `MySquare`. You can then make a text file (called, say, `myinput.txt`) and use it as input for `MySquare` on this input file.

Here is what happens after running `MySquare` on `myinput.txt`:

- ⋆ The first line receives the input and puts it in a variable p.
- ⋆ The second line sets q equal to p^2. (If the value of p is not a number, then the program terminates with an error message. See below for an example.)
- ⋆ The third line outputs the value of q, which is just p^2.
- ⋆ The fourth line tells the computer that the program has terminated, and it should stop.

Here are some examples of input files and what happens with running the program:

- ⋆ If `myinput.txt` contains the number `5`, then the programme terminates and outputs the number `25`.
- ⋆ If `myinput.txt` contains the number `.1`, then the programme terminates and outputs the number `.01`.
- ⋆ If `myinput.txt` contains the word `three`, then the programme still terminates, but outputs an error message similar to the following:

```
INVALID INPUT : CANNOT DO MATH OPERATION ON NON-NUMERIC VARIABLE.
```

Note that a programme may receive an input without really using it. For example, look at the following program:

```
01  INPUT{x}
02  OUTPUT{``I am done!''}
03  END
```

This program always generates the output `I am done!` and then terminates regardless of what the content of the input file is.

- **What is an infinite loop**? An infinite loop is simply a never-ending loop in a computer program. Look at the following program, called `BadMySquare`, which is just a minor modification of `MySquare`:

```
01  INPUT{p}
02  IF p = 1 THEN GO TO LINE 02
03  q := p * p
04  OUTPUT{q}
05  END
```

You can see that if the input file `myinput.txt` contains the number `1`, then the above program never terminates (because it will get trapped in the second line).

Let us summarize our setup so far

a. Every computer programme is stored in a `.txt` file, and receives an input file in the form of another `.txt` file.

b. Every `.txt` file can be used as the input to every program. Of course, certain input files may not yield the desired results. For example, the program may terminate with an error message, may terminate with an output which does not make sense, or may get trapped in an infinite loop.

We are now ready to pose our question more precisely:

Problem : existence of the ideal debugger

Can we write a computer application `IdealDebugger` which receives a program `Prog` and and an input for `Prog`, and determines if the execution of `Prog` on the given input evenually terminates or gets trapped in an infinite loop?

More precisely, what we expect from `idealDebugger` is the following:

a. `IdealDebugger` receives two text files: `prog.txt` and `input.txt`.

b. If the execution of `prog.txt`, with `input.txt` as input, terminates, then `IdealDebugger` terminates and outputs the word `TERMINATE`.

c. If the execution of `prog.txt`, with `input.txt` as input, gets trapped in an infinite loop, then `IdealDebugger` terminates and outputs the word `LOOP`.

Theorem 14.1.1. *The program* `IdealDebugger` *cannot exist!*

Proof. The proof is by contradiction. Assume, on the contrary, that `IdealDebugger` exists. Consider the following program (which we will call `Nasty`):

```
01  INPUT{p}
02  IF IdealDebugger{p,p} OUTPUTS ``TERMINATE'' THEN GO TO LINE 02
03  IF IdealDebugger{p,p} OUTPUTS ``LOOP'' THEN GO TO LINE 04
04  END
```

In other words,

- If running p with p as input terminates, then `Nasty` gets trapped in an infinite loop.

- If running p with p as input gets trapped in an infinite loop, then `Nasty` terminates.

Suppose that `Nasty` is stored in a `.txt` file called `nasty.txt`. We know that any text file can be used as the input of any program. In particular, we can run `Nasty` with `nasty.txt` as input. Now we ask the following question:

> What happens after we run `Nasty` with `nasty.txt` as input? Will the computer stop? Or does it get trapped in an infinite loop?

Look at the following reasoning:

- Assume that running `Nasty` with `nasty.txt` as input terminates. This means that the output of

  `IdealDebugger(nasty.txt,nasty.txt)`

  should be `LOOP`. Therefore if we run `Nasty` with `nasty.txt` as input the computer will get trapped in an infinite loop. This is clearly a contradiction.

- Now assume that running `Nasty` with `nasty.txt` as input gets trapped in an infinite loop. This means that the output of

  `IdealDebugger(nasty.txt,nasty.txt)`

  should be `TERMINATE`. Therefore if we run `Nasty` with `nasty.txt` as input the computer will eventually stop. Again, this is a contradiction.

We have shown that the assumption that the application `IdealDebugger` exists leads to contradiction. Therefore `IdealDebugger` cannot exist. □

Exercise.

Imagine that you are taking a course in computer programming, and in an assignment, your professor is asking you to write a computer program that (may not terminate but eventually) generates elements of a given subset of natural numbers. (The program is not allowed to receive input.) For example, the following program generates the set of even natural numbers (i.e., $2\mathbb{N}$):

```
01   i := 0
02   OUTPUT(i)
03   i := i + 2
04 GO TO LINE 02
```

The line $i := i + 1$ tells the computer to increase the value of the variable i by 2. (If you have ever done any programming, you know that this method of changing the value of a variable is standard and accepted in almost all computer languages.)

Similarly, the set

$$\{0, 1, 4, 9, 16, 25, 36, ...\} = \{n^2 \ : \ n \in \mathbb{N}\}$$

can be generated by the following program:

```
01   i := 0
02   OUTPUT(i*i)
03   i:=i+1
04 GO TO LINE 02
```

How about the set of prime numbers (i.e., $\{2, 3, 5, 7, 11, 13, 17, 19, ...\}$)? It is harder, but still possible to write a computer program that generates the set of prime numbers. Here is an example:

```
01   i := 1
02   i := i + 1
03   j := 1
04   j := j + 1
05   IF j < i AND (i MODULO j) = 0 THEN GO TO LINE 02
06   IF j < i AND (i MODULO j) ≠ 0 THEN GO TO LINE 04
07   IF j = i THEN OUTPUT(i)
08   GO TO LINE 02
```

Now, here is a natural question: can we generate *every* subset of natural numbers by a computer program? (Hint: show that the set of all possible computer programs is countable. Then use Theorem 13.3.4.)

15 The Continuum Hypothesis

Recall that:

- We have seen that $\mathbb{N} \prec \mathbb{R}$.
- We have also seen that many familiar subsets of $\mathbb{R}$ are either equinumerous to $\mathbb{N}$ (e.g., $\mathbb{Z} \sim \mathbb{N}$ and $\mathbb{Q} \sim \mathbb{N}$) or equinumerous to $\mathbb{R}$ (e.g., $[0, 1) \sim \mathbb{R}$).

This leads to the following natural question:

Does there exist a subset A of $\mathbb{R}$ such that $\mathbb{N} \prec A$ and $A \prec \mathbb{R}$?

15.1 A surprising answer

This question is a very difficult and subtle one. Cantor believed that the answer to it is "No". He devoted many years of his life (and eventually his mental health) to efforts in proving his conjecture.

Eventually, this question, which is now known as the *continuum hypothesis*, was answered by Kurt Gödel (1938) and Paul Cohen (1963). The final answer was quite unexpected. Let us first state the Continuum Hypothesis in the form it is usually mentioned:

Continuum Hypothesis. There does not exist a subset A of $\mathbb{R}$ such that $\mathbb{N} \prec A$ and $A \prec \mathbb{R}$.

Gödel proved the following statement:

Gödel (1938). In ZFC, it is impossible to disprove the Continuum Hypothesis.

Note the subtlety of this result: he could *not* prove the Continuum Hypothesis, nor could he prove that it can be proved. Gödel was not quite satisfied with his result. He hoped that someday someone could *prove* the continuum hypothesis.

After several years. Gödel's hopes were shattered by P. Cohen, who proved the following statement:

Cohen (1963). In ZFC, it is impossible to prove the Continuum Hypothesis.

In other words, the Continuum Hypothesis is an *undecidable* statement. It is totally independent of the other axioms of ZFC. Assuming that it is true would not lead to a contradiction, nor does the assumption that it is false!

But the Continuum Hypothesis seems so fundamental and natural that having a definite answer to it is very tantalizing. It is like someone tells you that the sentence *"God exists"* is undecidable. Even if we don't know how to prove (or disprove) that God exists, even if we are not convinced by any of the arguments for or against God's existence, the ultimate answer to this question should be a flat "Yes" or "No", shouldn't it?!

Gödel believed that the reason for the undecidability of the Continuum Hypothesis is that the axioms of the ZFC system are not chosen properly. Although ZFC is currently the only widely accepted system of axioms for set theory, the quest for other natural axiomatic systems which tell us more about the nature of the Continuum Hypothesis has become an active area of research for set theorists.